

# Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations

A proof of the remaining strict-growth conjecture  
and a two-term large-bound asymptotic

Research report prepared for Vladimir Reshetnikov

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## Abstract

Let  $a_n^{(m)}$  count permutations  $\pi \in S_n$  avoiding the pattern 132 and satisfying  $|\pi_{i+1} - \pi_i| \leq m$ . Nadler conjectured that their exponential growth constants increase strictly from 1 to 4. Mayama and Akita established a finite-state enumeration, the existence of the growth constants, non-strict monotonicity, and convergence to 4. We prove the missing strict inequality for every adjacency bound. The argument embeds each of the two irreducible cyclic state components at bound  $m$  into a proper principal submatrix of its counterpart at bound  $m + 1$  by increasing both finite endpoint thresholds by one. Perron–Frobenius comparison then gives strict growth in *each* component, independently of which component dominates.

A separate counting argument gives

$$4 - \alpha_m = \frac{2 \log m + 4 \log \log m + O(1)}{m}.$$

It compares the full class with a scalar Catalan majorant and an injectively encoded family of variable-length skew blocks. A truncated-series root lemma also bounds the possible constant term. Complete proofs, exact generating functions for  $m \leq 4$ , integer sequences, and independently checked rational Perron certificates accompany the report.

*Addition of 28 September 2026 (batch 38).* Part II, from a later manuscript, proves  $\alpha_m < \beta_m$  for the root  $\beta_m^{-1}$  of  $\sum_{k=1}^m C_{k-1}x^k = 1$  and  $\beta_m - \alpha_m = O((\log m)^2/m^2)$ ; hence the constant term converges,  $m(4 - \alpha_m) - 2 \log m - 4 \log \log m \rightarrow 2 \log \pi$ , and the growth deficit has an expansion to every inverse-logarithmic order.

*Addition of 29 September 2026 (batch 43).* Part III, from a further manuscript, takes the bound near its trivial value,  $m = n - d$ . The deficit  $n - M_n$  of the largest adjacent jump of a uniform 132-avoider converges in total variation, at the sharp rate  $n^{-1/2}$ , to an explicit algebraic law with  $\mathbb{P}(D > k) \sim 3/\sqrt{\pi k}$ ; uniformly,  $a_n^{(n-d)}/C_n = 3/\sqrt{\pi d} + O(d^{-3/2} + n^{-1/2})$ . This answers the joint limit in  $n$  and  $m$  near the boundary only; the bulk regime stays open.

*Addition of 29 September 2026 (batch 52).* Part IV, merged from two manuscripts, finds the limit profile of  $\sqrt{n} \mathbb{P}(n - M_n > un)$  and the leading constant of every moment of order  $p > 1/2$ ; for example  $\mathbb{E}(n - M_n) \sim 3\sqrt{n/\pi}$ .

**Scope and attribution.** The finite-state system and fixed- $m$  spectral description are established work, not claimed as new here. The contributions developed in this report are the all- $m$  strict comparison and the refined asymptotic bounds. A targeted literature check found no later resolution of the strict inequality in the sources inspected. This is a research proof with reproducible checks, not a claim of external peer review or an exhaustive novelty certification.

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# 1 The problem and the results

A permutation  $\pi \in S_n$  avoids 132 when there are no indices  $i < j < k$  for which  $\pi_i < \pi_k < \pi_j$ . Fix an integer  $m \geq 1$  and put

$$\mathcal{A}_n^{(m)} = \{\pi \in \text{Av}_n(132) : |\pi_{i+1} - \pi_i| \leq m \text{ for } 1 \leq i < n\}, \quad a_n^{(m)} = |\mathcal{A}_n^{(m)}|.$$

We set  $a_0^{(m)} = 1$  and use the ordinary generating function

$$A_m(x) = \sum_{n \geq 0} a_n^{(m)} x^n.$$

The growth constant is

$$\alpha_m = \lim_{n \rightarrow \infty} (a_n^{(m)})^{1/n}. \quad (1)$$

The order of limits matters:  $n \rightarrow \infty$  is taken for each fixed  $m$ , and only afterwards do we let  $m \rightarrow \infty$ . The trivial identity  $a_n^{(m)} = C_n$  for  $m \geq n - 1$  does not by itself settle this iterated limit. Here  $C_n = \binom{2n}{n}/(n+1)$  is the  $n$ th Catalan number. [Added 29 September 2026, batch 43: Part III studies the joint regime next to that identity,  $m = n - d$  with  $d$  fixed or  $d = o(n)$  (theorems 22.1 and 22.2); it does not bear on the iterated limit taken here.]

## 1.1 Which part was still open?

Nadler's Conjecture 2 in Section 5.2 of [1] asks for the existence of (1), strict increase  $1 = \alpha_1 < \alpha_2 < \dots < 4$ , and limit 4. Mayama and Akita subsequently proved rationality for all  $m$ , existence of the limit, non-strict increase, and  $4 - \alpha_m = O(\log m/m)$  [2, Theorems 2.4, 4.12, and 4.14]. Their monotonicity theorem states  $\alpha_m \leq \alpha_{m+1}$ , not strict inequality.

Merely finding additional permutations at bound  $m+1$  is insufficient: sequences  $2^n$  and  $2^n + 1$  differ strictly at every index but have identical exponential growth. The needed argument must show that the extra freedom changes the exponential rate.

**Theorem 1.1** (Strict increase). *For every integer  $m \geq 1$ ,*

$$\alpha_{m+1} > \alpha_m.$$

*In the two-component spectral representation described below, both component rates increase strictly: for every  $m \geq 2$ ,*

$$\lambda_U(m+1) > \lambda_U(m), \quad \lambda_V(m+1) > \lambda_V(m).$$

*Consequently Nadler's growth-constant conjecture holds in full.*

**Theorem 1.2** (Two-term approach to the Catalan limit). *As  $m \rightarrow \infty$  through integers,*

$$4 - \alpha_m = \frac{2 \log m + 4 \log \log m + O(1)}{m}. \quad (2)$$

*In particular,  $4 - \alpha_m \sim 2 \log m/m$ . All logarithms in this report are natural logarithms.*

**Theorem 1.3** (A window for the next constant). *Define  $E_m = m(4 - \alpha_m) - 2 \log m - 4 \log \log m$  for  $m \geq 2$ . Then*

$$2 \log \pi - 4 \log 2 \leq \liminf_{m \rightarrow \infty} E_m \leq \limsup_{m \rightarrow \infty} E_m \leq 2 \log \pi. \quad (3)$$

*No convergence of  $E_m$  is asserted.*

[Added 28 September 2026, batch 38: the last sentence describes the report at its pinned state 6c9175a54. Part II proves that  $E_m$  converges to the upper endpoint,  $E_m \rightarrow 2 \log \pi$  (theorem 10.2 and corollary 15.3).]

The strict comparison uses a finite-state embedding. The asymptotic results use scalar comparison series and do not require deciding which finite-state component has the larger rate. The distinction is important: the all- $m$  comparison of  $\lambda_U(m)$  with  $\lambda_V(m)$  is a separate question.

## 2 The exact endpoint recursion

This section develops the endpoint representation of Mayama and Akita [2, Section 2], including the counting details needed later. It is included to make the subsequent proofs independently readable.

### 2.1 Decomposition at the maximum

If  $\pi = L n R$  avoids 132, every entry of  $L$  exceeds every entry of  $R$ : otherwise an entry of  $L$ , then  $n$ , then an entry of  $R$  forms 132. Conversely, if this separation holds and both standardized sides avoid 132, their concatenation with  $n$  avoids 132.

Suppose  $R$  is nonempty and the maximum is at position  $k$ . Then  $L$  has length  $k - 1$ , its values are  $n - k + 1, \dots, n - 1$ , and  $R$  has values  $1, \dots, n - k$ . The jump from  $n$  to the first entry of  $R$  is at least  $k$ . Therefore

$$R \neq \emptyset \implies 1 \leq k \leq m. \quad (4)$$

The alternative  $R = \emptyset$  means appending the maximum to an avoider of length  $n - 1$ .

### 2.2 Cumulative endpoint thresholds

Let

$$B_m = \{0, 1, \dots, m - 1, \infty\}.$$

For  $p, q \in B_m$ , let  $T_{p,q}^{(m)}(n)$  count the permutations in  $\mathcal{A}_n^{(m)}$  satisfying

$$n - \pi_1 \leq p, \quad n - \pi_n \leq q.$$

A threshold  $\infty$  imposes no restriction. Define

$$F_{p,q}^{(m)}(x) = \sum_{n \geq 1} T_{p,q}^{(m)}(n) x^n.$$

A negative threshold gives the zero series, and  $\infty - k = \infty$ . We suppress the superscript  $(m)$  where it is fixed.

For  $k \geq 2$  define the nonnegative integer

$$c_{k,p} = |\{\sigma \in \text{Av}_{k-1}(132) : k - \sigma_1 \leq p\}|, \quad c_{1,p} = 1 \quad (5)$$

for nonnegative or infinite  $p$ ; for negative  $p$  set  $c_{k,p} = 0$ . These coefficients do not otherwise depend on  $m$ . Two properties will be crucial:

$$c_{k,p+1} \geq c_{k,p}, \quad c_{k,\infty} = C_{k-1}. \quad (6)$$

For finite  $p \geq 0$  and  $k \geq 2$ , an explicit formula is

$$c_{k,p} = \sum_{j=1}^{\min(p,k-1)} \frac{j}{k-1} \binom{2k-j-3}{k-2}. \quad (7)$$

A generating-function proof of this formula appears in section 5.3; no numerical fitting is involved.

**Proposition 2.1** (Endpoint recurrence). *For all  $p, q \in B_m$ ,*

$$F_{p,q} = x + \sum_{k=1}^m c_{k,p} x^k F_{m-k,q-k} + x F_{p-1,m-1}. \quad (8)$$

Moreover  $A_m(x) = 1 + F_{\infty,\infty}(x)$ .

*Proof.* The singleton permutation contributes  $x$  to every nonnegative threshold state. For length at least two, first suppose  $R \neq \emptyset$  in the maximum decomposition. By (4),  $k \leq m$ . The left block has  $k - 1$  values in an interval of that length, so its internal jumps, and its jump to  $n$ , are automatically at most  $m$ . Its first-entry restriction gives  $c_{k,p}$  choices; for  $k = 1$  the first entry is  $n$  and the left block is empty.

The right block has length  $n - k$ . Writing its first and last entries as  $R_1$  and  $R_{n-k}$ , the two remaining restrictions are exactly

$$(n - k) - R_1 \leq m - k, \quad (n - k) - R_{n-k} \leq q - k.$$

The first inequality is the jump bound from  $n$  to  $R_1$ ; the second is the last-entry threshold for the whole permutation. This gives  $c_{k,p} x^k F_{m-k, q-k}$ .

If  $R = \emptyset$ , remove the final maximum. The first deficiency increases by one when the maximum is appended, giving threshold  $p - 1$  for the smaller permutation. Its last deficiency must be at most  $m - 1$  to permit the last jump. The final entry of the whole permutation is  $n$ , so any nonnegative  $q$  is satisfied. This is  $x F_{p-1, m-1}$ . The cases are disjoint and reversible.  $\square$

Let  $W_m(x)$  have rows and columns indexed by  $B_m^2$ , with

$$(W_m)_{(p,q),(r,s)} = \sum_{k=1}^m c_{k,p} x^k \mathbf{1}_{(r,s)=(m-k, q-k)} + x \mathbf{1}_{(r,s)=(p-1, m-1)}. \quad (9)$$

Invalid negative thresholds are omitted. Rows are the left-hand states of (8); columns are the right-hand states. Thus

$$F = x\mathbf{1} + W_m(x)F, \quad F = x(I - W_m(x))^{-1}\mathbf{1}. \quad (10)$$

Every entry of  $W_m$  has zero constant term, so the inverse is well-defined as a formal power series. This also proves rationality and gives an exact algorithm for computing  $A_m$ .

### 3 Cyclic components and growth constants

The following fixed- $m$  structure is established in [2, Proposition 2.7 and Theorem 4.12]. We supply elementary component and transfer-graph arguments to make the spectral input explicit.

#### 3.1 The two relevant components

For  $m \geq 2$  put

$$U_m = \{(p, \infty) : 0 \leq p \leq m - 1\}, \quad (11)$$

$$V_m = \{(p, q) : 0 \leq q < p \leq m - 1\} \cup \{(p, m - 1) : 0 \leq p \leq m - 2\}. \quad (12)$$

Their sizes are  $m$  and  $(m - 1)(m + 2)/2$ . Write  $W_{U_m}$  and  $W_{V_m}$  for the corresponding principal matrices. A directed edge means a nonzero term in a row of (9), directed from its row state to its column state.

**Lemma 3.1** (Component structure). *At every real  $x > 0$ , both  $W_{U_m}(x)$  and  $W_{V_m}(x)$  are irreducible. The only other cyclic component of the full state graph is the singleton  $I_m = (\infty, m - 1)$ , whose internal matrix is  $(x)$ . Each of these three components is reachable from  $(\infty, \infty)$ .*

*Proof.* Within  $U_m$ , every state has a  $k = 1$  edge to  $(m - 1, \infty)$ . From that state, the  $k = 1, \dots, m$  edges reach every state of  $U_m$ , since  $c_{k, m-1} > 0$ . Hence  $U_m$  is strongly connected.

For  $V_m$ , call the states with  $q < p$  lower states and the states  $(p, m - 1)$  top states. Every lower state has an append edge to a top state. Every top state has a  $k = 1$  edge to

$$e = (m - 1, m - 2).$$

It remains to show that  $e$  reaches every state. For  $m = 2$  this is the immediate two-state cycle. Suppose  $m \geq 3$ . Appending at  $e$  reaches  $(m - 2, m - 1)$ ; its split edges reach all lower states  $(p, p - 1)$ ,  $1 \leq p \leq m - 1$ . Their append edges reach every top state. Inductively, suppose all lower states of gap at most  $d - 1$  are reachable, where  $2 \leq d \leq m - 1$ . The state  $(m - 1, m - d)$  is then reachable. Its split edge with  $k = m - p$  reaches  $(p, p - d)$  for every  $d \leq p \leq m - 1$ . Thus every lower gap is reached.

For finite  $q$ , every valid split edge ends at a lower state, because  $(m - k) - (q - k) = m - q > 0$ . Every valid append edge with finite  $p$  ends at a top state. Therefore a finite-finite state outside  $V_m$  cannot be on a cycle. A state  $(\infty, q)$  with finite  $q$  has append target  $I_m$  and split targets in  $V_m$ ; among these states only  $I_m$  has a self-loop. No edge returns to  $(\infty, \infty)$ . Edges with second coordinate  $\infty$  stay in  $U_m$  or leave it by an append edge, and cannot return from a finite second coordinate. This exhausts all possibilities.

Finally, the  $k = 1$  split from  $(\infty, \infty)$  reaches  $U_m$ , its append edge reaches  $I_m$ , and a  $k = 1$  split from  $I_m$  reaches  $e \in V_m$ .  $\square$

### 3.2 The strict matrix comparison used in the proof

We use the Perron–Frobenius theorem for finite nonnegative matrices: an irreducible matrix has positive left and right eigenvectors at its spectral radius; a primitive matrix has powers asymptotic to a positive rank-one matrix times the corresponding power of that radius.

**Lemma 3.2** (Proper principal comparison). *Let  $M$  be an irreducible nonnegative matrix and let  $S$  be a nonempty proper subset of its indices. If  $B = M[S, S]$ , then  $\text{spr}(B) < \text{spr}(M)$ . If  $0 \leq N \leq B$  entrywise, then  $\text{spr}(N) < \text{spr}(M)$  as well.*

*Proof.* Take a nonnegative nonzero Perron vector  $v$  of  $B$  and extend it by zeros to a vector  $w$  on all indices of  $M$ . Then  $Mw \geq \text{spr}(B)w$ . There is strict inequality in at least one coordinate: the support of  $w$  is nonempty and proper, and irreducibility supplies an edge from its complement to that support. If  $u$  is a positive left Perron vector of  $M$ , multiplication by  $u^\top$  gives

$$\text{spr}(M)u^\top w > \text{spr}(B)u^\top w.$$

Since  $u^\top w > 0$ , the first assertion follows. Entrywise monotonicity of the spectral radius gives  $\text{spr}(N) \leq \text{spr}(B)$ ; for example, this monotonicity follows by comparing powers and then taking matrix-norm roots.  $\square$

For  $C = U, V$ , there is a unique  $r_C(m) > 0$  satisfying

$$\text{spr}(W_{C_m}(r_C(m))) = 1, \quad \lambda_C(m) = r_C(m)^{-1}. \quad (13)$$

Indeed the spectral radius is continuous, tends to zero as  $x \downarrow 0$ , and tends to infinity as  $x \rightarrow \infty$  because the component has a positive cycle. If  $y > x > 0$ , every monomial has degree at least one, so  $W_{C_m}(y) \geq (y/x)W_{C_m}(x)$ . Its positive spectral radius is therefore strictly increasing.

**Proposition 3.3** (Fixed-bound growth). *For  $m \geq 2$  the limit (1) exists and*

$$\alpha_m = \max\{\lambda_U(m), \lambda_V(m)\}. \quad (14)$$

Also  $\lambda_U(m) > 1$ .

*Proof.* The state  $(m-1, \infty)$  has an internal self-loop of weight  $x$ . At  $x = 1$  its singleton principal matrix is  $(1)$ , and it is a proper principal matrix of  $W_{U_m}(1)$ . By lemma 3.2, the latter has spectral radius greater than one. Thus  $r_U(m) < 1$  and  $\lambda_U(m) > 1$ .

To justify the actual limit rather than only a limsup, replace each edge of weight  $cx^k$  by a directed path of  $k$  unit-length edges with total multiplicity  $c$ . Use fresh intermediate vertices for every edge. Let  $H$  be the resulting finite nonnegative adjacency matrix, and let  $b$  be one on original endpoint states and zero on intermediate states. With  $o$  the state  $(\infty, \infty)$ , (10) gives

$$a_n^{(m)} = e_o^T H^{n-1} b \quad (n \geq 1).$$

For an expanded irreducible component with Perron eigenvalue  $\lambda$ , eliminating the intermediate coordinates of a positive eigenvector gives  $z = W_{C_m}(1/\lambda)z$ . Its eigenvalue is therefore  $\lambda_C(m)$ . The expanded singleton component has eigenvalue one. The component classification and  $\lambda_U(m) > 1$  show that  $\text{spr}(H)$  is the right-hand side of (14).

The expanded  $U_m$  component is primitive, since it is strongly connected and has a length-one loop. For  $m \geq 3$ , the expanded  $V_m$  component has closed walks of lengths two and five. The first is

$$e \xrightarrow{1} (m-2, m-1) \xrightarrow{1} e.$$

For the second, split with  $k = 2$  on the middle state and then append and split with  $k = 1$ :

$$e \xrightarrow{1} (m-2, m-1) \xrightarrow{2} (m-2, m-3) \xrightarrow{1} (m-3, m-1) \xrightarrow{1} e.$$

The labels are edge lengths. All coefficients are positive, including when  $m = 3$ . Their greatest common divisor is one, so this component is primitive. When  $m = 2$ ,  $V_m$  is a two-cycle with rate one and is not dominant.

A dominant primitive component is reachable from  $o$ , and  $b$  is positive on its original states. Fixing an entrance path and then applying the primitive Perron theorem supplies  $a_n^{(m)} \geq c \text{spr}(H)^n$  for all sufficiently large  $n$ , with  $c > 0$ . A finite-matrix power bound gives  $a_n^{(m)} \leq Cn^D \text{spr}(H)^n$  for some fixed  $C, D$  (or, equivalently, an upper bound by  $(\text{spr}(H) + \varepsilon)^n$  for each  $\varepsilon > 0$ ). Taking  $n$ th roots proves (14) and the existence of the limit.  $\square$

## 4 A shift of endpoint thresholds proves strict increase

The key comparison preserves the split length  $k$ , not merely the underlying unweighted graph. Define

$$\phi(p) = \begin{cases} p+1, & p < \infty, \\ \infty, & p = \infty, \end{cases} \quad \Phi(p, q) = (\phi(p), \phi(q)).$$

The next lemma is the essential new observation.

**Lemma 4.1** (Coefficientwise shift embedding). *For every  $m \geq 2$  and  $C = U, V$ , the map  $\Phi$  embeds  $C_m$  into a proper subset of  $C_{m+1}$ . Under this identification,*

$$W_{C_m}(x) \leq W_{C_{m+1}}(x)[\Phi(C_m), \Phi(C_m)] \quad (15)$$

*coefficientwise in every matrix entry.*

*Proof.* A state  $(p, \infty)$  of  $U_m$  maps to  $(p+1, \infty)$  of  $U_{m+1}$ ; the state  $(0, \infty)$  of the latter is missing. A lower state  $(p, q)$  of  $V_m$  maps to a lower state  $(p+1, q+1)$  of  $V_{m+1}$ . A top state  $(p, m-1)$  maps to  $(p+1, m)$ , a top state at the new bound. The new top state  $(0, m)$  is missing, so that inclusion is also proper.

Consider a split term at row  $(p, q)$  of the old matrix, with column  $(m - k, q - k)$  and coefficient  $c_{k,p}x^k$ . At the new row  $\Phi(p, q)$ , the split with the same  $k$  has column

$$(m + 1 - k, \phi(q) - k) = \Phi(m - k, q - k),$$

whenever the old term is valid. Its coefficient is  $c_{k,\phi(p)}x^k \geq c_{k,p}x^k$ . For  $p = \infty$  the coefficients are equal. For a valid append term, the old column  $(p - 1, m - 1)$  maps to

$$(\phi(p) - 1, m),$$

which is exactly the new append column, and the weight remains  $x$ . Each term is preserved with at least its old multiplicity. Adding terms with the same row and column proves the coefficientwise assertion.  $\square$

*Proof of theorem 1.1.* Fix  $m \geq 2$ ,  $C \in \{U, V\}$ , and a positive real  $x$ . By lemmas 3.1, 3.2 and 4.1,

$$\begin{aligned} \text{spr}(W_{C_m}(x)) &\leq \text{spr}(W_{C_{m+1}}(x)[\Phi(C_m), \Phi(C_m)]) \\ &< \text{spr}(W_{C_{m+1}}(x)). \end{aligned}$$

At  $x = r_C(m)$  the first quantity is one. Strict increase of the new spectral radius as a function of  $x$  gives  $r_C(m + 1) < r_C(m)$ , hence  $\lambda_C(m + 1) > \lambda_C(m)$ .

Choose a component  $C$  realizing the maximum at bound  $m$ . Then

$$\alpha_{m+1} \geq \lambda_C(m + 1) > \lambda_C(m) = \alpha_m.$$

This works even if the dominant component changes between the two bounds.

For  $m = 1$ , any adjacent difference is  $\pm 1$ , and reversing direction would repeat a value. The only permutations of length  $n \geq 2$  are the increasing and decreasing ones, both avoiding 132. Thus  $\alpha_1 = 1$ . At  $m = 2$ , in the order  $(0, \infty), (1, \infty)$ ,

$$W_{U_2}(x) = \begin{pmatrix} 0 & x \\ x^2 & x \end{pmatrix}, \quad \det(I - W_{U_2}(x)) = 1 - x - x^3.$$

Its positive root is below one, so  $\alpha_2 \geq \lambda_U(2) > 1$ . This covers the remaining step.  $\square$

*Remark 4.2.* Strictness comes from a *proper principal extension of an irreducible component*. It does not require finding a strictly larger entry inside the old image. This is useful for the  $U$  component, where many coefficients are preserved exactly. Nor is a numerical lower bound on the gap between consecutive rates needed.

## 5 Scalar upper bounds and injective skew-block lower bounds

We next obtain two scalar comparison series whose root asymptotics have the same first two terms. Write  $P \leq Q$  for coefficientwise inequality of formal power series with nonnegative coefficients.

### 5.1 A Catalan majorant

Define

$$R_m(x) = x + \sum_{k=1}^m C_{k-1}x^k. \tag{16}$$

The separate initial  $x$  accounts for appending a maximum, whereas the  $k = 1$  term in the sum accounts for a maximum in the first position with a nonempty right block. Thus the coefficient of  $x$  in  $R_m$  is two.

**Proposition 5.1** (Scalar upper bound). *Let  $q_m > 0$  be the unique positive root of  $R_m(q_m) = 1$ . Then*

$$A_m(x) \leq 1 + \frac{x}{1 - R_m(x)}, \quad \alpha_m \leq q_m^{-1} < 4. \quad (17)$$

*Proof.* The maximum decomposition, with all endpoint restrictions discarded, gives for  $n \geq 2$

$$a_n^{(m)} \leq a_{n-1}^{(m)} + \sum_{k=1}^{\min(m, n-1)} C_{k-1} a_{n-k}^{(m)}.$$

The upper endpoint  $n - 1$  is essential: the right block in a split must be nonempty. Starting with  $a_1^{(m)} = 1$ , induction yields the generating-function majorant. Its radius is  $q_m$ : the denominator is nonzero for  $|x| < q_m$  by positivity, and it vanishes at  $q_m$  without numerator cancellation. Cauchy–Hadamard therefore gives  $\alpha_m \leq q_m^{-1}$ .

Finally,

$$R_m(1/4) < \frac{1}{4} + \sum_{k \geq 1} C_{k-1} 4^{-k} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} < 1.$$

Hence  $q_m > 1/4$  and the strict upper bound by four follows.  $\square$

## 5.2 Variable-length blocks with recoverable boundaries

Fix  $d \geq 1$  and  $m > d$ . An admissible block of length  $k$ , where  $1 \leq k \leq m - d$ , is a permutation  $\beta \in \text{Av}_k(132)$  such that

$$\beta_k = k, \quad k - \beta_1 \leq d.$$

There are  $c_{k,d}$  such blocks: for  $k \geq 2$ , deleting the last maximum leaves the permutations counted in (5); the singleton contributes  $c_{1,d} = 1$ .

For a sequence of blocks, assign consecutive value intervals in decreasing order. That is, every entry of an earlier block exceeds every entry of a later block. This operation is their *skew sum*. Let

$$K_{m,d}(x) = \sum_{k=1}^{m-d} c_{k,d} x^k. \quad (18)$$

**Proposition 5.2** (Injective block lower bound). *Let  $s_{m,d} > 0$  be the unique positive root of  $K_{m,d}(s_{m,d}) = 1$ . Then*

$$\frac{1}{1 - K_{m,d}(x)} \leq A_m(x), \quad s_{m,d}^{-1} \leq \alpha_m \leq q_m^{-1}. \quad (19)$$

*Proof.* A skew sum of 132-avoiding blocks avoids 132. If three indices are not all in one block, their first and last indices lie in different blocks, so the first value exceeds the last; this is incompatible with 132. Within a block, the largest possible jump is  $k - 1 < m$ .

For a boundary, let the earlier block have length  $k$  and let the next block have first-entry deficiency  $j \leq d$ . The earlier block ends at its maximum. Since the two value intervals are consecutive and decreasing, the boundary jump is exactly  $k + j \leq m$ . This also covers a singleton next block, where  $j = 0$ .

It remains to prove injectivity; this is not automatic for variable-length concatenations. A skew cut in a permutation is a proper cut for which every value on the left exceeds every value on the right. A block ending at its maximum has no internal skew cut, because its maximum lies to the right of every proper cut. Every actual boundary between the constructed blocks is a skew cut. Any other proposed cut is internal to a block and is not a skew cut. Consequently the complete set of skew cuts recovers exactly the block boundaries, and standardization recovers each block. Distinct block sequences cannot yield the same permutation.

The empty sequence and all nonempty sequences of admissible blocks are therefore counted by  $1 + K + K^2 + \dots = (1 - K)^{-1}$  and inject into the desired class. Its radius is  $s_{m,d}$  by the same positivity argument as above, which proves the growth lower bound.  $\square$

For  $d = 1$  this becomes particularly simple:

$$K_{m,1}(x) = x + \sum_{k=2}^{m-1} C_{k-2} x^k. \quad (20)$$

This one-parameter subfamily already proves theorem 1.2. Allowing arbitrary fixed  $d$  sharpens the bounded remainder in theorem 1.3.

### 5.3 Exact block weights and their Catalan tail

Let  $h(k, j)$  count blocks of length  $k \geq 2$  ending at their maximum and having first entry  $k - j$ , where  $1 \leq j \leq k - 1$ . Put

$$C(x) = \sum_{n \geq 0} C_n x^n = \frac{1 - \sqrt{1 - 4x}}{2x}, \quad C(x) = 1 + xC(x)^2.$$

**Lemma 5.3** (First-entry generating function). *For each fixed  $j \geq 1$ ,*

$$\sum_{k \geq j+1} h(k, j) x^k = x^{j+1} C(x)^j, \quad h(k, j) = \frac{j}{k-1} \binom{2k-j-3}{k-2}. \quad (21)$$

Consequently

$$K_{\infty,d}(x) := \sum_{k \geq 1} c_{k,d} x^k = x \sum_{j=0}^d (xC(x))^j. \quad (22)$$

*Proof.* Delete the final maximum from such a block. Its first entry is  $a = k - j$ . The  $j - 1$  larger entries  $a + 1, \dots, k - 1$  must appear in increasing order: a decreasing pair together with the initial  $a$  would form 132. The smaller entries lie in the  $j$  intervening slots, giving the form

$$a, \tau_0, a + 1, \tau_1, a + 2, \dots, k - 1, \tau_{j-1}.$$

Every value in an earlier small slot must exceed every value in a later small slot. Otherwise those two values and a separating large entry form 132. Each slot must itself avoid 132. Conversely these conditions are sufficient: triples meeting different small slots cannot form 132, the large entries are increasing, and any triple entirely in one small slot is excluded by its own avoidance condition. Once the slot lengths are specified, their value intervals are forced in decreasing order. There are therefore  $j$  independent Catalan slots and  $j + 1$  fixed entries when the deleted final maximum is restored. This gives the first identity.

For completeness, put  $Y = C(x) - 1$ , so  $Y = x(1 + Y)^2$ . For  $N \geq 1$ , Lagrange inversion gives

$$[x^N] C(x)^j = \frac{j}{N} [u^{N-1}] (1 + u)^{2N+j-1} = \frac{j}{2N+j} \binom{2N+j}{N}.$$

The same final expression is one for  $N = 0$ . Substituting  $N = k - j - 1$  and cancelling factorials gives the displayed formula for  $h(k, j)$ . Summing over  $1 \leq j \leq d$ , with the singleton added, proves (22) and (7).  $\square$

Since  $C(1/4) = 2$ , (22) yields

$$B_d := \sum_{k \geq 1} c_{k,d} 4^{-k} = \frac{1}{2} (1 - 2^{-(d+1)}). \quad (23)$$

To obtain the tail constant without singularity-analysis machinery, compare each first-entry count directly with a Catalan number:

$$\frac{h(k, j)}{C_{k-2}} = j \prod_{r=0}^{j-2} \frac{k-2-r}{2k-4-r} \longrightarrow \frac{j}{2^{j-1}} \quad (k \rightarrow \infty). \quad (24)$$

The empty product for  $j = 1$  is one. Stirling's formula gives  $C_n \sim 4^n / (\sqrt{\pi} n^{3/2})$ . For fixed  $d$  it follows that

$$c_{k,d} 4^{-k} \sim \frac{a_d}{4\sqrt{\pi}} k^{-3/2}, \quad a_d := \sum_{j=1}^d \frac{j}{2^{j+1}} = 1 - \frac{d+2}{2^{d+1}}. \quad (25)$$

Both  $a_d$  and  $B_d$  are strictly positive for  $d \geq 1$ , and  $B_d < 1$ .

## 6 A truncated-series root lemma

The following elementary estimate is the analytic step responsible for the  $\log \log m$  term. It applies to both sides of (19).

**Lemma 6.1** (A tilted  $k^{-3/2}$  tail). *Let  $b_k \geq 0$  satisfy*

$$b_k \sim c k^{-3/2}, \quad c > 0, \quad B := \sum_{k \geq 1} b_k < 1.$$

*Fix an integer  $h \geq 0$ . For all sufficiently large  $m$ , let  $r_m > 0$  be the unique positive solution of*

$$\sum_{k=1}^{m-h} b_k (4r_m)^k = 1.$$

*Then  $r_m > 1/4$  and*

$$m \log(4r_m) = \frac{1}{2} \log m + \log \log m + \log \frac{1-B}{2c} + o(1). \quad (26)$$

*Proof.* At  $r = 1/4$  the truncated sum is at most  $B < 1$ . For large enough  $m$  the polynomial is nonconstant with a positive coefficient, strictly increases on the positive real axis, and tends to infinity. Thus the stated root exists uniquely and exceeds  $1/4$ .

For a fixed real  $s$ , write

$$t = t_m(s) = \frac{1}{2} \log m + \log \log m + s.$$

We first prove the limiting profile

$$\sum_{k=1}^{m-h} b_k e^{tk/m} \longrightarrow B + 2ce^s. \quad (27)$$

The untilted truncated sum tends to  $B$ . It remains to evaluate the increment from exponential tilting.

For  $k \leq m/2$ , use  $e^u - 1 \leq ue^u$  for  $u \geq 0$  to obtain

$$\begin{aligned} 0 \leq \sum_{k \leq m/2} b_k (e^{tk/m} - 1) &\leq \frac{t}{m} e^{t/2} \sum_{k \leq m/2} kb_k \\ &= O\left(\frac{te^{t/2}}{\sqrt{m}}\right) = O\left(m^{-1/4}(\log m)^{3/2}\right) = o(1). \end{aligned}$$

Here  $b_k = O(k^{-3/2})$  implies  $\sum_{k \leq M} kb_k = O(\sqrt{M})$ .

For  $k > m/2$ , the untilted tail is  $O(m^{-1/2}) = o(1)$ . The assumed asymptotic for  $b_k$  holds uniformly in this range as  $m \rightarrow \infty$ . Put  $j = m - k$ . The tilted upper tail is therefore asymptotic to

$$cm^{-3/2}e^t \sum_{h \leq j < m/2} (1 - j/m)^{-3/2} e^{-tj/m}. \quad (28)$$

The sum in (28) is asymptotic to  $m/t$ . To check this precisely, for  $j \leq m/\sqrt{t}$  the factor  $(1 - j/m)^{-3/2}$  is  $1 + O(t^{-1/2})$  uniformly. The geometric sum without that factor is

$$\sum_{j \geq h} e^{-tj/m} = \frac{e^{-th/m}}{1 - e^{-t/m}} \sim \frac{m}{t}.$$

The omitted part  $j > m/\sqrt{t}$ , with  $j < m/2$ , is bounded by a constant times  $e^{-\sqrt{t}}/(1 - e^{-t/m}) = o(m/t)$ , since  $(1 - j/m)^{-3/2} \leq 2^{3/2}$ . The tail beyond  $m/2$  in the geometric sum is negligible by the same estimate. A fixed lower cutoff  $h$  has no effect on the asymptotic.

Thus (28) is asymptotic to

$$\frac{ce^t}{\sqrt{m}t} = \frac{ce^s \log m}{\frac{1}{2} \log m + \log \log m + s} \longrightarrow 2ce^s,$$

which proves (27).

Let  $s_0 = \log((1 - B)/(2c))$ . For each fixed  $\varepsilon > 0$ , the limiting value in (27) is below one at  $s = s_0 - \varepsilon$  and above one at  $s = s_0 + \varepsilon$ . Monotonicity of the polynomial in  $t$  traps  $m \log(4r_m)$  between  $t_m(s_0 - \varepsilon)$  and  $t_m(s_0 + \varepsilon)$  for all sufficiently large  $m$ . Since  $\varepsilon$  is arbitrary, (26) follows.  $\square$

*Remark 6.2* (Why the second logarithm appears). The effective large- $k$  contribution comes from a boundary layer of width about  $m/t$ . Each coefficient near the cutoff is of order  $m^{-3/2}$ , and the exponential factor is  $e^t$ . Its total mass is consequently  $e^t/(\sqrt{m}t)$ . Requiring a nonzero limiting mass gives  $t = \frac{1}{2} \log m + \log t + O(1)$ , hence the additional  $\log \log m$ . The proof above supplies the constants and controls the discarded ranges.

## 7 The refined asymptotic and its constant window

Put

$$L_m = \frac{1}{2} \log m + \log \log m, \quad t_m^* = m \log \frac{4}{\alpha_m}.$$

We apply lemma 6.1 with two different nonnegative coefficient sequences.

For the upper-growth majorant  $R_m$ , write

$$R_m(x) = \sum_{k=1}^m b_k (4x)^k, \quad b_1 = \frac{1}{2}, \quad b_k = C_{k-1} 4^{-k} \quad (k \geq 2).$$

Then

$$B = \frac{3}{4}, \quad c = \frac{1}{4\sqrt{\pi}}, \quad h = 0.$$

Therefore

$$m \log(4q_m) = L_m + \log \frac{\sqrt{\pi}}{2} + o(1). \quad (29)$$

For the block lower bound, take  $b_k = c_{k,d}4^{-k}$  and  $h = d$ , holding  $d$  fixed as  $m \rightarrow \infty$ . Equations (23) and (25) give

$$m \log(4s_{m,d}) = L_m + \gamma_d + o(1), \quad \gamma_d = \log \left( \frac{\sqrt{\pi}(1 + 2^{-(d+1)})}{1 - (d+2)2^{-(d+1)}} \right). \quad (30)$$

The  $o(1)$  in this formula is for each fixed  $d$ ; no uniformity in a growing  $d$  is used.

*Proof of theorem 1.2.* From (19),

$$m \log(4q_m) \leq t_m^* \leq m \log(4s_{m,d}). \quad (31)$$

Already  $d = 1$ , for which  $\gamma_1 = \log(5\sqrt{\pi})$ , shows  $t_m^* = L_m + O(1)$ . Hence  $\alpha_m = 4e^{-t_m^*/m}$  and

$$m(4 - \alpha_m) = 4m(1 - e^{-t_m^*/m}) = 4t_m^* + O\left(\frac{(t_m^*)^2}{m}\right) = 4t_m^* + o(1). \quad (32)$$

Since  $4L_m = 2 \log m + 4 \log \log m$ , the theorem follows. This also independently proves  $\alpha_m \rightarrow 4$ .  $\square$

*Proof of theorem 1.3.* The lower side of (31), together with (29) and (32), gives

$$\liminf_{m \rightarrow \infty} E_m \geq 4 \log(\sqrt{\pi}/2) = 2 \log \pi - 4 \log 2.$$

For every fixed  $d \geq 1$ , the upper side gives

$$\limsup_{m \rightarrow \infty} E_m \leq 4\gamma_d.$$

Now let  $d \rightarrow \infty$  after taking the limsup in  $m$ . Because  $\gamma_d \rightarrow \log \sqrt{\pi}$ , the upper bound becomes  $2 \log \pi$ . The middle inequality between liminf and limsup is automatic.  $\square$

*Remark 7.1* (The constants are not finite- $m$  error bars). The endpoints in (3) are asymptotic bounds, not a claim that every  $E_m$  lies in that interval. Numerically the scalar comparisons approach their limits slowly. Even at  $m = 10^6$ , the best of the four computed fixed- $d$  lower bounds gives a remainder upper bound above  $2 \log \pi$ . This is compatible with the theorem: a fixed  $d$  has limit  $4\gamma_d$ , and the root expansion also has a slowly vanishing error.

[Added 28 September 2026, batch 38: with the sharper upper bound and the growing cutoff of Part II, the computed corridor for  $E_m$  at  $m = 10^6$  is about  $[3.1495, 3.1503]$ , still far above the proved limit  $2 \log \pi$ ; see remark 17.1.]

## 8 Exact computations and reproducible certificates

The proofs above hold for all  $m$  and do not rest on a finite search. The computations serve three different purposes: checking the combinatorial implementation against direct enumeration, providing exact certificates for displayed finite cases, and illustrating the scalar bounds at large  $m$ . These evidence levels are kept separate.

Table 1: Initial counts computed from the exact endpoint recurrence. The empty permutation is included at  $n = 0$ .

| $n$ | $m = 1$ | $m = 2$ | $m = 3$ | $m = 4$ | $m = 5$ | $m = 6$ |
|-----|---------|---------|---------|---------|---------|---------|
| 0   | 1       | 1       | 1       | 1       | 1       | 1       |
| 1   | 1       | 1       | 1       | 1       | 1       | 1       |
| 2   | 2       | 2       | 2       | 2       | 2       | 2       |
| 3   | 2       | 5       | 5       | 5       | 5       | 5       |
| 4   | 2       | 8       | 14      | 14      | 14      | 14      |
| 5   | 2       | 12      | 28      | 42      | 42      | 42      |
| 6   | 2       | 18      | 55      | 95      | 132     | 132     |
| 7   | 2       | 26      | 108     | 216     | 322     | 429     |
| 8   | 2       | 37      | 214     | 494     | 799     | 1101    |
| 9   | 2       | 53      | 412     | 1135    | 2005    | 2892    |
| 10  | 2       | 76      | 787     | 2613    | 5054    | 7714    |
| 11  | 2       | 109     | 1497    | 5892    | 12758   | 20714   |
| 12  | 2       | 157     | 2841    | 13218   | 32140   | 55697   |

### 8.1 Small sequences and rational generating functions

For example, exact symbolic elimination gives

$$A_1(x) = \frac{1+x^2}{1-x}, \quad A_2(x) = \frac{1-2x+2x^2-x^4-x^6}{(1-x)^2(1-x-x^3)}. \quad (33)$$

Thus, with our convention  $a_0^{(2)} = 1$ ,

$$a_n^{(2)} = 3a_{n-1}^{(2)} - 3a_{n-2}^{(2)} + 2a_{n-3}^{(2)} - 2a_{n-4}^{(2)} + a_{n-5}^{(2)} \quad (n \geq 7).$$

The lower threshold on  $n$  comes from the numerator degree, not just the denominator degree. The archive includes exact numerator and denominator coefficient arrays for  $m = 1, 2, 3, 4$ , and coefficients through  $n = 200$  for  $m = 1, \dots, 12$ .

These rational functions are obtained by exact polynomial-matrix inversion, not by guessing recurrences from the initial terms. More importantly, *the verifier does not need to trust the symbolic solver*. For every  $m \leq 4$ , the supplied certificate gives a common denominator  $D(x)$  and numerator polynomials  $N_{p,q}(x)$  for all endpoint states, and verifies exactly that

$$(I - W_m(x))N(x) = xD(x)\mathbf{1}, \quad D(0) = 1. \quad (34)$$

Uniqueness of the formal solution then proves all coefficients, not merely the first 200 checked by an additional test. Integer cross multiplication also verifies the reported reduced expression for  $1 + F_{\infty, \infty}$ .

### 8.2 Certified spectral intervals

For each  $m = 2, \dots, 20$  and each component, a numerical eigensolver proposes an approximate positive Perron vector and an approximate radius. These proposals are converted to positive integer vectors and rational endpoints. They are accepted only when integer arithmetic verifies

$$W_{C_m}(\ell)v < v, \quad W_{C_m}(u)w > w \quad \text{coordinatewise}, \quad v, w > 0. \quad (35)$$

Positive left Perron vectors show that these imply  $\text{spr}(W_{C_m}(\ell)) < 1 < \text{spr}(W_{C_m}(u))$ . Consequently

$$\ell < r_C(m) < u, \quad \frac{1}{u} < \lambda_C(m) < \frac{1}{\ell}.$$

The verifier uses no floating-point operations for these assertions. If  $\ell = p/D$ , an edge of degree  $k$  contributes the integer factor  $p^k D^{m-k}$  after clearing the common denominator  $D^m$ .

For example, the  $m = 2$ ,  $U$  certificate gives

$$\frac{6823278035}{10^{10}} < r_U(2) < \frac{6823278042}{10^{10}}.$$

All 38 component certificates pass. Their disjoint intervals additionally certify the component comparison for every  $m = 2, \dots, 20$ :  $U$  dominates for  $m = 2, 3, 4$  and  $V$  for  $m = 5, \dots, 20$ . This finite conclusion is not extrapolated into an all- $m$  theorem.

Table 2: Component growth rates, rounded to six decimals. The component ordering is certified by rational intervals; displayed decimals are rounded numerical values.

| $m$ | $\lambda_U(m)$ | $\lambda_V(m)$ | $\alpha_m$ | Dominant |
|-----|----------------|----------------|------------|----------|
| 2   | 1.465571       | 1.000000       | 1.465571   | $U$      |
| 3   | 1.826516       | 1.691403       | 1.826516   | $U$      |
| 4   | 2.100456       | 2.091477       | 2.100456   | $U$      |
| 5   | 2.312119       | 2.351509       | 2.351509   | $V$      |
| 6   | 2.479693       | 2.535762       | 2.535762   | $V$      |
| 8   | 2.727583       | 2.785681       | 2.785681   | $V$      |
| 10  | 2.902175       | 2.952689       | 2.952689   | $V$      |
| 12  | 3.032016       | 3.074770       | 3.074770   | $V$      |
| 15  | 3.174821       | 3.208325       | 3.208325   | $V$      |
| 20  | 3.333400       | 3.356796       | 3.356796   | $V$      |

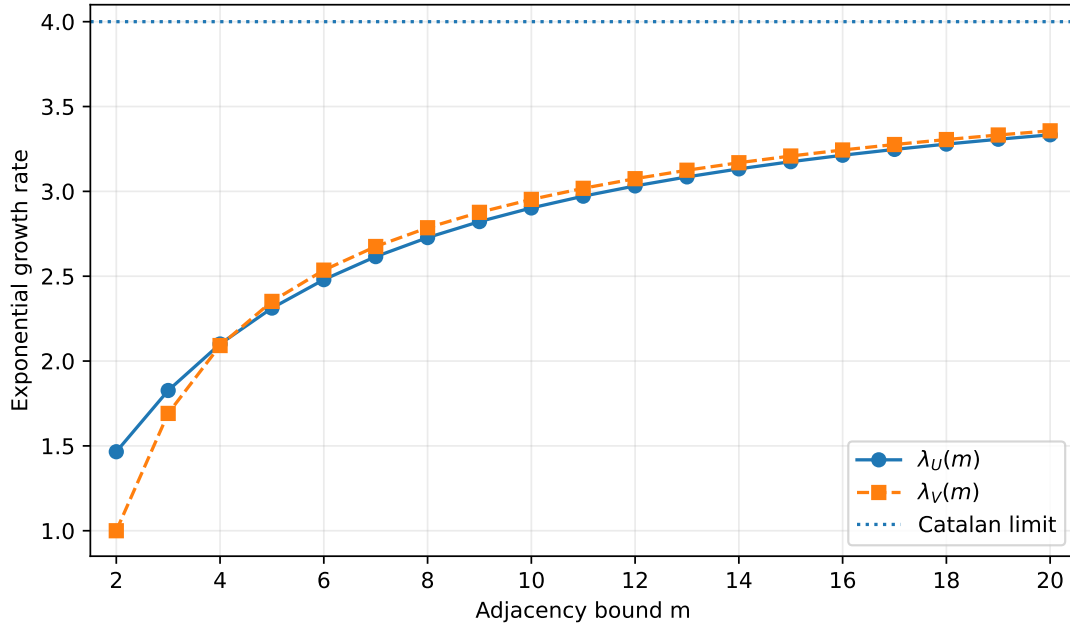


Figure 1: The two component rates for  $2 \leq m \leq 20$ . Both increase, and the component attaining the maximum changes between  $m = 4$  and  $m = 5$ . The proof of strict growth does not depend on this crossover pattern.

### 8.3 Large- $m$ scalar comparisons

The scalar bounds can be evaluated without matrices of order  $m^2$ . The implementation stores normalized Catalan coefficients  $C_k 4^{-k}$ , avoiding enormous intermediate Catalan integers in this

numerical calculation. It solves for  $t = m \log(4x)$ , not for a root whose distance from  $1/4$  is tiny.

Table 3: Approximate evaluation of the rigorous scalar root bounds. The lower column is  $\max_{d \in \{1,2,4,8\}} s_{m,d}^{-1}$ ; the upper is  $q_m^{-1}$ . These decimal evaluations are floating-point results, not outward-rounded interval certificates.

| $m$     | Lower-bound value | Upper-bound value |
|---------|-------------------|-------------------|
| 10      | 2.7001330054      | 3.2847596398      |
| 20      | 3.2358742549      | 3.5390478302      |
| 50      | 3.6448786614      | 3.7589497022      |
| 100     | 3.8065234928      | 3.8581087164      |
| 1000    | 3.9751398963      | 3.9790979985      |
| 10000   | 3.9969422752      | 3.9972999790      |
| 100000  | 3.9996392834      | 3.9996731933      |
| 1000000 | 3.9999586173      | 3.9999618975      |

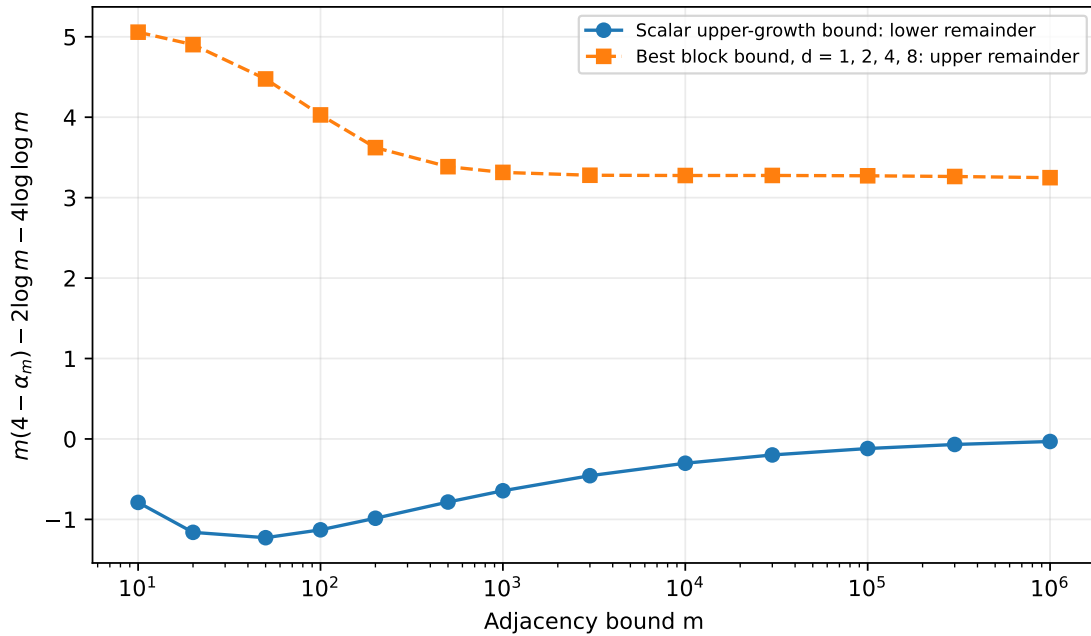


Figure 2: Numerical evaluations of the scalar bounds on the centered remainder  $E_m$ . The plotted upper envelope uses only  $d = 1, 2, 4, 8$ . Neither curve is the true remainder, and the asymptotic constants in (3) are not finite- $m$  interval bounds.

#### 8.4 What was actually checked

The bundled `verification_log.txt` records a successful complete run of the standard-library verifier. Its checks include direct triple-definition testing of all permutations through length seven against the Catalan generator; the first-entry formula for every Catalan avoider through length ten; 2,840 cumulative endpoint counts for  $m = 1, \dots, 8$ ; and 53,534 weighted-edge comparisons under the shift for  $m = 2, \dots, 30$ . Both component graphs are also tested for strong connectivity in those finite cases.

The verifier checks coefficientwise scalar majorants and block lower bounds through  $n = 100$  for  $m = 2, \dots, 12$  and the admissible tested values of  $d$ . It validates the 38 rational Perron intervals using 3,458 strict integer row inequalities, then compares their endpoints as exact fractions. Finally, it checks all four full polynomial-matrix certificates and compares the resulting

generating-function coefficients with the recurrence through  $n = 200$ .

For certificate checking, the verifier reconstructs states and weighted edges separately, using Catalan convolutions in place of the binomial formula used by the numerical matrix builder. It also checks agreement between these two implementations. This reduces the risk that a common implementation error merely reproduces itself. Nevertheless, finite computational checks are supplements to, not replacements for, the universal arguments in sections 4, 6 and 7.

## 9 What is resolved, and what remains

The strict inequality  $\alpha_{m+1} > \alpha_m$  is proved for every positive integer  $m$ . Together with the existing fixed-bound results, it completes the growth-constant conjecture selected here. The scalar comparison proof also determines the leading constant and next logarithmic term in the approach to four.

Several nearby questions are not resolved by these arguments. The first is whether  $\lambda_V(m) > \lambda_U(m)$  for *every*  $m \geq 5$ ; the reverse inequality holds for  $m = 2, 3, 4$ . This is the separate component-comparison problem formulated in [2, Problem P3]. Strict increase in both sequences is compatible with crossings, so it cannot settle their ordering. The finite certificates here establish the pattern only through  $m = 20$ . [Added 28 September 2026, batch 38: this question remains open. Part II shows only that a dominant  $V$  can lead by at most  $(\lambda_V(m) - \lambda_U(m))_+ = O((\log m)^2/m^2)$  (corollary 15.4), which is why the ordering is invisible to every inverse-logarithmic term; see section 19.]

A second question is whether  $E_m$  converges, and, if so, to which constant. The scalar majorant discards endpoint compatibility, while the lower construction permits only a special recoverable family of blocks. Their limiting constants leave a window of width  $4 \log 2$ . Closing that gap requires controlling more of the endpoint information, not merely taking more terms of the same scalar expansions. A growing- $d$  analysis could improve finite bounds, but our argument deliberately takes  $d$  fixed and then sends  $d$  to infinity only after a limsup. [Added 28 September 2026, batch 38: the preceding paragraph describes the report at its pinned state 6c9175a54. The question is answered in Part II:  $E_m \rightarrow 2 \log \pi$  (corollary 15.3). As anticipated here, the upper side keeps last-endpoint information, through a Catalan-weighted positive vector (lemma 12.2), and the lower side lets the cutoff grow,  $d = \lceil \log_2 m \rceil$ , using a tail bound uniform in  $d$  (lemma 13.2 and theorem 15.2).]

Finally, the minimal recurrence order and uniform denominator factorization of  $A_m(x)$  remain outside the scope of this report. The exact  $m \leq 4$  artifacts are useful examples, not a claimed solution of that general algebraic problem. The new strict-growth argument is valuable precisely because it avoids needing such a formula.

## Part II. The exact constant: an endpoint renewal comparison

Sections 1–9 and the appendices form the original report (Part I) of 19 September 2026, unchanged except for dated pointers to this addition. Part II was added on 28 September 2026 in batch 38 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Part I. It answers the question Part I left open in section 9:  $E_m$  converges, and its limit is  $2\log \pi$ . Section 10 records the provenance and fixes the notation of the addition.

### 10 Provenance, conventions and the answer

#### 10.1 One manuscript, one addition

Part II is batch-38 manuscript 04, *Closing the Catalan growth-constant gap: Endpoint renewal comparison for adjacency-bounded 132-avoiding permutations*, an AI-assisted research manuscript prepared for Vladimir Reshetnikov and dated September 28, 2026 (23 pages as delivered). It arrived as the archive `Closing_Catalan_Growth_Gap.zip` (inner directory `Adjacency_Growth_Constant/`) in ProveIt commit `fbba58593` and was placed in `938b2f74e`. It was written against ProveIt commit `6c9175a54`, at which Part I’s `article.tex` had the Git blob `5ec75ee4`; that blob is unchanged at the placement commit, so every statement the manuscript makes about Part I refers to the text printed above. Its code and data are shipped beside Part I’s files with the prefix `02-centered-remainder-` (Part I’s unprefix files count as 01); its manuscript, README and PDF are not shipped and survive in the arrival commit. Its two figures were added to `figures/` in the write phase (section 20.5).

The manuscript contributes, in the order printed here: a Catalan-weighted spectral upper bound  $\alpha_m < \beta_m$  (section 12); a uniform first-deficiency tail bound and a lower bound for the first component rate (section 13); a uniform endpoint expansion of the truncated Catalan root (section 14); a corridor of width  $O((\log m)^2/m^2)$  from a logarithmically growing cutoff, and with it the exact constant (section 15); an all-orders inverse-logarithmic expansion (section 16); exact checks and numerical illustrations (section 17); an  $m = 2$  example with a non-implication (section 18); nine research directions (section 19); and its proof boundaries and provenance (section 20). Every result, proof, example, remark, limitation and question of the manuscript is printed. The manuscript re-derives the finite endpoint structure, the first-entry lemma and the injective block construction of Part I; those are printed once, in Part I, and credited where Part II uses them.

**Status (manuscript 04).** The manuscript presents mathematical proofs of a repository-open asymptotic question and stronger approximation results. It is not externally refereed and contains no new Lean certification. Novelty is assessed against the inspected repository snapshot and the primary papers cited here, not against an exhaustive literature census. Nothing in Part II is formalized in any proof assistant.

#### 10.2 Notation of Part II

Part I’s symbols keep their meanings:  $a_n^{(m)}$ ,  $A_m$ ,  $\alpha_m$ ,  $C_j$ ,  $C$ ,  $B_m$ ,  $c_{k,p}$ ,  $W_m$ ,  $U_m$ ,  $V_m$ ,  $r_C(m)$ ,  $\lambda_C(m)$ ,  $h(k, j)$ ,  $K_{m,d}$ ,  $s_{m,d}$ ,  $\leq$ , and the centered remainder  $E_m$  of theorem 1.3. The manuscript uses the same symbols with the same meanings. Its new symbols, and the ones renamed here to avoid a clash with Part I, are listed in table 4.

No normalization changed: every renamed symbol denotes exactly the manuscript’s quantity. Logarithms are natural.

Table 4: Symbols of Part II. “Renamed” gives the manuscript’s own symbol. The last column prints the tempting false reading.

| Symbol                                                   | Meaning in Part II                                                                      | Not to be confused with                                                                                                                                                                       |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $K_m(x)$ (one index)                                     | $\sum_{k=1}^m C_{k-1} x^k$                                                              | Part I’s block polynomial $K_{m,d}$ (two indices). In particular $K_{m-d}$ is the Catalan polynomial truncated at $m-d$ , and $K_{m,d} \leq K_{m-d}$ . Part I’s majorant is $R_m = x + K_m$ . |
| $r_m, \beta_m = r_m^{-1}$                                | positive root of $K_m(r_m) = 1$                                                         | the component roots $r_U(m), r_V(m)$ , and the local $r_m$ inside lemma 6.1. Since $R_m = x + K_m$ , $q_m < r_m$ and $\beta_m < q_m^{-1}$ .                                                   |
| $t_m$                                                    | $m \log(4r_m) = m \log(4/\beta_m)$                                                      | Part I’s $t_m^* = m \log(4/\alpha_m)$ ; $t_m < t_m^*$ for $m \geq 2$ .                                                                                                                        |
| $b_k$                                                    | $C_{k-1} 4^{-k}$ for all $k \geq 1$ , so $b_1 = \frac{1}{4}$                            | the coefficients of $R_m$ in section 7, where $b_1 = \frac{1}{2}$ .                                                                                                                           |
| $\Lambda$ (renamed from $L$ )                            | $\frac{1}{2} \log m$                                                                    | Part I’s $L_m = \frac{1}{2} \log m + \log \log m$ .                                                                                                                                           |
| $z$                                                      | $\log \Lambda + \log(2\sqrt{\pi}) = \log \log m + \frac{1}{2} \log \pi$                 | —                                                                                                                                                                                             |
| $\mathcal{P}_j(z)$ (renamed from $P_j$ )                 | correction polynomials, proposition 16.1                                                | the numerator polynomials $P_3, P_4$ of section A.                                                                                                                                            |
| $\mathcal{B}(w), \mathcal{B}_N(t)$ (renamed from $B_N$ ) | the formal series $\sum_j (3/2)_j w^j$ and its truncation $\sum_{j < N} (3/2)_j t^{-j}$ | $B_m$ (thresholds), $B_d$ of (23), $B$ of lemma 6.1.                                                                                                                                          |
| $H_q(x)$                                                 | $\sum_{j=0}^q C_j x^j$                                                                  | the expanded adjacency matrix $H$ in the proof of proposition 3.3.                                                                                                                            |
| $\varepsilon_d$                                          | $(d+2)2^{-d}$                                                                           | Part I’s $a_d$ ; $1 - a_d = \varepsilon_d/2$ (remark 13.3).                                                                                                                                   |
| $d = d_m$                                                | $\lceil \log_2 m \rceil$ in section 15                                                  | Part I, where $d$ is fixed while $m \rightarrow \infty$ .                                                                                                                                     |
| $\Delta_m$ (renamed from $T_m$ )                         | mass of the last $d$ tilted terms                                                       | Part I’s endpoint counts $T_{p,q}^{(m)}(n)$ .                                                                                                                                                 |
| $C_*, c_*$ (renamed from $C, c$ )                        | absolute constants                                                                      | Catalan numbers $C_j$ ; the component letter $C \in \{U, V\}$ .                                                                                                                               |
| $W_{U_m}, W_{V_m}$ (manuscript: $W_U, W_V$ )             | component matrices of Part I                                                            | —                                                                                                                                                                                             |

### 10.3 The question and the answer

Part I proved  $4 - \alpha_m = (2 \log m + 4 \log \log m + O(1))/m$  (theorem 1.2) and, for the centered remainder  $E_m = m(4 - \alpha_m) - 2 \log m - 4 \log \log m$ , the window (3) of width  $4 \log 2$ . Its concluding discussion asks whether  $E_m$  converges and, if so, to which constant. That is the question resolved here. It was an open question in the inspected report (Part I), not a claim that every open problem in the original enumeration papers has been settled.

The older upper comparison was based on  $R_m(x) = x + \sum_{k=1}^m C_{k-1}x^k$ . The extra  $x$  accounts for appending a maximum when endpoint restrictions are discarded. One cannot simply delete that term from the counting recurrence. Instead, we retain the last-endpoint information and assign a different positive weight to each endpoint threshold. A Catalan convolution then incorporates the append transition without paying for it as an independent unconstrained scalar choice. Part I's diagnosis in section 9, that closing the window needs more endpoint information than the scalar series carry, is exactly what this weighting supplies on the upper side; on the lower side a cutoff  $d$  growing with  $m$  is used, made legitimate by a bound uniform in  $d$ .

Write

$$C_j = \frac{1}{j+1} \binom{2j}{j}, \quad K_m(x) = \sum_{k=1}^m C_{k-1}x^k, \quad K_m(r_m) = 1, \quad \beta_m = r_m^{-1}. \quad (36)$$

The polynomial  $K_m$  is strictly increasing on the positive real axis. Thus  $r_m$  is well defined. Since  $K_m(1/4) < \sum_{k \geq 1} C_{k-1}4^{-k} = 1/2$ , we have  $r_m > 1/4$ . For  $m \geq 2$ ,  $K_m(1) > 1$ , so  $r_m < 1$ .

**Theorem 10.1** (Finite renewal upper bound). *For every  $m \geq 2$ ,*

$$\alpha_m < \beta_m < 4.$$

For  $m = 1$ ,  $\alpha_1 = \beta_1 = 1$ .

**Theorem 10.2** (Quantitative scalar universality). *As  $m \rightarrow \infty$ ,*

$$0 \leq \beta_m - \alpha_m \leq \beta_m - \lambda_U(m) = O\left(\frac{(\log m)^2}{m^2}\right), \quad (37)$$

where  $\lambda_U(m)$  is the first cyclic-component growth rate of (13). In particular,

$$\lim_{m \rightarrow \infty} E_m = 2 \log \pi. \quad (38)$$

For the all-orders statement, put

$$\Lambda = \frac{1}{2} \log m, \quad z = \log \Lambda + \log(2\sqrt{\pi}) = \log \log m + \frac{1}{2} \log \pi. \quad (39)$$

**Theorem 10.3** (All inverse-logarithmic orders). *There are explicitly computable polynomials  $\mathcal{P}_j \in \mathbb{Q}[z]$ , of degree  $j$ , such that for every fixed integer  $J \geq 0$ ,*

$$m(4 - \alpha_m) = 4\Lambda + 4z + 4 \sum_{j=1}^J \frac{\mathcal{P}_j(z)}{\Lambda^j} + O_J\left(\frac{(1 + |z|)^{J+1}}{\Lambda^{J+1}}\right). \quad (40)$$

The first three are

$$\begin{aligned} \mathcal{P}_1(z) &= z - \frac{3}{2}, \\ \mathcal{P}_2(z) &= -\frac{1}{2}z^2 + \frac{5}{2}z - \frac{33}{8}, \\ \mathcal{P}_3(z) &= \frac{1}{3}z^3 - 3z^2 + \frac{43}{4}z - 15. \end{aligned} \quad (41)$$

The same expansion holds with  $\alpha_m$  replaced by  $\lambda_U(m)$  or  $\beta_m$ .

Since  $4\Lambda + 4z = 2 \log m + 4 \log \log m + 2 \log \pi$ , the case  $J = 0$  is again  $E_m \rightarrow 2 \log \pi$ : the upper endpoint of Part I's window (3). Part II refutes nothing in Part I; it sharpens the window to a point.

## 10.4 What is inherited and what is added

Table 5: Attribution and scope. “Here” means proved in Part II (manuscript 04).

| Ingredient or conclusion                                            | Provenance                                                   |
|---------------------------------------------------------------------|--------------------------------------------------------------|
| Endpoint recurrence and two cyclic components                       | Mayama–Akita; rederived in Part I and again in manuscript 04 |
| Strict increase of $\alpha_m$ and the window (3)                    | Part I; not claimed anew                                     |
| Injective skew blocks with fixed endpoint cutoff                    | Part I; recalled                                             |
| Catalan-weighted spectral upper bound $\alpha_m < \beta_m$          | Here                                                         |
| Uniform tail control and $O((\log m)^2/m^2)$ approximation          | Here                                                         |
| Exact constant $2\log \pi$ and the transferred all-orders expansion | Here                                                         |
| Ordering of $U$ and $V$ for every $m \geq 5$                        | Not resolved here                                            |

The inspected repository path, commit and source blob are recorded in section 20.4. This matters because the repository is an active research collection: an unpinned statement about what it contains would not be a reproducible novelty comparison.

## 11 The finite interface

This section collects the input to the new comparison. Its finite-state structure is established work [2] and is proved in Part I; manuscript 04 rederives it, to make its argument independent of undocumented implementation details, and its proofs coincide with Part I’s except where a second route is marked below.

The endpoint recurrence is proposition 2.1. In the language of edges,  $W_m(x)$  of (9) has a split edge

$$(p, q) \longrightarrow (m - k, q - k) \quad \text{with weight } c_{k,p}x^k, \quad (42)$$

and an append edge

$$(p, q) \longrightarrow (p - 1, m - 1) \quad \text{with weight } x, \quad (43)$$

invalid negative thresholds being omitted. By (6),

$$0 \leq c_{k,p} \leq C_{k-1}, \quad c_{k,p} \leq c_{k,p+1}, \quad c_{k,\infty} = C_{k-1}. \quad (44)$$

**Proposition 11.1** (Spectral interface). *For  $m \geq 2$ , both  $W_{U_m}(x)$  and  $W_{V_m}(x)$  are irreducible for  $x > 0$ . Their spectral radii are continuous and strictly increasing from zero to infinity. With  $r_C(m)$  and  $\lambda_C(m)$  as in (13),*

$$\alpha_m = \max\{\lambda_U(m), \lambda_V(m)\}, \quad \lambda_U(m) > 1. \quad (45)$$

*Proof.* This is lemma 3.1, the paragraph after lemma 3.2, and proposition 3.3. Manuscript 04’s Appendix A gives the same graph and transfer-matrix proof: the same strong-connectivity induction on the gap  $p - q$ , the same classification of cyclic components, the same expansion of every edge into unit-length edges, and the same closed walks of lengths two and five for primitivity. Its argument for  $\lambda_U(m) > 1$  differs and is printed next.  $\square$

*Second proof of  $\lambda_U(m) > 1$  (manuscript 04).* Let  $e_U = (m - 1, \infty)$ . At  $x = 1$ , the column of  $W_{U_m}$  indexed by  $e_U$  consists of ones, from the  $k = 1$  edges. Consequently  $W_{U_m}(1)\delta_{e_U} = \mathbf{1} \geq \delta_{e_U}$ , strictly in another coordinate when  $m \geq 2$ . A positive left Perron vector gives  $\text{spr}(W_{U_m}(1)) > 1$ , and hence  $r_U(m) < 1$ .  $\square$

*Remark 11.2* (Constants depend on  $m$ ). All constants in the proof of [proposition 3.3](#) (the  $c$ ,  $C$  and  $D$  of its last paragraph) may depend on  $m$ ; no joint-limit uniformity is claimed, in Part I or here. [Added 29 September 2026, batch 43: this remains true of Parts I and II. Part III proves a joint estimate near the boundary  $m = n - d$  by a separate route that does not use this interface ([theorem 22.2](#)).] The interface is all that Part II uses: not explicit determinants, a conjectural ordering of components, or a simple-pole assertion about  $A_m$ .

We will repeatedly use two elementary consequences of Perron–Frobenius positivity.

- Lemma 11.3** (Positivity tools). (i) *If  $M$  is irreducible and nonnegative, and  $v > 0$  satisfies  $Mv \leq v$  with a strict inequality in at least one coordinate, then  $\text{spr}(M) < 1$ .*  
(ii) *For any nonnegative matrix, the spectral radius is at least its minimum row sum, and it is monotone under entrywise increase and under principal extension.*

*Proof.* For (i), multiply by a positive left Perron vector  $u$ :  $\text{spr}(M)u^\top v = u^\top Mv < u^\top v$ . For (ii), the row-sum bound follows by iterating  $M\mathbf{1} \geq a\mathbf{1}$ ; the monotonicity follows by comparing powers. The mechanism of (i) is the one used in [lemma 3.2](#).  $\square$

## 12 The endpoint renewal comparison

The essential new finite observation is a positive vector adapted to the last threshold. Before applying it, it is useful to isolate its algebraic mechanism.

### 12.1 A renewal system with a reset

**Lemma 12.1** (Renewal–reset identity). *Let  $w_1, \dots, w_{m-1} \geq 0$  and  $a \geq 0$ . On indices  $0, \dots, m-1$ , let  $L$  be the strictly lower triangular matrix with  $L_{q,q-k} = w_k$  for  $1 \leq k \leq q$ , and put*

$$N = L + a\mathbf{1}e_{m-1}^\top.$$

*Define  $u_0 = 1$ ,  $u_n = \sum_{k=1}^n w_k u_{n-k}$  for  $1 \leq n < m$ , and  $H_q = \sum_{j=0}^q u_j$ . Then*

$$(I - L)H = \mathbf{1}, \tag{46}$$

$$NH = H + (aH_{m-1} - 1)\mathbf{1}, \tag{47}$$

$$\det(I - N) = 1 - aH_{m-1}. \tag{48}$$

*Proof.* For each  $q$ ,

$$\sum_{k=1}^q w_k H_{q-k} = \sum_{n=1}^q \sum_{k=1}^n w_k u_{n-k} = \sum_{n=1}^q u_n = H_q - 1.$$

This proves (46); the reset term gives (47). Since  $L$  is strictly triangular,  $\det(I - L) = 1$ . The determinant identity for a rank-one update, or direct elimination of  $(I - L)$ , gives

$$\det(I - N) = \det(I - L)(1 - ae_{m-1}^\top(I - L)^{-1}\mathbf{1}) = 1 - aH_{m-1}.$$

All identities also hold over a commutative coefficient ring when the nonnegativity assertions are omitted.  $\square$

For the Catalan application, take  $w_k = C_{k-1}x^k$  and  $a = x$ . The Catalan recurrence gives  $u_j = C_j x^j$ . Thus

$$H_q(x) = \sum_{j=0}^q C_j x^j, \quad xH_{m-1}(x) = K_m(x). \tag{49}$$

Equivalently, the formal identity  $C(x) = 1 + xC(x)^2$  gives  $(1 - xC(x))^{-1} = C(x)$ . The finite version we need is

$$\sum_{k=1}^q C_{k-1} x^k H_{q-k}(x) = H_q(x) - 1. \quad (50)$$

There is no convergence issue in this polynomial identity.

## 12.2 A positive vector for the true two-endpoint component

**Lemma 12.2** (Last-threshold supersolution). *For a state  $(p, q) \in V_m$ , set  $v_{p,q}(x) = H_q(x)$ . For every  $x > 0$ ,*

$$(W_{V_m}(x)v(x))_{p,q} \leq H_q(x) - 1 + K_m(x). \quad (51)$$

*At  $x = r_m$ , at least one coordinate inequality is strict.*

*Proof.* A valid split from  $(p, q)$  has  $k \leq q$ , and its target has last threshold  $q - k$ . We may bound  $c_{k,p}$  by  $C_{k-1}$  and include every  $1 \leq k \leq q$ . If an append edge exists, its target has last threshold  $m - 1$ ; including such an edge even when it is absent can only increase the result. Therefore

$$\begin{aligned} (W_{V_m}v)_{p,q} &\leq \sum_{k=1}^q C_{k-1} x^k H_{q-k}(x) + xH_{m-1}(x) \\ &= H_q(x) - 1 + K_m(x), \end{aligned}$$

by (50). At the state  $(0, m - 1)$ , which belongs to  $V_m$  for  $m \geq 2$ , there is no append edge. The positive term  $xH_{m-1}(x)$  was consequently added strictly. This proves strictness, in fact for every  $x > 0$ .  $\square$

*Proof of theorem 10.1.* At  $x = r_m$ , lemma 12.2 gives  $W_{V_m}v \leq v$ , with a strict coordinate and  $v > 0$ . Irreducibility and lemma 11.3(i) imply  $\text{spr}(W_{V_m}(r_m)) < 1$ .

In  $U_m$ , append edges leave the component. Its row at first threshold  $p$  has sum  $\sum_{k=1}^m c_{k,p} r_m^k \leq K_m(r_m) = 1$ . The row  $p = 0$  has sum  $r_m < 1$ , since only the  $k = 1$  coefficient survives. Thus  $\text{spr}(W_{U_m}(r_m)) < 1$  by the same positivity argument, now with vector  $\mathbf{1}$ .

Strict increase of each spectral radius gives  $r_U(m), r_V(m) > r_m$ . Taking reciprocals and using (45) proves  $\alpha_m < \beta_m$ . The already established  $r_m > 1/4$  gives  $\beta_m < 4$ . For  $m = 1$ , the only permutations of length at least two with jumps of absolute value one are the increasing and decreasing permutations; a change of direction would repeat a value. Hence  $\alpha_1 = 1 = \beta_1$ .  $\square$

**Remark 12.3** (Comparison with Part I's majorant). Since  $R_m = x + K_m$ , one has  $R_m(r_m) = 1 + r_m > 1$ , so  $q_m < r_m$  and  $\beta_m < q_m^{-1}$  for every  $m \geq 1$ : theorem 10.1 is strictly sharper than proposition 5.1. At  $m = 100$ , for instance, the upper bound drops from 3.8581... (table 3) to 3.8201... (table 7). Part I's bound is coefficientwise, theorem 10.1 only spectral; see the next remark.

**Remark 12.4** (What this comparison does not say). The scalar polynomial is not asserted to be a denominator of  $A_m$ , and  $\det(I - W_{V_m})$  is not asserted to equal  $1 - K_m$ . The determinant identity (48) belongs to the *majorizing reset system*, not to the full endpoint system. Nor does the proof assert the coefficientwise inequality  $A_m(x) \leq (1 - K_m(x))^{-1}$ ; it is false (section 18). Our comparison is a spectral one, witnessed by a positive vector. These distinctions prevent a valid growth bound from being misread as a solution of state minimization or exact enumeration.

## 13 A uniform lower approximation from recoverable blocks

The block construction itself is that of proposition 5.2 in Part I. The additional ingredients of manuscript 04 are a component version of the lower bound, an estimate uniform in the cutoff, and its use with a cutoff growing with  $m$ .

### 13.1 Block and component lower bounds

Fix integers  $1 \leq d < m$ , and let  $K_{m,d}$  and  $s_{m,d}$  be as in (18) and proposition 5.2:

$$K_{m,d}(x) = \sum_{k=1}^{m-d} c_{k,d} x^k, \quad K_{m,d}(s_{m,d}) = 1. \quad (52)$$

The coefficient of  $x$  is one, so a unique positive root exists.

**Proposition 13.1** (Block and component lower bounds). *For every  $1 \leq d < m$ ,*

$$\frac{1}{1 - K_{m,d}(x)} \leq A_m(x), \quad s_{m,d}^{-1} \leq \lambda_U(m) \leq \alpha_m < \beta_m. \quad (53)$$

*Proof.* The coefficientwise inequality is proposition 5.2: skew sums of blocks ending at their maximum, with the boundary jump  $k + j \leq m$ , and injectivity because the skew cuts recover the block boundaries. Manuscript 04 reproves it by the same argument.

For the sharper component statement, restrict  $W_{U_m}(x)$  to first thresholds  $p \in \{d, \dots, m-1\}$ . For every  $1 \leq k \leq m-d$ , the split target  $m-k$  is in that subset and its coefficient satisfies  $c_{k,p} \geq c_{k,d}$ . Every row sum of this principal submatrix is at least  $K_{m,d}(x)$ . At  $x = s_{m,d}$  its spectral radius is at least one. Principal comparison (lemma 11.3(ii)) implies  $\text{spr}(W_{U_m}(s_{m,d})) \geq 1$ , so  $r_U(m) \leq s_{m,d}$  and  $\lambda_U(m) \geq s_{m,d}^{-1}$ . The remaining inequalities are (45) and theorem 10.1.  $\square$

### 13.2 First-entry counts and a uniform tail

The first-entry formula is lemma 5.3:  $h(k, j) = \frac{j}{k-1} \binom{2k-j-3}{k-2}$  for  $1 \leq j < k$ , with generating function (21). Manuscript 04 reproves it with the same slot decomposition and the same Lagrange inversion. Consequently  $c_{k,d} = \sum_{j=1}^{\min(d, k-1)} h(k, j)$  for  $k \geq 2$ , as in (7).

**Lemma 13.2** (Uniform deficiency-tail estimate). *For all  $d \geq 1$  and  $k \geq 1$ ,*

$$0 \leq C_{k-1} - c_{k,d} \leq \varepsilon_d C_{k-1}, \quad \varepsilon_d = (d+2)2^{-d}. \quad (54)$$

Hence, whenever  $1 \leq d < m$  and  $\varepsilon_d < 1$ ,

$$(1 - \varepsilon_d)K_{m-d}(x) \leq K_{m,d}(x) \leq K_{m-d}(x) \quad (x \geq 0). \quad (55)$$

The corresponding derivative inequalities also hold for  $x \geq 0$ .

*Proof.* There is nothing to prove if  $k = 1$  or  $k \leq d+1$ . For the remaining cases, formula (21) gives

$$\frac{h(k, j)}{C_{k-2}} = j \prod_{r=0}^{j-2} \frac{k-2-r}{2k-4-r} \leq \frac{j}{2^{j-1}}. \quad (56)$$

The product is empty for  $j = 1$ . Each displayed factor is at most  $1/2$ . For a missing deficiency  $j \geq d+1 \geq 2$ , we have  $k \geq 3$  and

$$\frac{C_{k-2}}{C_{k-1}} = \frac{k}{4k-6} \leq \frac{1}{2}.$$

Thus  $h(k, j)/C_{k-1} \leq j2^{-j}$ . Summing missing deficiencies and extending the tail to infinity gives

$$\frac{C_{k-1} - c_{k,d}}{C_{k-1}} \leq \sum_{j=d+1}^{\infty} \frac{j}{2^j} = \frac{d+2}{2^d}.$$

For the final equality, differentiate the finite-tail geometric sum or subtract its successive terms. Multiplying coefficientwise by  $x^k$  and summing proves (55). Differentiation preserves a coefficientwise inequality between these nonnegative polynomials.  $\square$

The identity in (56) is Part I's (24); the inequality is new.

*Remark 13.3* (Merge note: the uniform bound against Part I's limit). By (25) and  $C_{k-1}4^{-k} \sim (4\sqrt{\pi})^{-1}k^{-3/2}$ , the missing fraction  $1 - c_{k,d}/C_{k-1}$  tends to  $1 - a_d = (d+2)2^{-(d+1)} = \varepsilon_d/2$  as  $k \rightarrow \infty$ . The uniform bound  $\varepsilon_d$  of lemma 13.2 therefore loses at most a factor two against the limit, the factor coming from  $C_{k-2}/C_{k-1} \leq \frac{1}{2}$  where the limit is  $\frac{1}{4}$ . This observation is the merge's, not the manuscript's.

The role of uniformity is decisive. A fixed- $d$  asymptotic for  $c_{k,d}$ , such as (25), would not justify substituting  $d = d(m)$ . Inequality (54) holds simultaneously for all  $k, d$ , so the growing cutoff used below requires no interchange of unproved uniform limits.

## 14 The truncated Catalan root at all orders

The remaining analysis begins with the scalar root alone. Normalize its coefficients by

$$b_k = C_{k-1}4^{-k}, \quad \sum_{k \geq 1} b_k = \frac{1}{2}, \quad b_k = \frac{1}{4\sqrt{\pi}}k^{-3/2}(1 + O(k^{-1})). \quad (57)$$

The last estimate follows from Stirling's formula. In particular,  $b_k \leq C_*k^{-3/2}$  with one absolute constant  $C_*$ . If

$$t_m = m \log(4r_m), \quad (58)$$

then

$$\sum_{k=1}^m b_k e^{t_m k/m} = 1, \quad \beta_m = 4e^{-t_m/m}. \quad (59)$$

The tail is summable but has total mass only  $1/2$ . A positive tilt must supply the missing half. It does so in a thin layer near  $k = m$ .

### 14.1 A uniform endpoint expansion

For  $a \in \mathbb{R}$  and  $j \geq 0$ , write  $(a)_0 = 1$  and  $(a)_j = a(a+1)\cdots(a+j-1)$ .

**Lemma 14.1** (Tilted Catalan endpoint expansion). *Fix  $M > 0$  and an integer  $N \geq 1$ . Uniformly for*

$$t = \frac{1}{2} \log m + \log \log m + s, \quad |s| \leq M, \quad (60)$$

as  $m \rightarrow \infty$ ,

$$\sum_{k=1}^m b_k e^{tk/m} = \frac{1}{2} + \frac{e^t}{4\sqrt{\pi m} t} \left( \sum_{j=0}^{N-1} \frac{(3/2)_j}{t^j} + O_{M,N}(t^{-N}) \right). \quad (61)$$

*Proof.* Split the sum at  $m/2$ . For the lower half, the unweighted partial sum is  $1/2 + O(m^{-1/2})$ . Since  $e^u - 1 \leq ue^u$  for  $u \geq 0$ , its tilt increment is at most

$$\begin{aligned} \sum_{k \leq m/2} b_k (e^{tk/m} - 1) &\leq \frac{C_* t}{m} e^{t/2} \sum_{k \leq m/2} k^{-1/2} \\ &= O\left(\frac{te^{t/2}}{\sqrt{m}}\right) = O_M\left(m^{-1/4}(\log m)^{3/2}\right). \end{aligned}$$

This is smaller than every fixed negative power of  $t$ . Thus the lower half contributes  $1/2$  plus an error that can be absorbed into the remainder in (61).

On the upper half, Stirling's estimate is uniform:

$$b_k = \frac{1}{4\sqrt{\pi}}k^{-3/2}(1 + O(m^{-1})).$$

A Riemann-sum comparison, using that  $y^{-3/2}e^{ty}$  is increasing on  $[1/2, 1]$  for all sufficiently large  $t$ , gives

$$\begin{aligned} \sum_{m/2 < k \leq m} b_k e^{tk/m} &= \frac{1}{4\sqrt{\pi m}} \int_{1/2}^1 y^{-3/2} e^{ty} dy + O\left(\frac{e^t}{m^{3/2}}\right) \\ &= \frac{e^t}{4\sqrt{\pi m} t} \int_0^{t/2} (1 - v/t)^{-3/2} e^{-v} dv + O\left(\frac{e^t}{m^{3/2}}\right). \end{aligned}$$

A possible half-integer lower endpoint changes only the same error term. For  $0 \leq v \leq t/2$ , Taylor's theorem with a uniform remainder gives

$$(1 - v/t)^{-3/2} = \sum_{j=0}^{N-1} \frac{(3/2)_j}{j!} (v/t)^j + O_N((v/t)^N).$$

Integrating each monomial against  $e^{-v}$  and extending its upper limit to infinity produces  $j!$ ; the extension error is exponentially small in  $t$ . The integral is therefore  $\sum_{j=0}^{N-1} (3/2)_j t^{-j} + O_N(t^{-N})$ .

In the window (60), the prefactor  $e^t/(\sqrt{m} t)$  is bounded above and below by positive constants depending on  $M$ . The relative Riemann error is  $O(t/m)$ , and all lower-half errors are smaller than any fixed  $t^{-N}$ . Combining the estimates proves the uniform expansion.  $\square$

**Corollary 14.2** (Scalar root location). *As  $m \rightarrow \infty$ ,*

$$t_m = \frac{1}{2} \log m + \log \log m + \frac{1}{2} \log \pi + o(1). \quad (62)$$

*In particular,  $t_m = O(\log m)$ ,  $e^{t_m} = O(\sqrt{m} \log m)$ , and  $r_m \rightarrow 1/4$ .*

*Proof.* For a fixed real  $s$ , the  $N = 1$  case of lemma 14.1 gives

$$\sum_{k=1}^m b_k \exp\left(\frac{k}{m} \left[\frac{1}{2} \log m + \log \log m + s\right]\right) \rightarrow \frac{1}{2} + \frac{e^s}{2\sqrt{\pi}}.$$

The limit equals one precisely at  $s_0 = \frac{1}{2} \log \pi$ . For every  $\delta > 0$ , its values at  $s_0 - \delta$  and  $s_0 + \delta$  lie strictly on opposite sides of one. The finite sum is strictly increasing in  $t$ , so its root lies between those trial values for all sufficiently large  $m$ . Let  $\delta \downarrow 0$ . The remaining assertions follow directly from (62) and (58).  $\square$

*Second proof (merge note, from Part I).* Apply lemma 6.1 to  $b_k = C_{k-1} 4^{-k}$ ,  $h = 0$ : then  $B = \frac{1}{2}$ ,  $c = 1/(4\sqrt{\pi})$ , and  $\log \frac{1-B}{2c} = \log \sqrt{\pi}$ , which is (62). Part I applied the same lemma to the coefficients of  $R_m$  ( $B = \frac{3}{4}$ ), which gave the lower endpoint of its window; the extra  $x$  of  $R_m$  alone accounts for the loss of  $4 \log 2$  there, as  $4 \log \frac{1-3/4}{1-1/2} = -4 \log 2$ .  $\square$

**Remark 14.3** (An immediate resolution using the earlier window). Already theorem 10.1 and corollary 14.2 imply  $\liminf E_m \geq 2 \log \pi$ . Combined with the upper bound on  $\limsup E_m$  in (3), this closes Part I's question. The next section proves a substantially stronger quantitative result and provides an independent lower argument rather than relying on that asymptotic window.

## 15 A logarithmically growing cutoff gives a sharp corridor

Take, for all sufficiently large  $m$ ,

$$d = d_m = \lceil \log_2 m \rceil, \quad \varepsilon_d = (d+2)2^{-d} = O(\log m/m). \quad (63)$$

In particular,  $d < m$ ,  $d = O(\log m)$ , and  $\varepsilon_d < 1$ .

**Lemma 15.1** (Value deficit and slope). *At the scalar root  $r_m$ , with  $d$  as in (63),*

$$0 \leq 1 - K_{m,d}(r_m) = O\left(\frac{(\log m)^2}{m}\right), \quad (64)$$

$$K'_{m,d}(r_m) \geq c_* m \quad \text{for all sufficiently large } m, \quad (65)$$

for an absolute constant  $c_* > 0$ .

*Proof.* Set  $t = t_m$ . The lost last  $d$  terms of  $K_m(r_m)$  satisfy

$$\begin{aligned} \Delta_m &:= K_m(r_m) - K_{m-d}(r_m) = \sum_{k=m-d+1}^m b_k e^{tk/m} \\ &\leq C_* d(m-d)^{-3/2} e^t = O\left(\frac{d \log m}{m}\right) = O\left(\frac{(\log m)^2}{m}\right), \end{aligned} \quad (66)$$

using corollary 14.2. Since  $K_m(r_m) = 1$ , lemma 13.2 gives

$$K_{m,d}(r_m) \geq (1 - \varepsilon_d)(1 - \Delta_m), \quad 1 - K_{m,d}(r_m) \leq \varepsilon_d + \Delta_m.$$

This proves (64).

The lower-half estimate in the proof of lemma 14.1, now evaluated at  $t_m$ , shows

$$\sum_{k \leq m/2} b_k e^{t_m k/m} \rightarrow \frac{1}{2}.$$

The whole sum is one, and  $\Delta_m \rightarrow 0$ . Consequently

$$\sum_{m/2 < k \leq m-d} b_k e^{t_m k/m} \rightarrow \frac{1}{2}. \quad (67)$$

Every term in this range has  $k > m/2$ , so

$$K'_{m-d}(r_m) = \frac{1}{r_m} \sum_{k=1}^{m-d} k b_k e^{t_m k/m} \geq \frac{m}{2r_m} \sum_{m/2 < k \leq m-d} b_k e^{t_m k/m}.$$

As  $r_m \rightarrow 1/4$ , the right side is bounded below by a positive constant times  $m$ . The derivative version of (55), together with  $\varepsilon_d \rightarrow 0$ , proves (65).  $\square$

**Theorem 15.2** (Quantitative scalar corridor). *With  $d = \lceil \log_2 m \rceil$ ,*

$$0 \leq s_{m,d} - r_m = O\left(\frac{(\log m)^2}{m^2}\right), \quad 0 \leq \beta_m - s_{m,d}^{-1} = O\left(\frac{(\log m)^2}{m^2}\right). \quad (68)$$

In particular, (37) holds.

*Proof.* Since  $K_{m,d} \leq K_m$ , we have  $s_{m,d} \geq r_m$ . Its derivative is nondecreasing on the positive axis, since all coefficients are nonnegative. Writing  $s = s_{m,d}$ , integration or the mean-value inequality gives

$$1 - K_{m,d}(r_m) = K_{m,d}(s) - K_{m,d}(r_m) \geq K'_{m,d}(r_m)(s - r_m).$$

Divide the estimate (64) by (65) to obtain the first part of (68). Because  $s, r_m \geq 1/4$ ,

$$0 \leq r_m^{-1} - s^{-1} = \frac{s - r_m}{r_m s} \leq 16(s - r_m),$$

which proves the second part. Finally, use the entire chain (53):

$$0 \leq \beta_m - \alpha_m \leq \beta_m - \lambda_U(m) \leq \beta_m - s_{m,d}^{-1}. \quad \square$$

**Corollary 15.3** (The exact constant). *The centered remainder of theorem 1.3 satisfies  $E_m \rightarrow 2 \log \pi$ . Equivalently,*

$$4 - \alpha_m = \frac{2 \log m + 4 \log \log m + 2 \log \pi + o(1)}{m}. \quad (69)$$

*Proof.* By theorem 15.2 and  $\beta_m = 4e^{-t_m/m}$ ,

$$\begin{aligned} m(4 - \alpha_m) &= 4m(1 - e^{-t_m/m}) + O((\log m)^2/m) \\ &= 4t_m + O((\log m)^2/m). \end{aligned} \quad (70)$$

Here Taylor's theorem applies since  $t_m = O(\log m)$ . Substitute (62). This proves (69) and theorem 10.2 in full.  $\square$

**Corollary 15.4** (A one-sided constraint on component competition). *Regardless of which component dominates,*

$$0 \leq \alpha_m - \lambda_U(m) = (\lambda_V(m) - \lambda_U(m))_+ = O\left(\frac{(\log m)^2}{m^2}\right). \quad (71)$$

This is deliberately a one-sided conclusion. We have not proved a corresponding lower approximation for  $\lambda_V(m)$  when it is below  $\lambda_U(m)$ , and do not claim an absolute bound on  $|\lambda_V(m) - \lambda_U(m)|$  in that case. The result does show that whenever  $V$  wins, its advantage is invisible to every inverse-logarithmic term in the scaled deficit.

## 16 Inverting the scalar expansion to every logarithmic order

The quantitative corridor is what makes a scalar asymptotic expansion a statement about the actual permutation class. We now derive the coefficients, including their error interpretation.

### 16.1 Finite asymptotic equations, not a convergent series

Introduce the *formal* power series

$$\mathcal{B}(w) = \sum_{j \geq 0} (3/2)_j w^j. \quad (72)$$

Its coefficients grow factorially. The series has radius of convergence zero and is never evaluated as an ordinary convergent series at  $w > 0$ . For each fixed  $N \geq 1$ , instead use the finite polynomial in  $1/t$

$$\mathcal{B}_N(t) = \sum_{j=0}^{N-1} (3/2)_j t^{-j}.$$

Substituting  $t = t_m$  into lemma 14.1 gives

$$t_m - \log t_m + \log \mathcal{B}_N(t_m) = \frac{1}{2} \log m + \log(2\sqrt{\pi}) + O_N(t_m^{-N}). \quad (73)$$

Indeed the missing mass is  $1/2$ , so  $e^{t_m} \mathcal{B}_N(t_m)/t_m = 2\sqrt{\pi m} (1 + O_N(t_m^{-N}))$ . The polynomial  $\mathcal{B}_N(t)$  is positive for  $t > 0$ , and its logarithm is unambiguous.

Formal logarithms are used below only as a finite coefficient-extraction device. To compute a coefficient of degree  $j$ , only coefficients through degree  $j$  of (72) are needed. Thus the divergent notation organizes an unlimited sequence of finite, justified asymptotic statements; it is not an analytic summation prescription.

## 16.2 A triangular recursion for all coefficients

**Proposition 16.1** (Coefficient recursion). *Set  $\mathcal{P}_0$  aside and define  $\mathcal{P}_j$  recursively as follows. At step  $j \geq 1$ , put*

$$T_{j-1}(u, z) = 1 + zu + \sum_{h=1}^{j-1} \mathcal{P}_h(z) u^{h+1},$$

and define

$$\mathcal{P}_j(z) = [u^j] \left\{ \log T_{j-1}(u, z) - \log \mathcal{B}\left(\frac{u}{T_{j-1}(u, z)}\right) \right\}. \quad (74)$$

These are well-defined rational polynomials of degree  $j$ , with leading coefficient  $(-1)^{j+1}/j$ . For every fixed  $J \geq 0$ ,

$$t_m = \Lambda + z + \sum_{j=1}^J \frac{\mathcal{P}_j(z)}{\Lambda^j} + O_J\left(\frac{(1+|z|)^{J+1}}{\Lambda^{J+1}}\right), \quad (75)$$

with  $\Lambda, z$  as in (39).

*Proof.* Write  $u = 1/\Lambda$  and seek  $t = \Lambda T$ , with

$$T = 1 + zu + \sum_{j=1}^J \mathcal{P}_j(z) u^{j+1}.$$

Since  $z = \log \Lambda + \log(2\sqrt{\pi})$ , substitution into the formal version of (73) leaves the residual

$$\sum_{j=1}^J \mathcal{P}_j(z) u^j - \log T + \log \mathcal{B}(u/T). \quad (76)$$

The coefficient of  $u^j$  in the last two terms depends only on  $\mathcal{P}_1, \dots, \mathcal{P}_{j-1}$ :  $\mathcal{P}_j$  first enters  $T$  at order  $u^{j+1}$ . Cancelling the coefficient of  $u^j$  therefore gives exactly (74), proving existence and uniqueness of the recursion.

Induction shows  $\deg \mathcal{P}_j \leq j$ . The only term of degree  $j$  in  $z$  at order  $u^j$  comes from  $\log(1 + zu)$  and equals  $(-1)^{j+1} z^j / j$ . Terms involving an earlier  $\mathcal{P}_h u^{h+1}$  spend at least one more power of  $u$  than their possible degree in  $z$ ; terms from  $\log \mathcal{B}(u/T)$  have the same strict deficit. Thus the degree is exactly  $j$  and its leading coefficient has the stated value.

To justify the error, fix  $J$  and take  $N \geq J + 2$  in the finite equation (73). For the trial value  $t_*$  obtained by truncating (75), we have  $T = 1 + O_J((1 + |z|)/\Lambda)$  and  $z = O(\log \Lambda)$ . Finite Taylor expansions of the logarithms, with remainders uniform for  $|T - 1| \leq 1/2$ , turn the formal cancellations into the numerical residual estimate

$$t_* - \log t_* + \log \mathcal{B}_N(t_*) - \Lambda - \log(2\sqrt{\pi}) = O_J((1 + |z|)^{J+1} \Lambda^{-J-1}).$$

Both  $t_*$  and  $t_m$  lie in a window with bounded shift of the form (60), by corollary 14.2 and  $\sum_{j \leq J} \mathcal{P}_j(z) \Lambda^{-j} = o(1)$ . The derivative of  $t - \log t + \log \mathcal{B}_N(t)$  is

$$1 - \frac{1}{t} + O_N(t^{-2}),$$

so it is at least  $1/2$  throughout the interval between them for large  $m$ . Apply the mean-value theorem to the residuals of  $t_*$  and  $t_m$ . The  $O_N(t_m^{-N})$  term is smaller than the claimed bound. This proves (75).  $\square$

The first four polynomials are

$$\begin{aligned} \mathcal{P}_1(z) &= z - \frac{3}{2}, \\ \mathcal{P}_2(z) &= -\frac{1}{2}z^2 + \frac{5}{2}z - \frac{33}{8}, \\ \mathcal{P}_3(z) &= \frac{1}{3}z^3 - 3z^2 + \frac{43}{4}z - 15, \\ \mathcal{P}_4(z) &= -\frac{1}{4}z^4 + \frac{10}{3}z^3 - \frac{153}{8}z^2 + \frac{223}{4}z - \frac{4461}{64}. \end{aligned} \tag{77}$$

For an independent algebraic check, the formal logarithm begins

$$\log \mathcal{B}(w) = \frac{3}{2}w + \frac{21}{8}w^2 + \frac{69}{8}w^3 + \frac{2529}{64}w^4 + O(w^5). \tag{78}$$

The bundled symbolic verifier constructs the polynomials from the recursion and confirms that (76) vanishes through degree four using exact rational polynomial arithmetic.

### 16.3 Transfer to the true growth constant

*Proof of theorem 10.3.* Equation (70) gives  $m(4 - \alpha_m) = 4t_m + O((\log m)^2/m)$ . For each fixed  $J$ , this error is smaller than every fixed power of  $1/\log m$ , and hence is absorbed by the remainder in (75). Multiply that expansion by four. The same proof applies to  $\lambda_U(m)$  by (37). For  $\beta_m$ , the direct Taylor expansion of  $4e^{-t_m/m}$  supplies the transfer.  $\square$

With  $z = \log \log m + \frac{1}{2} \log \pi$ , the first three corrections can also be written in powers of  $1/\log m$  (the manuscript abbreviates  $\log m$  as  $H$ , which Part II avoids):

$$\begin{aligned} m(4 - \alpha_m) &= 2 \log m + 4 \log \log m + 2 \log \pi + \frac{8z - 12}{\log m} \\ &\quad + \frac{-8z^2 + 40z - 66}{(\log m)^2} \\ &\quad + \frac{\frac{32}{3}z^3 - 96z^2 + 344z - 480}{(\log m)^3} + O\left(\frac{(1 + |\log \log m|)^4}{(\log m)^4}\right). \end{aligned} \tag{79}$$

The assertion is asymptotic for each fixed truncation order. It does not promise that adding another term improves a numerical approximation at a particular small or moderate value of  $m$ .

### 16.4 Two different error scales

There are two conceptually distinct approximations. The scalar surrogate  $\beta_m$  differs from  $\alpha_m$  by at most order  $(\log m)^2/m^2$ . A fixed truncation of the logarithmic expansion generally has a much larger error, of order a power of  $1/\log m$  after multiplying the growth deficit by  $m$ . Thus a scalar root can be numerically extremely accurate while its first few asymptotic terms are still inaccurate.

This also explains the component issue. The proof does not need to decide whether  $\alpha_m = \lambda_U(m)$  or  $\lambda_V(m)$ . When  $V$  is dominant, corollary 15.4 confines its extra growth to a scale smaller than all the terms in (79). A calculation retaining only those terms has discarded the very information needed to settle that component ordering.

## 17 Exact checks and numerical evidence

The universal results above are proved analytically. The companion computations have three separate roles: exact checks of finite algebra and combinatorics, exact root enclosures in finite examples, and numerical illustrations. They are deliberately not merged into a single ambiguous notion of verification. File names below are the shipped ones; the scripts themselves still read and write their delivered names, so they are rerun on a copy (section 20.2).

### 17.1 Exact finite checks

The exact verifier, the standard-library program code/02-centered-remainder-verify.py, uses integers and fractions.Fraction; it needs no floating-point library. Its successful run is recorded in data/02-centered-remainder-verification\_report.json. The row check verifies coefficient-

Table 6: Executed exact checks (manuscript 04). Finite ranges do not replace the all- $m$  proofs.

| Check                                                                  | Executed range or count                |
|------------------------------------------------------------------------|----------------------------------------|
| Direct triple-definition avoidance versus recursive Catalan generation | All permutations for $0 \leq n \leq 7$ |
| First-entry formula against explicit maximum-ending blocks             | 55 counts, $1 \leq k \leq 10$          |
| Actual $V$ -row polynomial domination by the Catalan witness           | 4,930 rows, $2 \leq m \leq 30$         |
| Uniform deficiency-tail inequality                                     | 9,000 pairs, $k \leq 300, d \leq 30$   |
| Restricted $U$ -row coefficient comparisons                            | 67,425 comparisons, $m \leq 30$        |
| Exact scalar-root corridors                                            | Five $(m, d)$ pairs                    |

twice that  $H_q - (W_{V_m} v)_{p,q} - (1 - K_m)$  has nonnegative coefficients. This is stronger than testing the inequality at a list of decimal values of  $x$ . The tail check clears the denominator  $2^d$  and verifies an integer inequality. The direct avoidance test provides a separate combinatorial check of the recursive generator before its blocks are used to test the first-entry formula. The first two rows repeat, with independent code and the same ranges, checks of Part I's verifier (section 8); the other four are new.

The symbolic program code/02-centered-remainder-verify\_expansion.py uses SymPy but no floating-point coefficients. It verifies (77), (78), and the vanishing residual through degree four. Its output is data/02-centered-remainder-expansion\_verification.json. Symbolic cancellation proves these finite coefficient identities, not the analytic error estimate; that estimate is the content of lemma 14.1 and proposition 16.1.

### 17.2 Certified finite corridors

For a positive polynomial  $P$  with zero constant term, exact bisection encloses the root of  $P(x) = 1$  by fractions  $\ell, u$  satisfying  $P(\ell) \leq 1 \leq P(u)$ . Apply this separately to  $K_m$  and  $K_{m,d}$ . If the enclosures are  $[\ell_r, u_r]$  and  $[\ell_s, u_s]$ , then

$$\frac{1}{u_s} \leq \alpha_m \leq \frac{1}{\ell_r}. \quad (80)$$

The rational certificate file data/02-centered-remainder-rational\_certificates.json retains the full fractions and can be regenerated by the exact verifier. The following decimal endpoints are rounded *outwards* from those fractions, not nearest-rounded approximations to unknown

Table 7: Rigorous finite intervals from exact scalar bisection.

| $m$ | $d$ | Lower endpoint for $\alpha_m$ | Upper endpoint for $\alpha_m$ |
|-----|-----|-------------------------------|-------------------------------|
| 5   | 1   | 2.0000000000                  | 2.4580325859                  |
| 10  | 2   | 2.7001330054                  | 2.9828910906                  |
| 20  | 3   | 3.2335773659                  | 3.3677623965                  |
| 50  | 5   | 3.6481708456                  | 3.6845734707                  |
| 100 | 7   | 3.8074760932                  | 3.8201232724                  |

values of  $\alpha_m$ . These intervals are not advertised as optimal small- $m$  estimates. Their purpose is to provide simple, independently checkable examples of the new all- $m$  scalar corridor without computing a large characteristic polynomial. For  $m \leq 20$ , Part I's rational Perron certificates (section 8) are much sharper.

### 17.3 Floating-point component and large-bound calculations

The separate program code/02-centered-remainder-compute.py evaluates sparse component matrices and scalar roots. Its component values reproduce the known small-bound pattern (compare table 2) and illustrate the new upper bound. For larger  $m$ , no matrix eigensolver is

Table 8: Numerical growth rates; displayed digits are not interval certificates. The source file data/02-centered-remainder-component\_numerics.csv also has a row for  $m = 30$ .

| $m$ | $\lambda_U(m)$ | $\lambda_V(m)$ | $\alpha_m$  | $\beta_m$   |
|-----|----------------|----------------|-------------|-------------|
| 2   | 1.465571232    | 1.000000000    | 1.465571232 | 1.618033989 |
| 3   | 1.826515772    | 1.691402554    | 1.826515772 | 2.000000000 |
| 4   | 2.100456128    | 2.091476748    | 2.100456128 | 2.263408655 |
| 5   | 2.312119493    | 2.351509000    | 2.351509000 | 2.458032586 |
| 10  | 2.902174743    | 2.952689154    | 2.952689154 | 2.982891091 |
| 20  | 3.333400105    | 3.356795518    | 3.356795518 | 3.367762396 |
| 50  | 3.675866778    | 3.681914231    | 3.681914231 | 3.684573471 |

needed to evaluate the scalar corridor. Normalized coefficients  $b_k$  are generated by

$$b_1 = \frac{1}{4}, \quad \frac{b_{k+1}}{b_k} = \frac{2k-1}{2(k+1)}.$$

Solving for  $t = m \log(4x)$  avoids subtracting a tiny root displacement from  $1/4$ . The block coefficients are also normalized, using the consecutive first-deficiency ratio obtained from (21).

Table 9: Floating-point evaluations with  $d = \lceil \log_2 m \rceil$ . The mathematical endpoints are rigorous bounds; these nearest-rounded numerical evaluations are not themselves certified intervals. From data/02-centered-remainder-scalar\_numerics.csv (63 values of  $m$ ).

| $m$    | $d$ | $s_{m,d}^{-1}$ | $\beta_m$      | Corridor width         |
|--------|-----|----------------|----------------|------------------------|
| $10^2$ | 7   | 3.807476093297 | 3.820123272395 | $1.265 \cdot 10^{-2}$  |
| $10^3$ | 10  | 3.975171751563 | 3.975423985146 | $2.522 \cdot 10^{-4}$  |
| $10^4$ | 14  | 3.996950887245 | 3.996955062448 | $4.175 \cdot 10^{-6}$  |
| $10^5$ | 17  | 3.999640264705 | 3.999640325443 | $6.074 \cdot 10^{-8}$  |
| $10^6$ | 20  | 3.999958715503 | 3.999958716333 | $8.301 \cdot 10^{-10}$ |

At  $m = 10^6$ , the two numerical bounds for  $E_m$  are approximately 3.149478 and 3.150308, although  $2 \log \pi \approx 2.289460$ . This is compatible with the theorem: the leading correction to the

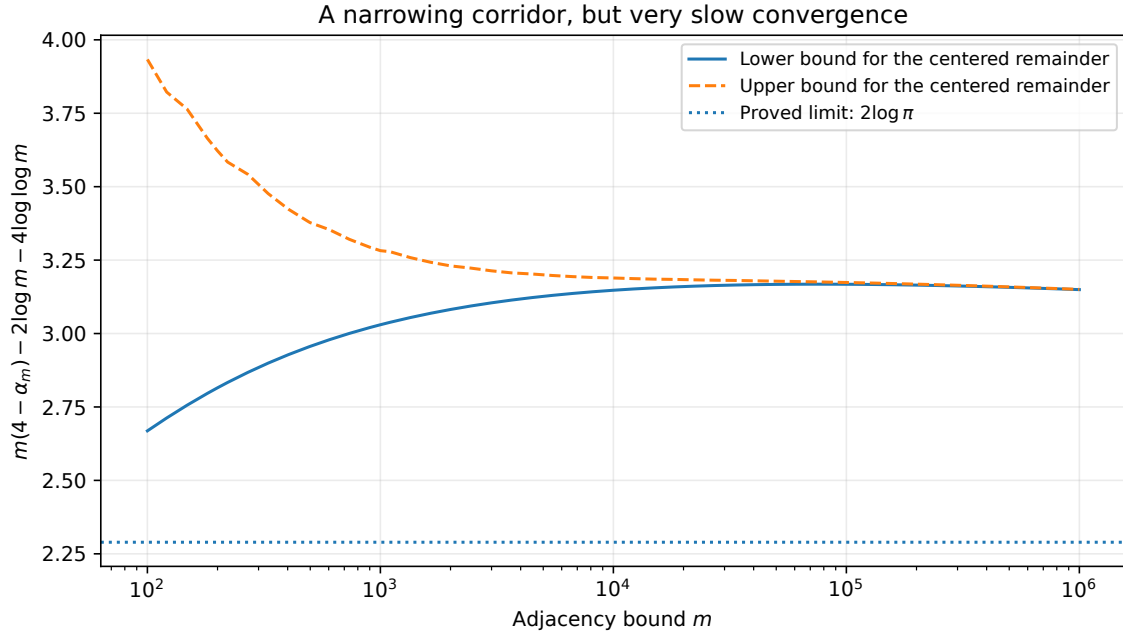


Figure 3: Numerical evaluations of the lower and upper bounds for  $E_m$ , not direct evaluations of the unknown large- $m$  component maximum. The dotted line is the proved limiting constant. The asymptotic bounds need not be monotone, and the visible range has not reached the limit. (Manuscript 04’s figure; compare figure 2, whose bounds are Part I’s.)

constant is  $(8z - 12)/\log m$ , which decays very slowly. A close upper and lower corridor is not evidence that its center is already near its limit.

*Remark 17.1* (Merge note: the numerics are not evidence for the value of the limit). The corridor in figure 3 sits about 0.86 above  $2 \log \pi$  at  $m = 10^6$ , and its lower endpoint is not even monotone in the computed range: it rises to about 3.168 near  $m = 73,920$  and then decreases slowly. At  $m = 10^6$  the first correction term  $(8z - 12)/\log m$  of (79) is about 0.983, and the three displayed corrections together give  $2 \log \pi + 0.983 - 0.104 - 0.005 \approx 3.164$ , within 0.015 of the corridor. That agreement shows that the computations are consistent with the inverse-logarithmic corrections. It is *not* evidence for the value  $2 \log \pi$  of the limit: no computation in the range  $m \leq 10^6$  distinguishes that constant from nearby ones, and a slowly drifting corridor centred near 3.15 would look the same whatever the limit were. The value  $2 \log \pi$  rests on the proofs of sections 12 to 15 alone. The decimals quoted in this remark were computed for the merge from data/02-centered-remainder-scalar\_numerics.csv.

The computations do not estimate an exact leading constant for the scalar defect, certify component ordering beyond the tested values, or validate an infinite asymptotic expansion by numerical fitting.

## 18 A complete small example and a useful non-implication

At  $m = 2$ , order the  $U$  states by  $p = 0, 1$ , and the  $V$  states as  $(1, 0), (0, 1)$ . Then

$$W_{U_2}(x) = \begin{pmatrix} 0 & x \\ x^2 & x \end{pmatrix}, \quad W_{V_2}(x) = \begin{pmatrix} 0 & x \\ x & 0 \end{pmatrix}.$$

Thus  $\lambda_U(2)$  is the positive root of  $\lambda^3 - \lambda^2 - 1 = 0$ , while  $\lambda_V(2) = 1$  (the matrix  $W_{U_2}$  also appears in the proof of theorem 1.1). The scalar polynomial is  $K_2(x) = x + x^2$ , and  $\beta_2 = (1 + \sqrt{5})/2$ .

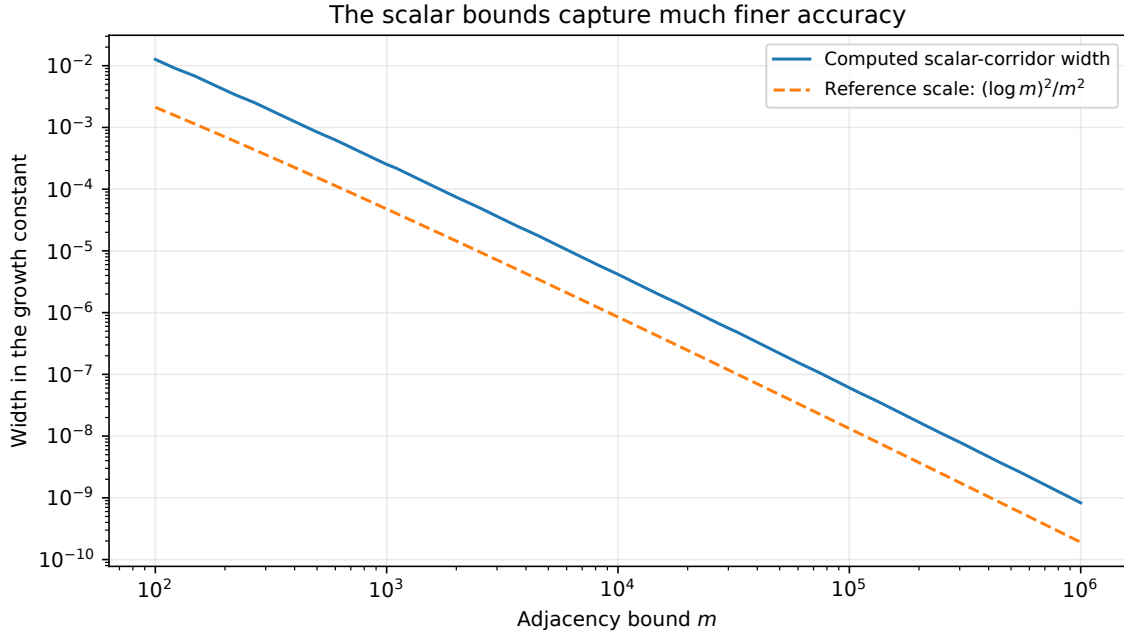


Figure 4: The numerical width  $\beta_m - s_{m,d}^{-1}$  and the comparison scale  $(\log m)^2/m^2$ . The dashed curve has coefficient one solely for scale comparison; no claim is made that it is a finite- $m$  upper bound or the optimal asymptotic equivalent.

The last-threshold vector is  $v = (1, 1 + x)^\top$ . At the root  $K_2(r_2) = 1$ ,

$$W_{V_2}(r_2)v = \begin{pmatrix} r_2 + r_2^2 \\ r_2 \end{pmatrix} = \begin{pmatrix} 1 \\ r_2 \end{pmatrix} <_{\text{some}} \begin{pmatrix} 1 \\ 1 + r_2 \end{pmatrix} = v,$$

where  $<_{\text{some}}$  means coordinatewise inequality with at least one strict coordinate. The separate reset comparator of lemma 12.1 is

$$N(x) = \begin{pmatrix} 0 & x \\ x & x \end{pmatrix}, \quad \det(I - N(x)) = 1 - x - x^2.$$

This differs from  $\det(I - W_{V_2}(x)) = 1 - x^2$ , illustrating exactly why a spectral comparison is not an equality of state systems.

There is also an immediate counterexample to the tempting coefficientwise interpretation. For length three and  $m = 2$ , every 132-avoider meets the adjacency bound, so  $[x^3]A_2(x) = C_3 = 5$  (table 1). But

$$[x^3] \frac{1}{1 - K_2(x)} = [x^3] \frac{1}{1 - x - x^2} = 3.$$

The upper bound on exponential growth is valid even though this proposed coefficientwise majorant is false. This example is a useful diagnostic when transporting the renewal argument to another combinatorial class.

## 19 Further research questions and proposed approaches

The exact constant is no longer the first obstruction. The quantitative corridor changes which questions are mathematically distinguishable and which tools are likely to be useful. The second question below is Part I's first open question (section 9), now constrained by corollary 15.4; the eighth continues Part I's remark on minimal recurrence orders and denominators.

### 19.1 The first term beyond scalar universality

Determine the true order and leading behavior of  $\beta_m - \alpha_m$ . Our upper bound is  $O((\log m)^2/m^2)$ , obtained by discarding a logarithmic number of terminal block lengths and controlling a deficiency tail uniformly. Those losses need not be optimal. A sharper approach would keep the compatibility between a block length and the next first deficiency rather than impose a single worst-case cutoff. A small matrix of deficiency classes, with a positive-vector comparison analogous to lemma 12.2, is a concrete candidate. No asymptotic equivalent for this defect is asserted here.

### 19.2 The eventual ordering of the two components

Resolve whether  $\lambda_V(m) > \lambda_U(m)$  for every  $m \geq 5$ , the ordering suggested by finite computations and related to Problem P3 in [2]. By corollary 15.4, a proof cannot be obtained merely by comparing their shared inverse-logarithmic expansion with the scalar root. It must detect finer endpoint losses or construct a comparison preserving more information than the present supersolution. One possibility is to compare first-return operators after isolating the large-threshold states, where long blocks are compatible with nearly all first entries.

### 19.3 The geometry of near-critical Perron vectors

Describe the positive eigenvectors of  $W_{U_m}$  and  $W_{V_m}$  at their critical roots as  $m \rightarrow \infty$ . The scalar tilted mass is concentrated in a layer of width about  $m/t_m$  near the maximal block length, while the chosen cutoff is only of order  $\log m$ . A useful question is whether endpoint coordinates admit a scaling limit connecting these two regimes. Such a limit could explain the loss of the scalar upper bound and supply an analytic route to the sign of the component gap. The present test vector is a supersolution, not a proved approximation to the true eigenvector.

### 19.4 Poles, amplitudes, and a uniform spectral gap

For each fixed  $m$ , determine when the dominant pole of  $A_m$  is simple and obtain a usable expression for the leading amplitude. An equality of component rates could interact with paths between components and alter pole order. Our argument identifies only their maximal exponential rate. A uniform analysis of subdominant singularities would be needed to turn fixed- $m$  spectral information into estimates with useful error constants as  $m$  grows. This is distinct from the scalar root's own simple zero, which follows immediately from  $K'_m(r_m) > 0$ .

### 19.5 Joint growth of length and adjacency bound

Study  $a_n^{(m(n))}/C_n$  when  $m = m(n)$  grows with  $n$ . The fixed- $m$  limit used here is not automatically uniform, so substituting  $m(n)$  into  $\alpha_m^n$  is not justified. Candidate regimes should compare the exponential penalty  $n \log(4/\alpha_m)$  with the finite-state mixing scale and the Catalan polynomial factor. A satisfactory result would establish an explicit range of uniformity before proposing a transition law. The identity  $a_n^{(m)} = C_n$  when  $m \geq n - 1$  provides an exact boundary condition for such a theory. [Added 29 September 2026, batch 43: the boundary end of this direction is answered in Part III. For fixed  $d$ ,  $a_n^{(n-d)}/C_n$  converges to an explicit rational limit, and uniformly for  $n \geq 8$ ,  $2 \leq d \leq n - 1$ ,  $a_n^{(n-d)}/C_n = 3/\sqrt{\pi d} + O(d^{-3/2} + n^{-1/2})$ , an equivalent whenever  $d \rightarrow \infty$  and  $d = o(n)$  (theorem 22.2 and equation (136)). The direction stays open for  $n - m \asymp n$  (remark 29.3) and for any interpolation with the fixed- $m$  regime (section 32.10).] [Added 29 September 2026, batch 52: Part IV answers the linear regime  $m = \lfloor \theta n \rfloor$  for  $1/2 < \theta < 1$  to first order,  $a_n^{(\lfloor \theta n \rfloor)}/C_n \sim \frac{3(2\theta-1)}{\sqrt{\pi\theta(1-\theta)}} n^{-1/2}$  (theorem 38.1); for  $0 < \theta \leq 1/2$  it proves only  $o(n^{-1/2})$  and polynomial lower bounds (theorem 45.1). The interpolation with the fixed- $m$  regime stays open.]

## 19.6 Renewal comparisons for other constrained tree classes

The reset identity lemma 12.1 is not specifically Catalan. It applies whenever a monotone threshold is lowered by weighted steps and restored by a reset. Identify other pattern-avoidance or tree-enumeration problems where the associated renewal sequence is explicitly controlled. If a truncated scalar coefficient law has a summable power tail, an analogous endpoint tilt should be investigated. The necessary research task is to prove a matching combinatorial lower construction; a scalar majorant alone does not make its asymptotic constant universal.

## 19.7 Weighted statistics and multivariate perturbations

Refine the endpoint recurrence with variables recording descents, records, or block statistics, in the direction of Problem P4 of [2]. The central question is whether a positive last-threshold witness persists under a parameter perturbation. A successful extension might permit derivative bounds for a weighted growth rate and limit laws at fixed  $m$ . Positivity and irreducibility must be checked in the perturbed system; formal differentiation of an unproved multivariate analogue of  $K_m$  would be insufficient.

## 19.8 State minimization versus a spectral majorant

Investigate whether the renewal witness can guide an exact reduction of the endpoint system without confusing domination with equality. Problems P1 and P2 in [2] concern issues such as state complexity and rational denominators. Our  $m$ -state reset comparator has determinant  $1 - K_m$ , but it forgets first-endpoint compatibility. One concrete goal is a characterization of the lost information as a correction operator of small rank or controlled support. An exact operator identity would be a substantive advance beyond the present scalar comparison.

## 19.9 Certified computation and formalization

Build an adaptive exact or interval algorithm choosing  $d$  to meet a requested error tolerance, using  $K_m$  and  $K_{m,d}$  directly. This would avoid large component determinants and would certify large- $m$  growth intervals with clear cost accounting. For formalization, the finite renewal polynomial identity, the spectral supersolution argument, the skew-cut injection, and the uniform deficiency-tail inequality are natural first targets. The analytic tail splitting and all-orders inversion can then be formalized as separate lemmas with explicit uniform quantifiers. None of this formalization has been started.

# 20 Proof boundaries, reproducibility and provenance

## 20.1 What the manuscript has established

The proof route can be read as a dependency chain:

endpoint system  $\implies \alpha_m < \beta_m \implies$  sharp scalar corridor  $\implies$  exact constant and all orders.

The lower arrow uses both recoverable skew blocks and the uniform first-entry tail; the analytic arrow uses the finite endpoint expansion with its stated remainder. No repository assertion about strict monotonicity is needed for the new theorems: theorem 1.1 is not used. The finite-state growth interface is proved in Part I and was rederived in the manuscript, so it is also not being accepted solely on the authority of a software claim.

## 20.2 Reproducing Part II's computations

Part II's files are

|                                      |                                            |
|--------------------------------------|--------------------------------------------|
| code/02-centered-remainder-model.py  | Exact Catalan, endpoint, and block objects |
| code/02-centered-remainder-verify.py | Standard-library exact verification        |

Table 10: Trust and scope ledger of Part II.

| Statement or artifact                                   | Validation status                                                                                                                               |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| All- $m$ upper bound and asymptotic theorems            | Written mathematical proofs here                                                                                                                |
| Finite row, block, and tail tests                       | Executed exact integer/rational checks                                                                                                          |
| First four inverse-log polynomials                      | Executed exact symbolic checks                                                                                                                  |
| Large- $m$ numerical figures and component table        | Floating-point illustrations only                                                                                                               |
| New Lean or Rocq theorem declarations                   | None supplied or claimed                                                                                                                        |
| External peer review or exhaustive novelty verification | Not performed                                                                                                                                   |
| Component ordering, optimal scalar defect, joint limits | Open directions (section 19) [joint limits near $m = n$ : Part III, added 29 September 2026; for $1/2 < m/n < 1$ : Part IV, added the same day] |

|                                                |                                                                                                     |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| code/02-centered-remainder-verify_expansion.py | Exact symbolic coefficient check                                                                    |
| code/02-centered-remainder-compute.py          | Floating-point numerical illustrations                                                              |
| code/02-centered-remainder-plot_results.py     | The two figures from the CSV data                                                                   |
| data/02-centered-remainder-*                   | Check reports, rational certificates, CSVs, environment, requirements, records of the delivered PDF |
| figures/02-centered-remainder-*.pdf            | The two figures of this part                                                                        |

The scripts are shipped byte-identical to the delivery and still use the delivered names. They write

|                                  |                                      |
|----------------------------------|--------------------------------------|
| data/verification_report.json    | data/rational_certificates.json      |
| data/expansion_verification.json | data/scalar_numerics.csv             |
| data/component_numerics.csv      | figures/remainder_corridor.{pdf,png} |
|                                  | figures/corridor_width.{pdf,png}     |

and the verifier and the numerical program execute from `model import`. Run in place, that import would load Part I's different `code/model.py` and fail with `ImportError`, and the outputs would land beside Part I's unprefix files. The report's README gives a recipe that restores the delivered layout in a scratch copy. There, the exact checks need Python 3.10 or newer and no third-party packages (`python code/verify.py`; do not use `-O`, the verifier uses assertions); the symbolic check needs SymPy; the numerical programs need NumPy, SciPy and, for the plots, Matplotlib (versions in `data/02-centered-remainder-environment.json`). A POSIX `OPENBLAS_NUM_THREADS=1` prefix on the numerical command is optional; it limits numerical-library threading rather than changing any mathematical assertion. The retained figures suffice to rebuild the article without rerunning any computation.

### 20.3 Points checked in the written argument

The sharp upper bound is not obtained by dropping a legitimate append term: that term is incorporated by the last-threshold vector. The first-entry tail estimate is uniform in both parameters, which permits a growing cutoff. The conversion from a polynomial value deficit to a root displacement uses a proved derivative lower bound, not a heuristic linearization. The order of inequalities is preserved when reciprocal roots become growth bounds and when growth bounds become deficit bounds. Finally, the factorially divergent formal series is used only to define finite coefficient calculations, while each analytic statement has its own fixed-order remainder.

These checks identify the parts of the proof most worth independent scrutiny. They do not replace that scrutiny, but they make its targets explicit and keep finite experiments distinct from universal arguments.

## 20.4 Repository provenance and novelty boundary

The manuscript inspected Part I at the repository-relative path

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/
adjacency-bounded-132-avoiders/article.tex
```

at the ProveIt commit

```
6c9175a54bcaad3bfc2d416257d2b7d3dff2c1e0
```

where its Git blob identifier was

```
5ec75ee48df94b6d12804a26ba109c08564df5c7
```

Its section “What is resolved, and what remains” (section 9) explicitly asks whether  $E_m$  converges and describes the unresolved constant window; the opening theorem statements specify the window in (3). The finite endpoint system and lower block construction used here are attributed to their prior sources, with independent proofs supplied in the manuscript.

The manuscript’s primary-paper check used the arXiv version-one records and HTML sources for [1, 2], accessed on September 28, 2026. The inspected materials did not contain the last-threshold supersolution, the sharp scalar corridor, or the exact constant proved here. This is a bounded provenance statement. It does not certify the absence of an independent unpublished argument, a later repository change, or an uninspected paper.

## 20.5 Where the merge had to choose

- *Placement.* Part II follows Part I’s last main section and precedes Part I’s appendices, which contain no numbered equations, tables or figures; no number of Part I changed.
- *Duplicates.* The endpoint recurrence, the component classification and spectral interface (the manuscript’s Section 2 and Appendix A), the first-entry lemma and the injective skew-block construction are printed once, in Part I, and credited in sections 11 and 13. The manuscript’s different argument for  $\lambda_U(m) > 1$  is kept as a second proof. The second proof of corollary 14.2 and remarks 12.3, 13.3 and 17.1 are the merge’s.
- *Notation.*  $L \rightarrow \Lambda$ ,  $P_j \rightarrow \mathcal{P}_j$ ,  $B_N \rightarrow \mathcal{B}_N$ ,  $T_m \rightarrow \Delta_m$ ,  $C, c \rightarrow C_*, c_*$  (absolute constants),  $W_U, W_V \rightarrow W_{U_m}, W_{V_m}$ , and  $H = \log m$  written out (table 4). No normalization changed.
- *References to Part I.* The manuscript cites Part I as a repository report; here those citations are internal cross-references, and the manuscript’s bibliography entries for [1, 2] are Part I’s.
- *Reproduction.* The manuscript’s archive listing and commands are replaced by the shipped names and the copy recipe (section 20.2); its `article.tex`, `article.pdf`, `README.md` and the PNG versions of its figures are not shipped.
- *Figures.* In the write phase the manuscript’s two figure PDFs were copied byte-identically into `figures/`, under their delivered names with the prefix `02-centered-remainder-` (figures 3 and 4).
- *Stale sentences in Part I.* The abstract, theorem 1.3, the remark after theorem 1.3’s proof, section 9 and section B carry dated pointers to Part II; their original wording is kept.

## Part III. The largest jump near the boundary

*Part III was added on 29 September 2026 in batch 43 of ProveIt's incoming-report intake, from one manuscript written against the state of Parts I and II printed above. Parts I and II take  $n \rightarrow \infty$  with the adjacency bound  $m$  fixed. Part III takes the other side of the joint problem: the bound sits a distance  $d$  below its trivial value,  $m = n - d$ , with  $d$  fixed or growing slower than  $n$ . It answers Part II's research direction "Joint growth of length and adjacency bound" (section 19) in that boundary regime only; the bulk regime  $n - m \asymp n$  and the fixed- $m$  spectral questions stay open (section 21.3).*

### 21 Provenance, conventions and the answer

#### 21.1 One manuscript, one addition

Part III is batch-43 manuscript 02, *The Largest Jump Is Almost Maximal: An algebraic boundary law for random 132-avoiding permutations*, an AI-assisted research manuscript prepared for Vladimir Reshetnikov and dated September 29, 2026 (26 pages, US Letter, as delivered). It arrived as the archive ProveIt\_Largest\_Jump\_Research.zip (inner directory ProveIt\_Largest\_Jump/, main file article.tex) in ProveIt commit 06bcc37a8 and was placed in faef2ed2a. It was written against ProveIt commit 8d936ee23, at which this report's README.md and article.tex had the Git blobs 7f56b878 and f24ea59a. Both blobs were unchanged at the placement commit, so every statement the manuscript makes about this report refers to Parts I and II as printed above. Its code, data and audit notes are shipped beside the earlier parts' files with the prefix 03-largest-jump- (Part I's unprefixed files count as 01, Part II's 02-centered-remainder- files as 02); its manuscript, README and PDF are not shipped and survive in the arrival commit. Its two figures were added to figures/ in the write phase (section 36.2).

The manuscript contributes, in the order printed here: its main theorems and their scope (section 22); first- and last-entry identities for 132-avoiders (section 23); a countable probability space of finite absorbing skeletons (section 24); a deterministic reconstruction lemma (section 25); a total-variation bound by counting domination (section 26); the exact algebraic probability generating function of the limit law (section 27); singularity analysis and a sharp rate (section 28); the joint limits for the adjacency bound (section 29); a moment transition (section 30); exact probabilities and finite checks (section 31); ten research questions (section 32); a proof audit (section 33); and, as the last three sections of this part, its coefficient recursions, an algebraic certificate and its provenance (sections 34 to 36), which were appendices in the manuscript. Every result, proof, example, remark, limitation and question of the manuscript is printed.

**Status (manuscript 02).** The theorems of Part III have written proofs and reproducible exact finite checks. They are not Lean- or Rocq-verified and have not undergone external peer review. The contribution is an extension of the report's subject, not a verified claim of publication priority. No asymptotic theorem from an unrefereed repository report is assumed: Part III uses none of the growth-constant theorems of Parts I and II. Nothing in Part III is formalized in any proof assistant.

#### 21.2 Notation of Part III

The symbols  $a_n^{(m)}$ ,  $C_n$ ,  $C$  and  $\text{Av}_n(132)$  keep the meanings of Parts I and II, and  $m$  is again the adjacency bound and  $n$  the length. Part III is otherwise self-contained, and several of its letters are used in Parts I and II for other objects. Table 11 lists them, with the tempting false reading in the last column; five symbols are renamed from the manuscript.

No normalization changed: every renamed symbol denotes exactly the manuscript's quantity. Probability, expectation, total variation and law are written  $\mathbb{P}$ ,  $\mathbb{E}$ ,  $d_{\text{TV}}$ ,  $\mathcal{L}$ , as in the manuscript.

Table 11: Symbols of Part III. “Renamed” gives the manuscript’s own symbol. The last column prints the tempting false reading.

| Symbol                                                                            | Meaning in Part III                                                                            | Not to be confused with                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $M(\pi), M_n$                                                                     | largest adjacent absolute difference; that of a uniform $\Pi_n \in \text{Av}_n(132)$           | the matrix $M$ of lemma 11.3.                                                                                                                                                                |
| $D(\pi), D_n, D$                                                                  | the deficit $n - M(\pi)$ , its value $n - M_n$ , and the limit law of $D_n$                    | Part I’s common denominator $D(x)$ of (34), its denominators $D_U(x), D_V(x)$ (section A), the integer $D$ of its Perron certificates, and the exponent $D$ in the proof of proposition 3.3. |
| $d$                                                                               | a deficit threshold, $m = n - d$ ; also the first coordinate of the endpoint state $(d, \ell)$ | Part I’s fixed cutoff $d$ and Part II’s $d_m = \lceil \log_2 m \rceil$ ; neither is tied to $n$ .                                                                                            |
| $F(\pi) = \pi_1$                                                                  | first entry                                                                                    | Part I’s series $F_{p,q}$ . Part I’s first deficiency is $n - F(\pi)$ .                                                                                                                      |
| $N_{N,f}$                                                                         | avoiders of size $N$ with first entry $f$                                                      | the same count as Part I’s $h(k, j)$ , with $k = N + 1, j = N + 1 - f$ (remark 23.2).                                                                                                        |
| $P(x, z)$                                                                         | bivariate first-entry series                                                                   | Part I’s numerators $P_3, P_4$ ; Part II’s $\mathcal{P}_j$ .                                                                                                                                 |
| $R_\beta, A, T_\tau$                                                              | skeleton operations; $\beta, \tau$ are avoiders                                                | $R_m = x + K_m, A_m(x)$ , the counts $T_{p,q}^{(m)}(n)$ , and Part II’s number $\beta_m = r_m^{-1}$ .                                                                                        |
| $t, T$                                                                            | cost of a skeleton; the random cost                                                            | Part II’s $t_m = m \log(4r_m)$ ; $T_{p,q}^{(m)}(n)$ .                                                                                                                                        |
| $b_t, B(x)$                                                                       | number of skeletons of cost $t$ ;<br>$B = (C - 1)^2$                                           | Part II’s $b_k = C_{k-1}4^{-k}$ ; Part I’s threshold set $B_m$ and $B_d$ .                                                                                                                   |
| $c_j$ (one index)                                                                 | $C_j/4^j$                                                                                      | Part I’s $c_{k,p}$ (two indices).                                                                                                                                                            |
| $h$ (no argument)                                                                 | the cutoff $\lfloor n/4 \rfloor$ , or a cutoff below $n/2$                                     | Part I’s first-entry count $h(k, j)$ .                                                                                                                                                       |
| $L$                                                                               | random last-entry state, in $\mathbb{N}_{\geq 0} \cup \{\infty\}$                              | the left block $L$ of $\pi = L n R$ (also used here), Part I’s $L_m$ .                                                                                                                       |
| $\omega_n$ (renamed from $\varepsilon_n$ )                                        | $n(n + 1)(2n + 1)2^{-n/2}$                                                                     | Part II’s $\varepsilon_d = (d + 2)2^{-d}$ , a different bound for a related fraction (remark 23.2).                                                                                          |
| $z; s, r$                                                                         | generating-function variable; $\sqrt{1 - z}$ , $\sqrt{(1 + s)/2}$                              | Part II’s $z = \log \log m + \frac{1}{2} \log \pi$ ; Part I’s $s_{m,d}$ ; Part II’s root $r_m$ and $r_U, r_V$ .                                                                              |
| $q = Q(z)$                                                                        | $(1 - \sqrt{1 - z})/2$                                                                         | Part I’s root $q_m$ and threshold index $q$ .                                                                                                                                                |
| $G, H, J$                                                                         | generating function of $D$ ; its parts with $L < \infty$ and $L = \infty$                      | Part II’s $H_q(x)$ , the matrix $H$ in the proof of proposition 3.3, Part II’s order index $J$ .                                                                                             |
| $\mathcal{K}(z)$                                                                  | clipping series                                                                                | $K_{m,d}$ (Part I), $K_m$ (Part II).                                                                                                                                                         |
| $\Upsilon(z)$ (renamed from $\mathcal{A}$ )                                       | clipping correction                                                                            | Part I’s class $\mathcal{A}_n^{(m)}$ , the manuscript’s own glyph.                                                                                                                           |
| $\kappa_{1-\varepsilon}$ (renamed from $q_{1-\varepsilon}$ )                      | quantile of $D$                                                                                | $q = Q(z)$ .                                                                                                                                                                                 |
| $C_{\text{TV}}$ (renamed from $C, C_0$ ), $\underline{c}_p$ (renamed from $c_p$ ) | constants in two proofs                                                                        | the Catalan numbers, $C_0 = 1$ ; $c_j$ ; $c_{k,p}$ ; Part II’s $C_*, c_*$ .                                                                                                                  |
| $p$                                                                               | a moment order                                                                                 | Part I’s threshold $p \in B_m$ .                                                                                                                                                             |

The equations of Part III are numbered consecutively with those of Parts I and II, not by section as in the manuscript.

### 21.3 Which question is answered, and how far

Part II's research direction "Joint growth of length and adjacency bound" (section 19) asks for  $a_n^{(m(n))}/C_n$  when  $m = m(n)$  grows with  $n$ , warns that the fixed- $m$  limit is not automatically uniform, asks that "a satisfactory result" establish an explicit range of uniformity before a transition law, and names the identity  $a_n^{(m)} = C_n$  for  $m \geq n - 1$  as a boundary condition. The report's README lists "no joint limit in  $n$  and  $m$ " among the things not claimed. Part III settles the regime next to that boundary condition:

- for each fixed  $d$ ,  $a_n^{(n-d)}/C_n \rightarrow \mathbb{P}(D \geq d)$ , an explicit rational number (equation (136));
- uniformly for  $n \geq 8$  and  $2 \leq d \leq n - 1$ ,  $a_n^{(n-d)}/C_n = 3/\sqrt{\pi d} + O(d^{-3/2} + n^{-1/2})$  with an absolute constant (theorem 22.2), which is an explicit range of uniformity of the kind Part II asks for;
- hence  $a_n^{(n-d)}/C_n \sim 3/\sqrt{\pi d}$  whenever  $d \rightarrow \infty$  and  $d = o(n)$ .

It does not settle the question. When  $n - m$  is a fixed fraction of  $n$ , the additive error is as large as the main term and no equivalent is claimed (remark 29.3); for fixed  $m$  the growth-constant regime of Parts I and II is untouched; and nothing connects the two regimes yet (section 32.10). Part II's direction is therefore re-scoped, not closed: its boundary end is answered here, and its bulk and interpolation questions remain open. Part III refutes nothing in Parts I and II and uses none of their theorems. [Added 29 September 2026, batch 52: Part IV settles the bulk regime for  $1/2 < m/n < 1$  to first order and determines the exact supercritical moment constants (section 37.3); the regime  $m/n \leq 1/2$  and the interpolation with fixed  $m$  remain open.]

## 22 The question, its context, and the main results

### 22.1 A different order of limits

As in Part I, a permutation  $\pi \in S_n$  avoids 132 if no three indices  $i < j < k$  satisfy  $\pi_i < \pi_k < \pi_j$ . Write  $\text{Av}_n(132)$  for this class, and let

$$C_n = \frac{1}{n+1} \binom{2n}{n}, \quad C(x) = \sum_{n \geq 0} C_n x^n = \frac{1 - \sqrt{1 - 4x}}{2x}.$$

For  $n \geq 2$ , set

$$M(\pi) = \max_{1 \leq i < n} |\pi_{i+1} - \pi_i|, \quad D(\pi) = n - M(\pi). \quad (81)$$

If  $\Pi_n$  is uniform on  $\text{Av}_n(132)$ , write  $M_n = M(\Pi_n)$  and  $D_n = D(\Pi_n)$ . Thus  $1 \leq D_n \leq n - 1$ .

Let  $a_n^{(m)}$  count the avoiders with every adjacent difference at most  $m$ . The identity connecting enumeration and probability is

$$\frac{a_n^{(m)}}{C_n} = \mathbb{P}(M_n \leq m) = \mathbb{P}(D_n \geq n - m). \quad (82)$$

The fixed- $m$  problem studies  $n \rightarrow \infty$  with  $m$  unchanged. Part III studies the boundary  $m = n - d$ , and then allows  $d$  to grow with  $n$ . Neither problem is a substitution into the other.

Nadler initiated the adjacency-constrained enumeration and analyzed  $m = 2$  [1]. Mayama and Akita supplied a uniform endpoint-state system, rational generating functions for every fixed  $m$ , and a spectral characterization of growth [2]. The ProveIt report *Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations*, including its September 28 continuation, that is, Parts I and II above, addresses strict growth and increasingly precise fixed-bound growth

asymptotics. Its README explicitly excludes a joint limit in  $n$  and  $m$  from its claims (at the pin recorded in section 36).

Our connection to that report is the same counting class and the same maximum decomposition, not a dependence on its claimed growth-constant estimates. The proofs below derive their own finite combinatorics. The infinite-tree viewpoint is standard background in conditioned tree probability [3]; here it is made into an explicit finite counting argument. Other statistics on random 132-avoiders have been studied through tree and excursion methods [4]; those results are not substituted for the extreme-jump analysis.

## 22.2 The law and the rate

For real  $0 \leq z \leq 1$ , put

$$s = \sqrt{1-z}, \quad r = \sqrt{\frac{1+s}{2}}. \quad (83)$$

Both square roots are the nonnegative ones. Around  $z = 0$ , they denote the analytic branches with  $s(0) = r(0) = 1$ .

**Theorem 22.1** (Algebraic boundary law). *There is a positive integer-valued random variable  $D$  with probability generating function*

$$G(z) = \mathbb{E}z^D = \frac{(1-s)(4+8s+3s^2-s^2r)}{4(1+2s)(1+s)^2}. \quad (84)$$

Moreover,

$$d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) = \Theta(n^{-1/2}). \quad (85)$$

With  $h = \lfloor n/4 \rfloor$ , an explicit bound for  $n \geq 8$  is

$$d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq \min \left\{ 1, \frac{2(h+1)(2h+1)}{h+2} \frac{C_h}{4^h} + n(n+1)(2n+1)2^{-n/2} \right\}. \quad (86)$$

In particular,  $D_n = O_{\mathbb{P}}(1)$  and  $M_n/n \rightarrow 1$  in probability.

The convention for total variation is  $d_{\text{TV}}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$ , equivalently half the  $\ell^1$  distance on a countable space. The explicit bound is designed for a transparent proof rather than a small finite- $n$  constant. The matching lower bound follows from the limiting heavy tail and the finite support of  $D_n$ , as shown in corollary 28.2.

**Theorem 22.2** (Tail and joint enumeration). *As  $k \rightarrow \infty$ ,*

$$\mathbb{P}(D = k) = \frac{3}{2\sqrt{\pi}} k^{-3/2} + O(k^{-5/2}), \quad (87)$$

$$\mathbb{P}(D > k) = \frac{3}{\sqrt{\pi}} k^{-1/2} + O(k^{-3/2}). \quad (88)$$

Uniformly for integers  $n \geq 8$  and  $2 \leq d \leq n-1$ ,

$$\frac{a_n^{(n-d)}}{C_n} = \frac{3}{\sqrt{\pi d}} + O(d^{-3/2} + n^{-1/2}), \quad (89)$$

with an absolute implied constant. Consequently, if  $d \rightarrow \infty$  and  $d = o(n)$ , then

$$\frac{a_n^{(n-d)}}{C_n} \sim \frac{3}{\sqrt{\pi d}}, \quad a_n^{(n-d)} \sim \frac{3}{\pi} \frac{4^n}{n^{3/2} \sqrt{d}}. \quad (90)$$

**Theorem 22.3** (Moment transition). *For each fixed real  $p > 0$ ,*

$$0 < p < \frac{1}{2} : \mathbb{E}D_n^p = \mathbb{E}D^p + O(n^{p-1/2}), \quad (91)$$

$$p = \frac{1}{2} : \mathbb{E}\sqrt{D_n} = \frac{3}{2\sqrt{\pi}} \log n + O(1), \quad (92)$$

$$p > \frac{1}{2} : \mathbb{E}D_n^p = \Theta(n^{p-1/2}). \quad (93)$$

*The constants in the first and third lines may depend on  $p$ . In particular,*

$$\mathbb{E}M_n = n - \Theta(\sqrt{n}), \quad \text{Var}(D_n) = \Theta(n^{3/2}). \quad (94)$$

### 22.3 What is and is not resolved

The results determine the entire fixed-deficit boundary law, give a uniform quantitative approximation to it, and establish a growing-deficit regime. They do not determine the asymptotic equivalent of  $\mathbb{E}D_n$ , nor the fraction  $a_n^{(\lfloor \theta n \rfloor)} / C_n$  to relative precision for fixed  $0 < \theta < 1$ . The additive error in (89) has the same order as its displayed term when  $d$  is proportional to  $n$ ; it must not be advertised as an equivalent there. The component-ordering question in the fixed- $m$  finite-state system is also untouched.

## 23 Catalan decompositions and endpoint identities

### 23.1 Decomposition at the maximum

If  $\pi = L n R$  avoids 132, every value of  $L$  is greater than every value of  $R$ . Conversely, two avoiding standardized blocks with this separation give an avoiding permutation. If  $|R| = b$ , the left block is obtained by adding  $b$  to each entry of an avoider of size  $n - b - 1$ . This proves

$$C_n = \sum_{b=0}^{n-1} C_b C_{n-b-1}, \quad C(x) = 1 + xC(x)^2.$$

All the reconstruction operations used below are instances of this bijection; no approximate counting is involved. (The decomposition is the one of Part I, section 2, where the separation property is also proved; manuscript 02 re-derives it independently.)

### 23.2 First entries and first deficiencies

For a nonempty avoider  $\pi$ , let  $F(\pi) = \pi_1$ . Define

$$P(x, z) = \sum_{n \geq 1} \sum_{\pi \in \text{Av}_n(132)} x^n z^{F(\pi)}.$$

When the left block at the maximum is empty, the first value is the maximum. When it is nonempty, its first value is shifted by the size of the right block. Hence

$$P(x, z) = xzC(xz) + xC(xz)P(x, z),$$

and therefore

$$P(x, z) = \frac{xzC(xz)}{1 - xC(xz)}. \quad (95)$$

Replacing  $x$  by  $xu$  and  $z$  by  $u^{-1}$  gives the useful deficiency form

$$\sum_{n \geq 1} \sum_{\pi \in \text{Av}_n(132)} x^n u^{n-F(\pi)} = \frac{x C(x)}{1 - u x C(x)}. \quad (96)$$

In particular, for every  $j \geq 0$ ,

$$\sum_{n \geq j+1} \#\{\pi \in \text{Av}_n(132) : n - F(\pi) = j\} x^n = (xC(x))^{j+1}. \quad (97)$$

**Lemma 23.1** (First-entry count). *For  $1 \leq f \leq N$ ,*

$$N_{N,f} := \#\{\pi \in \text{Av}_N(132) : F(\pi) = f\} = \frac{N-f+1}{N} \binom{N+f-2}{N-1}. \quad (98)$$

Moreover, for  $1 \leq t \leq N$ ,

$$\frac{\#\{\pi \in \text{Av}_N(132) : F(\pi) \leq t\}}{C_N} \leq t(N+1)(2N+1)2^{t-N-2}. \quad (99)$$

*Proof.* The coefficient formula

$$[x^a]C(x)^b = \frac{b}{2a+b} \binom{2a+b}{a}$$

follows by applying Lagrange inversion to  $C = 1 + xC^2$ . Apply it to (97), with  $j = N - f$ , to obtain (98); the two binomial forms are equal by cancellation of factorials. For the bound, discard the prefactor, use  $\binom{u}{v} \leq 2^u$ , and sum over  $f \leq t$ :

$$\sum_{f \leq t} N_{N,f} \leq t2^{N+t-2}.$$

Since the largest binomial coefficient in the expansion of  $(1+1)^{2N}$  is central,  $\binom{2N}{N} \geq 4^N/(2N+1)$ . Dividing proves the claim.  $\square$

*Remark 23.2* (Merge note: the same count in Parts I and II). Deleting the final maximum of a block of length  $k$  that ends at its maximum and has first entry  $k - j$  leaves an avoider of size  $N = k - 1$  with first entry  $f = N + 1 - j$ . Hence  $N_{N,f} = h(N+1, N+1-f)$ , and (98) is Part I's formula (21) for  $h(k, j)$ , reindexed. Part I proves it by a slot decomposition at the first entry (lemma 5.3), manuscript 02 by the maximum decomposition and the bivariate series (95); both finish with Lagrange inversion. The manuscript's route is kept here, because (95) and (97) are used again in section 27.

The fraction bounded in (99) is also bounded by Part II's uniform tail estimate. With  $k = N + 1$  and  $d = N - t \geq 1$ , the difference  $C_N - c_{N+1, N-t}$  counts exactly the avoiders of size  $N$  with first entry at most  $t$ , so (54) gives

$$\frac{\#\{\pi \in \text{Av}_N(132) : F(\pi) \leq t\}}{C_N} \leq (N-t+2)2^{-(N-t)} \quad (1 \leq t \leq N-1).$$

This is sharper than (99) by a factor of order  $N^3$ . In the setting of lemma 26.1,  $N-t = n-2t \geq n/2$ , and  $x \mapsto (x+2)2^{-x}$  decreases for  $x \geq 0$ , so the exceptional fraction is at most  $(n/2+2)2^{-n/2}$ , which is below 1 for every  $n \geq 8$ ; the term  $n(n+1)(2n+1)2^{2h-n}$  in the proof of corollary 28.2 can likewise be replaced by  $(n-2h+2)2^{2h-n}$ . Neither replacement changes an order of magnitude or any statement of Part III, which prints the manuscript's bounds. This remark is the merge's, not the manuscript's; the bound was checked by exhaustive enumeration for  $N \leq 8$ .

### 23.3 Last entries

A particularly simple decomposition occurs at the last value  $\ell$ . Every entry greater than  $\ell$  must precede every entry smaller than  $\ell$ : otherwise the smaller entry, the greater entry, and the final  $\ell$  form 132. Thus, uniquely,

$$\pi = (\tau + \ell) \sigma \ell, \quad |\tau| = n - \ell, \quad |\sigma| = \ell - 1, \quad (100)$$

where  $\tau$  and  $\sigma$  are arbitrary avoiders of those sizes. The converse is verified directly, or by the maximum decomposition. Consequently,

$$\#\{\pi \in \text{Av}_n(132) : \pi_n = \ell\} = C_{n-\ell} C_{\ell-1}. \quad (101)$$

An additional consequence will control the clipping operation: if the last value is not the maximum, then the first value is greater than the last value. Indeed,  $\tau$  in (100) is then nonempty.

## 24 A finite absorbing decomposition

### 24.1 Three operations with one unexpanded block

We use a finite decorated word, called a *skeleton*, whose outer operations end at one terminal operation. The unexpanded block will later be filled with a large avoiding permutation. There are three operation types:

**$R_\beta$ : a large right block.**

Here  $\beta$  is any avoider of size  $b \geq 0$ . Applied to a core  $\sigma$  of size  $N$ , it returns  $(\beta + N)(N + b + 1)\sigma$ . Its cost is  $b + 1$ .

**$A$ : append a maximum.**

Applied to a core of size  $N$ , it returns  $\sigma(N + 1)$ . Its cost is 1.

**$T_\tau$ : a terminal finite right block.**

Here  $\tau$  is any nonempty avoider of size  $b \geq 1$ . Applied to the hole  $\sigma$  of size  $N$ , it returns  $(\sigma + b)(N + b + 1)\tau$ . Its cost is  $b + 1$ .

A skeleton is a word in  $R_\beta$  and  $A$ , followed by exactly one  $T_\tau$ . Operations act from the terminal outward. Its cost  $t$  is the sum of their costs, so inserting a hole of size  $N$  gives a permutation of size  $N + t$ . All resulting permutations avoid 132.

For example, the word  $AR_{(1)}T_{(2,1)}$  has cost  $1 + 2 + 3 = 6$ . The terminal will initialize the pair  $(2, 1)$ ,  $R_{(1)}$  changes it to  $(4, 1)$ , and  $A$  changes it to  $(1, \infty)$ . Thus this skeleton produces a maximal possible jump, once a sufficiently large hole with a sufficiently large first value is inserted.

### 24.2 A probability measure on skeletons

Give each individually decorated operation of cost  $c$  weight  $4^{-c}$ . The total weights are

$$\sum_{\beta} 4^{-(|\beta|+1)} = \frac{1}{2}, \quad w(A) = \frac{1}{4}, \quad \sum_{\tau \neq \emptyset} 4^{-(|\tau|+1)} = \frac{1}{4}. \quad (102)$$

Thus one can repeatedly choose an operation using these weights, stopping when a terminal is chosen. The probability of not stopping after  $j$  choices is  $(3/4)^j$ . Every skeleton is finite almost surely, and a specified decorated skeleton of cost  $t$  has probability  $4^{-t}$ .

This is a probability measure on a countable set, not a heuristic “typical giant block” assertion. The finite comparison in the next two sections will connect it to the uniform permutation measure.

### 24.3 Exact cost distribution

Let  $b_t$  be the number of skeletons of cost  $t$ . Their ordinary generating function is

$$B(x) = \sum_{t \geq 0} b_t x^t = \frac{x(C(x) - 1)}{1 - xC(x) - x} = (C(x) - 1)^2. \quad (103)$$

For the last equality, note that  $1 - xC(x) - x = C(x)^{-2}$  and  $xC(x)^2 = C(x) - 1$ . For  $t \geq 1$ ,

$$b_t = C_{t+1} - 2C_t = \frac{2(t-1)}{t+2}C_t, \quad b_0 = b_1 = 0. \quad (104)$$

If  $T$  denotes the random cost, then

$$\mathbb{E}z^T = B(z/4) = \left( \frac{1 - \sqrt{1-z}}{1 + \sqrt{1-z}} \right)^2. \quad (105)$$

**Lemma 24.1** (Exact cost tail). *For every integer  $h \geq 0$ ,*

$$\mathbb{P}(T > h) = \frac{2(h+1)(2h+1)}{h+2} \frac{C_h}{4^h} \sim \frac{4}{\sqrt{\pi h}}. \quad (106)$$

*Proof.* Put  $c_j = C_j/4^j$ . The Catalan ratio is  $c_{j+1}/c_j = (2j+1)/(2j+4)$ , and induction gives  $\sum_{j=0}^h c_j = 2 - (2h+1)c_h$ . Using  $b_t 4^{-t} = 4c_{t+1} - 2c_t$  for  $t \geq 1$ , with the constant-term adjustment at  $t = 0$ , yields

$$\sum_{t=0}^h b_t 4^{-t} = 1 - \frac{2(h+1)(2h+1)}{h+2} c_h.$$

Stirling's formula gives the equivalent. □

## 24.4 The endpoint state and its transformations

The state is a pair  $(d, \ell)$ , where  $d$  is a positive integer and  $\ell \in \mathbb{N}_{>0} \cup \{\infty\}$ . The symbol  $\infty$  records a last entry close to the high end of a large block; it is not an actual entry of a finite permutation. Starting at the terminal, define

$$T_\tau : (d, \ell) = (F(\tau), \tau_{|\tau|}), \quad (107)$$

$$R_\beta : (d, \ell) \mapsto (d + |\beta| + 1, \ell), \quad (108)$$

$$A : (d, \ell) \mapsto (\min\{d+1, \ell\}, \infty), \quad (109)$$

where  $\min(d+1, \infty) = d+1$ . Let  $(D, L)$  be the resulting pair on the probability space of skeletons. It is finite in its first coordinate almost surely and obeys

$$1 \leq D \leq T - 1. \quad (110)$$

This defines the candidate law probabilistically before any generating function is simplified; coefficient nonnegativity will therefore not need to be guessed from the radical expression.

## 25 The deterministic reconstruction lemma

The state rules need proof: an endpoint heuristic would not justify a law for the largest jump. The following statement is the central finite lemma.

**Lemma 25.1** (A large hole determines the extreme jump). *Fix a skeleton  $\mathfrak{s}$  of cost  $t$  and state  $(d, \ell)$ . Insert an avoider  $\sigma$  of size  $N > t$  whose first entry satisfies  $F(\sigma) > t$ . Let  $\pi$  be the resulting permutation of size  $n = N + t$ . Then*

$$D(\pi) = d. \quad (111)$$

*If  $\ell < \infty$ , then  $\pi_n = \ell$ . If  $\ell = \infty$ , then  $\pi_n \geq N + 1$ . At every intermediate reconstruction stage, the first entry is greater than  $t$  and the largest jump is at least  $N + 1$ .*

*Proof.* Proceed from the terminal outward. Suppose the terminal decoration has size  $b \geq 1$ , first entry  $f$ , and last entry  $\ell$ . The reconstructed terminal block is

$$(\sigma + b)(N + b + 1)\tau.$$

The jump from its maximum to  $f$  has size  $N + b + 1 - f \geq N + 1$ . The jumps internal to  $\sigma$  are at most  $N - 1$ ; the jump into the maximum from the left is at most  $N$ ; and the jumps internal to  $\tau$  are at most  $b - 1 < N + 1$ . Thus that downward jump is largest, its deficit is exactly  $f$ , and the last value is  $\ell$ . The first value is  $F(\sigma) + b > t$ .

Maintain the following stronger invariants. If  $c$  is the cost used so far, the current length is  $N + c$ , its deficit  $d_c$  satisfies  $d_c \leq c - 1$ , its first entry exceeds  $t$ , and its last entry is either the recorded finite value, at most  $t$ , or at least  $N + 1$  when the state records  $\infty$ . The largest jump is then at least  $N + 1$ .

For  $R_\beta$ , let  $b = |\beta|$  and let the old length be  $a$ . The old internal maximum jump has new deficit  $d_c + b + 1$ . The finite left block contributes only jumps of size at most  $b$ , including its boundary with the new maximum. These cannot exceed the old maximum jump. The jump from the new maximum to the old first value is strictly less than  $a + b + 1 - t$ , whereas the old maximum is

$$a - d_c = (a + b + 1) - (d_c + b + 1) \geq (a + b + 1) - t + 1,$$

because the new accumulated cost is at most  $t$ . So the old internal maximum remains largest, giving (108). The last entry is unchanged. The new first entry is either the new maximum or a value exceeding  $a$ , hence exceeds  $t$  since  $a \geq N > t$ .

For  $A$ , appending the new maximum changes the deficit of the old largest jump from  $d_c$  to  $d_c + 1$ . The new boundary jump has deficit equal to the old last value. When that value is finite, the new deficit is precisely  $\min(d_c + 1, \ell)$ . When it is recorded as  $\infty$ , the actual value is at least  $N + 1 > t$ , while  $d_c + 1 \leq t - 1$ ; the new jump cannot win. The last value becomes the new maximum, at least  $N + 1$ , and the first value is unchanged. These are (109) and all the invariants.  $\square$

**Lemma 25.2** (Disjoint reconstruction families). *Fix a total size  $n$  and a cutoff  $h < n/2$ . Reconstructions from distinct skeletons of cost  $t \leq h$ , with holes of size  $n - t$ , are disjoint. For each fixed skeleton, reconstruction is injective in the hole.*

*Proof.* At every maximum decomposition, the block containing the hole is larger than its finite sibling: its size is at least  $n - t > t$ , while the sibling has size at most  $t$ . Hence it is identifiable from the final permutation. A large right block specifies an  $R$  operation. A large left block with empty right block specifies  $A$ . A large left block with nonempty finite right block specifies the terminal  $T$ , where parsing stops. Each finite decoration is recovered from the actual sibling values by standardization. The remaining large left block at that terminal recovers the hole. Thus the whole decorated word and the hole are uniquely determined.  $\square$

*Remark 25.3.* The first-entry hypothesis cannot simply be deleted. A very small first entry in the hole can create an additional large downward jump at an outer  $R$  operation. The next section bounds precisely this exceptional set. Likewise, using only the deficit and omitting the last-entry state would lose the clipping at an append operation.

## 26 Counting domination and total variation

### 26.1 Why each short skeleton has enough finite mass

The normalized Catalan numbers  $c_j = C_j/4^j$  decrease strictly, because  $c_{j+1}/c_j = (2j+1)/(2j+4) < 1$ . Consequently, whenever  $0 \leq t < n$ ,

$$\frac{C_{n-t}}{C_n} = 4^{-t} \frac{c_{n-t}}{c_n} \geq 4^{-t}. \quad (112)$$

The left side is the probability mass of the reconstruction family of one fixed skeleton of cost  $t$  under the uniform size- $n$  measure, before the first-entry restriction is imposed. The right side is exactly that skeleton's probability under the limiting construction. This one-sided comparison is stronger for our purpose than an unquantified local convergence statement.

**Lemma 26.1** (Exceptional first entries). *For  $n \geq 8$ ,  $t \leq \lfloor n/4 \rfloor$ , and  $N = n - t$ , the fraction of holes in  $\text{Av}_N(132)$  with first entry at most  $t$  is at most*

$$\omega_n = n(n+1)(2n+1)2^{-n/2}. \quad (113)$$

*Proof.* Apply (99). Since  $t - N = 2t - n \leq -n/2$  and  $t \leq n$ ,  $N \leq n$ , its right side is at most the displayed bound. The deliberately enlarged prefactor avoids unnecessary finite-size casework. If it exceeds 1, the assertion is still valid.  $\square$

## 26.2 A common subprobability measure

For probability measures on a countable space, a common subprobability measure of mass  $1 - \delta$  implies total-variation distance at most  $\delta$ . This follows immediately from  $d_{\text{TV}}(\mu, \nu) = 1 - \sum_x \min(\mu\{x\}, \nu\{x\})$ .

*Proof of the rate in theorem 22.1.* Let  $h = \lfloor n/4 \rfloor$  and suppose first that  $\omega_n \leq 1$ . For each skeleton  $s$  of cost  $t \leq h$ , the good reconstruction family has probability at least

$$\frac{C_{n-t}}{C_n}(1 - \omega_n) \geq 4^{-t}(1 - \omega_n).$$

The families are disjoint by lemma 25.2. Every good member has deficit  $d(s)$  by lemma 25.1. The limiting deficit law also puts the skeleton mass  $4^{-t}$  at that same value. After summing over skeletons, even when multiple skeletons yield the same deficit, the two deficit laws have a common subprobability measure of mass at least

$$(1 - \omega_n) \sum_{\text{cost}(s) \leq h} 4^{-\text{cost}(s)} = (1 - \omega_n) \mathbb{P}(T \leq h).$$

It follows that

$$d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq \mathbb{P}(T > h) + \omega_n. \quad (114)$$

If  $\omega_n > 1$ , the same displayed upper bound is trivial. Insert the exact tail from lemma 24.1 and combine with the universal bound  $d_{\text{TV}} \leq 1$ . Stirling's formula gives

$$\mathbb{P}(T > \lfloor n/4 \rfloor) = \frac{8}{\sqrt{\pi n}} + O(n^{-3/2}),$$

while  $\omega_n$  is exponentially small up to its polynomial factor. This proves the upper bound in (85) and proves (86); the lower bound is given in corollary 28.2. Since  $D$  is a proper finite-valued random variable, the laws of  $D_n$  are tight, and  $M_n/n = 1 - D_n/n \rightarrow 1$  in probability.  $\square$

*Remark 26.2* (The order is sharp; the leading constant remains open). The cutoff  $n/4$  makes every inequality transparent. After deriving the limiting tail, corollary 28.2 will prove the matching order and improve the upper constant by taking a cutoff approaching  $n/2$ . These arguments do not identify an exact asymptotic constant for the total-variation error.

## 27 Evaluation of the algebraic probability generating function

### 27.1 Renewal notation

For  $0 \leq z \leq 1$ , write

$$q = Q(z) = \frac{z}{4}C(z/4) = \frac{1 - \sqrt{1-z}}{2}. \quad (115)$$

Then

$$z = 4q(1-q), \quad \frac{1}{1-q} = C(z/4), \quad s = 1 - 2q. \quad (116)$$

An  $R_\beta$  operation adds  $|\beta| + 1$  to the deficit and has weight  $4^{-(|\beta|+1)}$ . Its deficit-increment generating function is therefore  $q$ . A sequence of such operations has generating function

$$\frac{1}{1-q} = \sum_{b \geq 0} \frac{C_b}{4^b} z^b. \quad (117)$$

Here  $b$  records the *sum of increments*, not the number of operations and not the size of a single decoration.

Define two subprobability generating functions

$$H(z) = \mathbb{E}[z^D; L < \infty], \quad J(z) = \mathbb{E}[z^D; L = \infty], \quad G = H + J. \quad (118)$$

The last-entry state is finite exactly when there is no  $A$  operation: it starts finite at the terminal,  $R$  preserves it, and  $A$  changes it permanently to  $\infty$ .

### 27.2 The finite-last part

The terminal first-entry series is  $\frac{1}{4}P(\frac{1}{4}, z)$ . By (95),

$$P(\frac{1}{4}, z) = \frac{z}{1 + 2\sqrt{1-z}} = \frac{z}{3 - 4q}.$$

Adding an arbitrary outer sequence of  $R$  operations gives

$$H(z) = \frac{P(\frac{1}{4}, z)}{4(1-q)} = \frac{q}{3 - 4q}. \quad (119)$$

We also need the last-entry marginal, rather than its joint relation with  $D$ . From (101),

$$\sum_{n \geq 1} \sum_{\pi \in \text{Av}_n(132)} x^n z^{\pi_n} = C(x) x z C(xz).$$

At  $x = 1/4$  the terminal last-entry series is  $q/2$ . An arbitrary sequence of  $R$  operations has total weight  $1/(1 - Q(1)) = 2$ . Therefore

$$\mathbb{E}[z^L; L < \infty] = q. \quad (120)$$

In particular,  $H(1) = J(1) = 1/2$  and  $\mathbb{P}(L = \ell) = C_{\ell-1}/4^\ell$  for finite  $\ell \geq 1$ .

### 27.3 Why the append operation is not independent

Let

$$\mathcal{K}(z) = \mathbb{E}[z^{\min(D+1, L)}; L < \infty]. \quad (121)$$

The outermost-operation decomposition gives

$$J(z) = qJ(z) + \frac{z}{4}J(z) + \frac{1}{4}\mathcal{K}(z).$$

The first term is an outer  $R$ , the second an outer  $A$  applied to an infinite-last state, and the third an outer  $A$  applied to a finite-last state. Hence

$$J(z) = \frac{\mathcal{K}(z)}{4(1-q-z/4)} = \frac{\mathcal{K}(z)}{4(1-q)^2}. \quad (122)$$

The dependence between  $D$  and  $L$  is concentrated in  $\mathcal{K}$ . Replacing  $L$  by an independent variable with its marginal law would be incorrect. The next calculation retains the full dependence.

## 27.4 An exact clipping calculation

If a terminal permutation  $\tau$  does not end in its maximum, then  $F(\tau) > \tau_{|\tau|}$  by (100). Every outer  $R$  only increases the first state coordinate, so clipping returns the terminal last value. The only correction to the marginal (120) arises from terminal permutations ending in their maximum.

Write such a non-singleton terminal as  $\sigma(k+1)$ , where  $|\sigma| = k \geq 1$ ,  $F(\sigma) = f$ , and  $j = k - f$ . If the outer  $R$  sequence has total increment  $b$ , the state before clipping is  $(b+f, k+1)$ . Clipping is strict exactly when  $b < j$ . The singleton terminal produces no correction. Its non-singleton correction therefore has generating function

$$\mathcal{K}(z) - q = \frac{z}{16} \sum_{j \geq 1} q^{j+1} \sum_{0 \leq b < j} \frac{C_b}{4^b} (z^{b-j} - 1). \quad (123)$$

To see the factor  $1/16$ , the terminal size is  $k+1$ , hence its weight is  $4^{-(k+2)}$ . Formula (97), evaluated at  $x = z/4$ , sums its possible  $\sigma$  to  $q^{j+1}$ . The factor  $z$  records the final maximum; (117) supplies  $C_b/4^b$ .

For  $0 < z < 1$ , the geometric sums converge absolutely; in particular  $q/z = 1/(4(1-q)) < 1$ . Interchanging the sums in (123) gives

$$\begin{aligned} \mathcal{K}(z) - q &= \frac{z}{16} \sum_{b \geq 0} \frac{C_b}{4^b} q^{b+2} \left( \frac{1}{z-q} - \frac{1}{1-q} \right) \\ &= \frac{z(1-z)q^2 C(q/4)}{16(z-q)(1-q)} \\ &= (1-z) \frac{q^2 C(q/4)}{4(3-4q)}. \end{aligned} \quad (124)$$

Both sides extend analytically to  $z = 0$ . As  $qC(q/4) = 2(1 - \sqrt{1-q})$ , define

$$\Upsilon(z) = \frac{q(1 - \sqrt{1-q})}{2(3-4q)}. \quad (125)$$

We have proved the particularly useful representation

$$\boxed{\mathcal{K}(z) = q + (1-z)\Upsilon(z)}. \quad (126)$$

The series  $\Upsilon$  also has the direct tail interpretation

$$[z^k]\Upsilon(z) = \mathbb{P}(L > k, D \leq k-1, L < \infty). \quad (127)$$

Indeed, for finite  $L$ , the tail of  $\min(D+1, L)$  at  $k$  differs from the tail of  $L$  precisely on this event; use the standard probability-generating-function tail identity.

## 27.5 Completion of the formula

Combining (119), (122), and (126),

$$G(z) = \frac{q}{3-4q} + \frac{q + (1-z)q(1-\sqrt{1-q})/(2(3-4q))}{4(1-q)^2}. \quad (128)$$

Substitute  $q = (1-s)/2$  and  $\sqrt{1-q} = r$ . The result is (84), proving the generating-function assertion of theorem 22.1. This derivation also verifies  $G(0) = 0$  and  $G(1) = 1$ . The probabilistic construction has already proved that its coefficients are nonnegative and sum to one.

*Remark 27.1* (Algebraicity, but not a finite constant-coefficient recurrence). The expression lies in the field obtained from  $\mathbb{Q}(z)$  by adjoining  $s$  and then  $r$ , so its algebraic degree is at most four. It is not rational: the nonzero square-root term at  $z = 1$ , computed next, excludes rationality. Thus the limiting probability sequence does not satisfy an eventual constant-coefficient linear recurrence. This is compatible with rationality of each fixed- $m$  generating function  $A_m(x)$ : it concerns a different sequence and a different limit.

## 28 Singularity analysis and the heavy tail

### 28.1 The dominant singularity is genuinely unique

The square-root branches in (84) extend to  $\mathbb{C} \setminus [1, \infty)$ . On this domain the principal value of  $s = \sqrt{1-z}$  has positive real part. Consequently  $1+s$  and  $1+2s$  do not vanish, and  $(1+s)/2$  never meets the negative real axis or zero. Thus  $r$  is analytic there as well. In particular,  $G$  is analytic in a slit neighborhood of the closed unit disk except at  $z = 1$ . There are no competing unit-circle singularities or periodicity factors hidden in the coefficient asymptotics.

Expansion of the explicit formula at  $s = 0$  gives

$$\begin{aligned} G(z) = 1 - 3s + \left(\frac{23}{4} - \frac{\sqrt{2}}{8}\right)s^2 + \left(-\frac{43}{4} + \frac{9\sqrt{2}}{16}\right)s^3 \\ + \left(\frac{81}{4} - \frac{99\sqrt{2}}{64}\right)s^4 + \left(-\frac{155}{4} + \frac{461\sqrt{2}}{128}\right)s^5 + O(s^6). \end{aligned} \quad (129)$$

This calculation uses only  $r = 2^{-1/2}(1+s)^{1/2}$  and ordinary Taylor expansion. An optional symbolic checker in the package (code/03-largest-jump-verify\_symbolic.py) verifies the displayed coefficients. The term  $-3\sqrt{1-z}$  is the source of both the infinite mean of  $D$  and the critical exponent  $1/2$ .

### 28.2 Coefficient extraction, including the second term

For a fixed nonintegral  $\gamma$ , the exact generalized binomial formula and the asymptotic expansion of the gamma quotient give

$$[z^k](1-z)^\gamma = \frac{k^{-\gamma-1}}{\Gamma(-\gamma)} \left(1 + \frac{\gamma(\gamma+1)}{2k} + O(k^{-2})\right). \quad (130)$$

Analytic continuation in the slit neighborhood just established permits termwise singularity transfer; this is the standard univariate transfer principle of [5]. Integer powers of  $1-z$  are locally analytic terms and do not contribute the displayed algebraic orders. For completeness, the nonanalytic terms needed for the first two orders are  $(1-z)^{1/2}$  and  $(1-z)^{3/2}$ , whose leading coefficient constants are respectively  $-1/(2\sqrt{\pi})$  and  $3/(4\sqrt{\pi})$ .

**Proposition 28.1** (Two-term probability and survival expansions). *As  $k \rightarrow \infty$ ,*

$$\mathbb{P}(D = k) = \frac{3}{2\sqrt{\pi}} k^{-3/2} + \frac{-480 + 27\sqrt{2}}{64\sqrt{\pi}} k^{-5/2} + O(k^{-7/2}), \quad (131)$$

$$\mathbb{P}(D > k) = \frac{3}{\sqrt{\pi}} k^{-1/2} + \frac{-184 + 9\sqrt{2}}{32\sqrt{\pi}} k^{-3/2} + O(k^{-5/2}). \quad (132)$$

*Proof.* For the point probabilities, apply (130) to the  $s$  and  $s^3$  terms of (129). The coefficient of  $k^{-5/2}$  is

$$\frac{9}{16\sqrt{\pi}} + \frac{3}{4\sqrt{\pi}} \left( -\frac{43}{4} + \frac{9\sqrt{2}}{16} \right) = \frac{-480 + 27\sqrt{2}}{64\sqrt{\pi}}.$$

For the survival probabilities use the exact identity

$$\sum_{k \geq 0} \mathbb{P}(D > k) z^k = \frac{1 - G(z)}{1 - z}. \quad (133)$$

Writing  $a_2 = 23/4 - \sqrt{2}/8$  and  $a_3 = -43/4 + 9\sqrt{2}/16$ , its expansion begins  $3(1 - z)^{-1/2} - a_2 - a_3(1 - z)^{1/2} + O(1 - z)$ . The second coefficient is

$$-\frac{3}{8\sqrt{\pi}} + \frac{a_3}{2\sqrt{\pi}} = \frac{-184 + 9\sqrt{2}}{32\sqrt{\pi}}.$$

Continuing the expansion two half-orders further gives the stated remainders. The analyticity argument above supplies the hypotheses of singularity transfer throughout.  $\square$

**Corollary 28.2** (Sharp total-variation order). *The error in the boundary law has order exactly  $n^{-1/2}$ . More precisely,*

$$\frac{3}{\sqrt{\pi}} \leq \liminf_{n \rightarrow \infty} \sqrt{n} d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq \limsup_{n \rightarrow \infty} \sqrt{n} d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq \frac{4\sqrt{2}}{\sqrt{\pi}}. \quad (134)$$

*Proof.* Because  $D_n \leq n - 1$ , the event  $\{n, n + 1, \dots\}$  gives

$$d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \geq \mathbb{P}(D \geq n) = \frac{3}{\sqrt{\pi n}} + O(n^{-3/2}).$$

For the improved upper bound, repeat the common-submeasure proof with any integer  $2 \leq h < n/2$ . The first-entry estimate gives

$$d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq \mathbb{P}(T > h) + n(n + 1)(2n + 1)2^{2h-n}.$$

For all sufficiently large  $n$ , choose  $h = \lfloor n/2 - 2 \log_2 n \rfloor$ . The second term is  $O(n^{-1})$ , while the exact cost tail is  $(4\sqrt{2}/\sqrt{\pi})n^{-1/2}(1 + o(1))$ . This proves the three inequalities and completes the sharp-rate claim in theorem 22.1.  $\square$

**Corollary 28.3** (Quantiles and limiting moments). *The variable  $D$  has finite moment of order  $p > 0$  exactly when  $p < 1/2$ . If  $\kappa_{1-\epsilon} = \min\{k : \mathbb{P}(D \leq k) \geq 1 - \epsilon\}$ , then*

$$\kappa_{1-\epsilon} \sim \frac{9}{\pi\epsilon^2} \quad (\epsilon \downarrow 0). \quad (135)$$

*Proof.* The moment criterion follows by summing  $k^p \mathbb{P}(D = k) \sim (3/(2\sqrt{\pi}))k^{p-3/2}$ . For the quantile statement, bracket the survival function between  $(1 \pm \eta)3/\sqrt{\pi k}$  for all sufficiently large  $k$ , invert these inequalities, and then let  $\eta \downarrow 0$ . Rounding the resulting bounds to integers has no effect on the equivalent.  $\square$

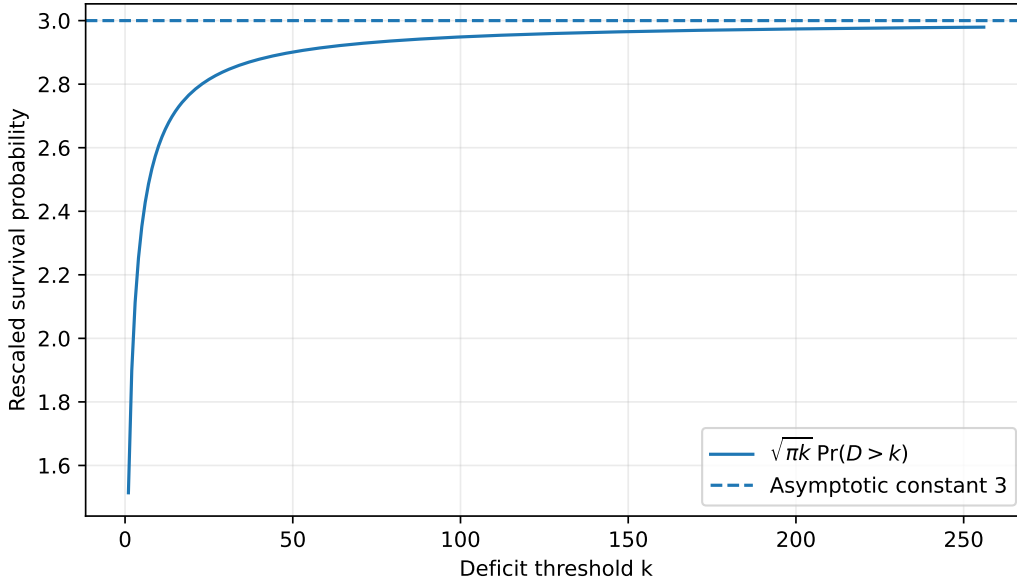


Figure 5: The scaled survival function  $\sqrt{\pi k} \mathbb{P}(D > k)$ , computed from exact rational coefficients of  $G$ . The horizontal reference is 3. The graph illustrates (132); it is not used as evidence in its proof.

## 29 Joint limits for the adjacency bound

### 29.1 A fixed distance from the maximal possible jump

For every fixed integer  $d \geq 1$ , (82) and total-variation convergence give

$$\lim_{n \rightarrow \infty} \frac{a_n^{(n-d)}}{C_n} = \mathbb{P}(D \geq d) = 1 - \sum_{k=1}^{d-1} [z^k]G(z). \quad (136)$$

Thus the limiting profile across the boundary consists of explicit rational numbers. At  $d = 1$  the fraction is identically one. At  $d = 2$ , its limit is  $41/48$ : asymptotically  $7/48$  of all avoiders contain a jump of the largest possible size, namely  $n - 1$ .

This is not a smooth limiting distribution for  $M_n/n$ . That normalized statistic tends to the constant 1. The nontrivial distribution is visible only after retaining the integer distance  $n - M_n$ .

### 29.2 A uniform estimate and its domain of relative validity

*Proof of theorem 22.2.* The point and survival estimates follow from proposition 28.1. For  $2 \leq d \leq n - 1$ , total variation gives

$$\left| \frac{a_n^{(n-d)}}{C_n} - \mathbb{P}(D \geq d) \right| \leq C_{TV} n^{-1/2}$$

for an absolute constant  $C_{TV}$ . Also

$$\mathbb{P}(D \geq d) = \mathbb{P}(D > d - 1) = \frac{3}{\sqrt{\pi d}} + O(d^{-3/2}).$$

Combining these inequalities proves (89). Dividing the error by the main term gives

$$\frac{a_n^{(n-d)}}{C_n} = \frac{3}{\sqrt{\pi d}} \left( 1 + O\left(d^{-1} + \sqrt{d/n}\right) \right). \quad (137)$$

When  $d \rightarrow \infty$  and  $d = o(n)$  the relative error tends to zero. Finally  $C_n \sim 4^n / (\sqrt{\pi} n^{3/2})$  gives the second equivalent in (90).  $\square$

*Example 29.1* (Polynomial and logarithmic boundary windows). For any fixed  $0 < \alpha < 1$  and  $d_n = \lfloor n^\alpha \rfloor$ ,

$$\frac{a_n^{(n-d_n)}}{C_n} \sim \frac{3}{\sqrt{\pi}} n^{-\alpha/2}.$$

Taking  $d_n = \lfloor \log n \rfloor$  gives the equivalent  $3/\sqrt{\pi \log n}$ . More generally, the same formula with  $d_n$  in place of these choices holds along every unbounded sublinear integer sequence. Neither monotonicity nor regular variation of  $d_n$  is required.

**Corollary 29.2** (All diverging boundary deficits are negligible). *If  $d_n \rightarrow \infty$  and  $1 \leq d_n \leq n-1$ , then*

$$a_n^{(n-d_n)} / C_n \rightarrow 0.$$

For every fixed  $0 < \theta < 1$ ,

$$a_n^{(\lfloor \theta n \rfloor)} / C_n = O(n^{-1/2}).$$

*Proof.* The first assertion follows either from tightness of  $D_n$  or directly from the absolute estimate. In the second,  $d_n = n - \lfloor \theta n \rfloor$  is of order  $n$ , and both terms of (89) are  $O(n^{-1/2})$ .  $\square$

*Remark 29.3* (The unresolved bulk regime). When  $d$  is a fixed positive fraction of  $n$ , the total-variation error is of the same order as the proposed leading term in (89). The equation is still a valid absolute estimate, but it does *not* prove a relative equivalent there. In particular, Part III does not assert  $a_n^{(\lfloor \theta n \rfloor)} / C_n \sim 3/\sqrt{\pi(1-\theta)n}$ . An additional finite-size scaling analysis would be needed, and it could yield a different function of  $\theta$ . [Added 29 September 2026, batch 52: it does. Part IV proves  $a_n^{(\lfloor \theta n \rfloor)} / C_n \sim \frac{3(2\theta-1)}{\sqrt{\pi\theta(1-\theta)}} n^{-1/2}$  for  $1/2 < \theta < 1$  and  $o(n^{-1/2})$  for  $\theta \leq 1/2$  (theorem 38.1); the formula above is therefore false for every  $0 < \theta < 1$ .]

## 30 The moment transition and its consequences

### 30.1 Truncation converts total variation into moment bounds

For two probability measures  $\mu, \nu$  and a measurable function with  $0 \leq f \leq B$ , the layer-cake representation gives

$$\left| \int f \, d\mu - \int f \, d\nu \right| \leq B \, d_{\text{TV}}(\mu, \nu). \quad (138)$$

Apply this to  $f(x) = \min(x, n)^p$ . Since  $D_n < n$ ,

$$|\mathbb{E} D_n^p - \mathbb{E} \min(D, n)^p| = O(n^{p-1/2}). \quad (139)$$

The finite support of  $D_n$  matters: total-variation convergence without a rate or a truncation bound would not control these unbounded moments.

For any positive integer-valued  $X$ , summation by parts gives

$$\mathbb{E} \min(X, n)^p = \sum_{k=0}^{n-1} ((k+1)^p - k^p) \mathbb{P}(X > k). \quad (140)$$

Using (132) and  $(k+1)^p - k^p = pk^{p-1} + O(k^{p-2})$  for  $k \geq 1$ , we obtain all three moment regimes.

### 30.2 Subcritical and critical moments

*Proof of theorem 22.3, for  $p \leq 1/2$ .* If  $0 < p < 1/2$ , the tail of the sum (140) is  $O(n^{p-1/2})$ , and therefore

$$\mathbb{E} \min(D, n)^p = \mathbb{E} D^p + O(n^{p-1/2}).$$

Equation (139) proves (91).

If  $p = 1/2$ , the summand in (140) is

$$(\sqrt{k+1} - \sqrt{k})\mathbb{P}(D > k) = \frac{3}{2\sqrt{\pi}} \frac{1}{k} + O(k^{-2}).$$

Consequently

$$\mathbb{E} \min(D, n)^{1/2} = \frac{3}{2\sqrt{\pi}} \log n + O(1).$$

The error in (139) is now  $O(1)$ , proving the critical statement.  $\square$

### 30.3 Supercritical moments: the order, not the constant

*Completion of the proof of theorem 22.3.* Let  $p > 1/2$ . The sum in (140) is  $O(n^{p-1/2})$ , so (139) gives the upper bound. For the lower bound choose a constant  $C_{TV}$  such that  $d_{TV}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq C_{TV} n^{-1/2}$  for all sufficiently large  $n$ . Fix a sufficiently small  $\epsilon > 0$  such that

$$\frac{3}{2\sqrt{\pi\epsilon}} > C_{TV} + 1.$$

Set  $d = \lfloor \epsilon n \rfloor$ . The limiting tail estimate implies, for all sufficiently large  $n$ ,

$$\mathbb{P}(D \geq d) \geq \frac{3}{2\sqrt{\pi\epsilon}} n^{-1/2}.$$

Hence  $\mathbb{P}(D_n \geq d) \geq n^{-1/2}$ , and

$$\mathbb{E} D_n^p \geq d^p \mathbb{P}(D_n \geq d) \geq \underline{c}_p n^{p-1/2}$$

for some  $\underline{c}_p > 0$ . This proves the claimed two-sided order.  $\square$

**Corollary 30.1** (Mean and variance of the largest-jump deficit). *As  $n \rightarrow \infty$ ,*

$$\mathbb{E} D_n = \Theta(\sqrt{n}), \quad \mathbb{E} M_n = n - \Theta(\sqrt{n}), \quad \text{Var}(D_n) = \Theta(n^{3/2}). \quad (141)$$

*The same variance identity holds for  $M_n$ .*

*Proof.* Use theorem 22.3 with  $p = 1, 2$ . The squared mean is  $O(n)$ , whereas the second moment is bounded above and below by positive constants times  $n^{3/2}$ . Subtracting the squared mean preserves this order. Finally  $M_n = n - D_n$ .  $\square$

*Remark 30.2* (Why a bounded typical deficit and an unbounded mean coexist). The limit has median 9, but an infinite mean. Rare permutations with much larger deficits carry an increasing fraction of the finite-size mean. There is therefore no contradiction between  $D_n = O_{\mathbb{P}}(1)$  and  $\mathbb{E} D_n = \Theta(\sqrt{n})$ . Nor can the constant in the mean be read off from the tail constant  $3/\sqrt{\pi}$ : the range  $D_n$  of order  $n$  can contribute to the mean at the same order, and that bulk range has not been identified asymptotically in Part III. [Added 29 September 2026, batch 52: Part IV identifies it (theorem 38.1); the mean constant is  $3/\sqrt{\pi}$ , and half of it comes from deficits above  $n/5$  (theorems 38.3 and 38.4). The value coincides with the tail constant, but it is not read off from it: extrapolating the boundary tail  $3/\sqrt{\pi x}$  over  $0 < x < 1$  would give  $6/\sqrt{\pi}$ . This comparison is the merge's.]

Table 12: Initial point probabilities and cumulative probabilities of the limiting deficit. The full first 256 point probabilities are included as exact numerator–denominator pairs in the package (data/03-largest-jump-limit\_probabilities.csv).

| $k$ | $\mathbb{P}(D = k)$                 | $\mathbb{P}(D \leq k)$ |
|-----|-------------------------------------|------------------------|
| 1   | $7/48 = 0.1458333333$               | 0.1458333333           |
| 2   | $223/2304 = 0.0967881944$           | 0.2426215278           |
| 3   | $7657/110592 = 0.0692364728$        | 0.3118580006           |
| 4   | $139913/2654208 = 0.0527136532$     | 0.3645716538           |
| 5   | $10681477/254803968 = 0.0419203715$ | 0.4064920253           |
| 6   | 0.0344074271                        | 0.4408994524           |
| 7   | 0.0289238487                        | 0.4698233011           |
| 8   | 0.0247727295                        | 0.4945960306           |
| 9   | 0.0215382164                        | 0.5161342470           |
| 10  | 0.0189582188                        | 0.5350924658           |
| 11  | 0.0168600735                        | 0.5519525393           |
| 12  | 0.0151257982                        | 0.5670783376           |

## 31 Exact probabilities, finite checks, and reproducibility

### 31.1 The first probabilities and the median

The probability coefficients are rational: the branches of the two square roots at  $z = 0$  have rational Taylor coefficients, as do all the rational operations in (84). Table 12 shows the first twelve values. The decimals are rounded displays of exact fractions, not floating-point coefficients used in the proof.

The exact inequalities  $\mathbb{P}(D \leq 8) < 1/2 < \mathbb{P}(D \leq 9)$ , checked with rational arithmetic, show that 9 is the unique integer median. Total-variation convergence also implies that, for every sufficiently large  $n$ , the finite deficit  $D_n$  has the same integer median. This is an eventual statement; the article does not give a minimal such  $n$ .

### 31.2 An independent exact check of the first mass

**Proposition 31.1** (The largest possible jump). *For  $n \geq 3$ ,*

$$\#\{\pi \in \text{Av}_n(132) : M(\pi) = n - 1\} = C_{n-2} + \sum_{j=0}^{n-2} C_j. \quad (142)$$

Consequently  $\mathbb{P}(D_n = 1) \rightarrow 7/48$ .

*Proof.* A jump of size  $n - 1$  is exactly an adjacency of the values 1 and  $n$ . If it is the ascent 1,  $n$ , this pair must be at the end: any later entry would form 132 with the pair. The preceding entries, standardized, are an arbitrary avoider of length  $n - 2$ , giving  $C_{n-2}$  choices.

If it is the descent  $n, 1$ , decompose at the maximum. The right block starts with 1 and must be increasing, since a later descent together with that first 1 would form 132. For each right-block length  $r \in \{1, \dots, n - 1\}$ , the left block is an arbitrary avoider of length  $n - r - 1$ . This gives the sum in (142). The two cases are disjoint.

Finally  $C_{n-2}/C_n \rightarrow 1/16$ . Also  $\sum_{j=0}^{n-2} C_j \sim (4/3)C_{n-2}$ , as follows either by reversing this sum and using the Catalan ratio, or by Stolz–Cesàro applied to  $\sum_{j=0}^N C_j$  and  $C_N$ . Thus the second term divided by  $C_n$  tends to  $1/12$ . Their sum is  $7/48$ .  $\square$

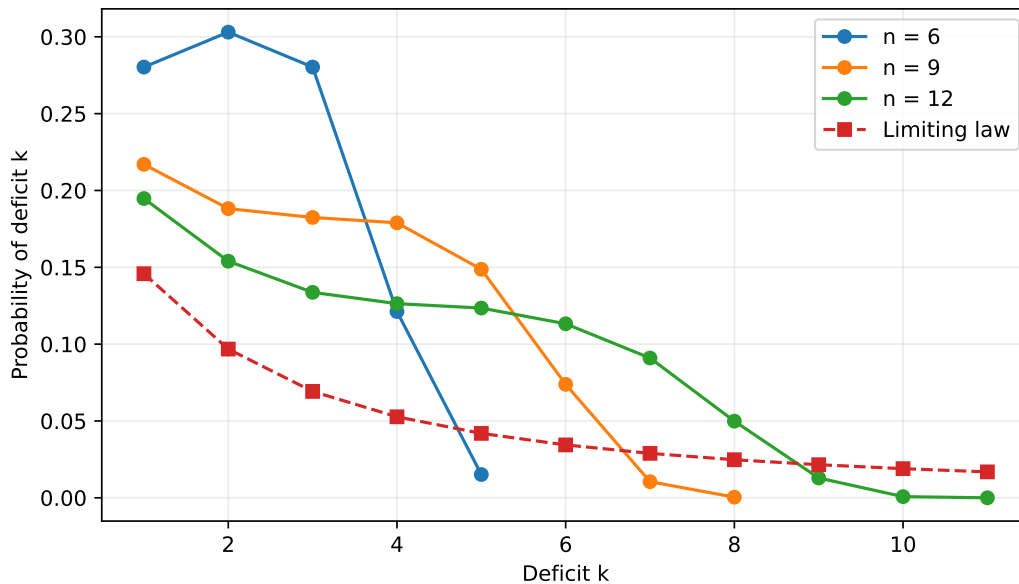


Figure 6: Exact finite point probabilities of  $D_n$  for selected small sizes, compared with the limiting coefficients. Finite support strongly affects the far end of each finite distribution. The curves are illustrations generated from exhaustive Catalan enumeration, not Monte Carlo samples.

### 31.3 What the verifier actually checks

The standard-library verifier performs the following separate tests. The wording is deliberately finite: passing them supports implementation correctness, not the infinite or asymptotic claims by itself.

1. It computes the first 40 point probabilities exactly, verifies positivity, the first five printed fractions, and the median. It compares 31 clipping-correction coefficients computed from the radical formula with a separate finite-endpoint tail summation.
2. For sizes 1 through 8, it compares the Catalan recursive generator with all ordinary permutations filtered by a direct three-index 132 test. This checks 2,055 avoiders and verifies every first-entry count and the extreme-jump formula in that range.
3. For every one of the 804 decorated skeletons of costs 2 through 7, it tests every hole of size  $t + 1$  with first entry greater than  $t$ . There are 269,402 reconstruction tests. Each checks the exact deficit, the last-entry state, and absence of collisions within the fixed-cost reconstruction family. The counts of skeletons and their probability masses are checked against the exact Catalan formulas as well.

All tests passed in the recorded run. The cross-cost disjointness needed in the proof is established by lemma 25.2; the finite collision check is not described as a test of every pair of costs.

### 31.4 Rebuilding and rerunning

Part III's files are

|                                         |                                                            |
|-----------------------------------------|------------------------------------------------------------|
| code/03-largest-jump-model.py           | Exact Catalan, endpoint, skeleton and rational-series code |
| code/03-largest-jump-verify.py          | Standard-library exact finite verifier                     |
| code/03-largest-jump-verify_symbolic.py | Optional SymPy identity and expansion checks               |
| code/03-largest-jump-make_data.py       | Regenerates the three CSV files                            |
| code/03-largest-jump-plot_results.py    | Optional Matplotlib figures                                |
| code/03-largest-jump-build.sh           | The manuscript's three-pass build                          |

|                                                      |                                                                                                                                |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| data/03-largest-jump-*                               | Recorded runs, 256 exact limiting probabilities, finite histograms<br>n = 2..12, extreme-jump checks,<br>optional requirements |
| figures/03-largest-jump-*.pdf                        | The two figures of this part                                                                                                   |
| 03-largest-jump-{PROOF_STATUS,SOURCES,VALIDATION}.md | Audit notes                                                                                                                    |

The scripts are shipped byte-identical to the delivery and still use the delivered names: they write `data/verification.json`, `data/symbolic_verification.json`, the three unprefix CSV files and `figures/tail_scaling.*`, `figures/finite_probabilities.*`, and the verifier and the data generator execute from `model import`. Run in place, that import would load Part I's different `code/model.py` and fail with `ImportError`, and `code/03-largest-jump-build.sh` would look for an `article.tex` in `code/`. The report's README gives a recipe that restores the delivered layout in a scratch copy. There, the manuscript's instructions apply unchanged. The exact verification needs Python 3.10 or later and no third-party package:

```
python code/verify.py
python code/make_data.py
```

Do not use Python's `-O` option, which disables the verifier's assertions. The optional symbolic checks need SymPy, and optional plot regeneration needs Matplotlib:

```
python code/verify_symbolic.py
python code/plot_results.py
```

The retained figures make Python unnecessary for rebuilding the article. The manuscript's build script placed auxiliary files in a temporary directory and copied back only its `article.pdf`; this report is built as described in its README. The included data files record exact fractions; rounded display columns and plots are not interval-arithmetic certificates.

## 32 Further research questions

The results above settle the bounded and unbounded sublinear deficit windows. They also reveal several questions whose answers would extend the theory rather than merely recompute its constants.

### 32.1 Bulk finite-size scaling

For  $0 < u < 1$ , determine whether the limit

$$\Psi(u) = \lim_{n \rightarrow \infty} \sqrt{n} \mathbb{P}(D_n > \lfloor un \rfloor) \quad (143)$$

exists, and if so compute it, including its endpoint behavior. This is a proposed problem, not a theorem or an asserted formula. Part III proves an upper bound on the expression, and a positive lower bound for sufficiently small fixed  $u$ . It does not establish a limit. A likely starting point is to retain the finite Catalan ratio  $C_{n-t}/C_n$  instead of replacing it by  $4^{-t}$ , and to analyze the regime where both sides at a maximum decomposition are macroscopic. [Added 29 September 2026, batch 52: answered in Part IV. The limit exists for every  $0 < u \leq 1$ , uniformly on  $[\delta, 1]$  for each  $\delta > 0$ , and  $\Psi(u) = \frac{3}{\sqrt{\pi}} \frac{1-2u}{\sqrt{u(1-u)}}$  for  $u < 1/2$ ,  $\Psi(u) = 0$  for  $u \geq 1/2$  (theorem 38.1). At the endpoints,  $\Psi(u)\sqrt{u} \rightarrow 3/\sqrt{\pi}$  as  $u \rightarrow 0$ , matching theorem 22.2, and  $\Psi(1/2 - \varepsilon) \sim 12\varepsilon/\sqrt{\pi}$  (section 43.1). Both manuscripts behind Part IV analyze exactly the regime suggested here, where both sides of a maximum decomposition are macroscopic.]

### 32.2 Exact constants for the mean and higher moments

Determine whether  $n^{1/2-p}\mathbb{E}D_n^p$  converges for each  $p > 1/2$ , especially whether  $\mathbb{E}D_n/\sqrt{n}$  has a limit. A bulk profile with sufficient uniform integrability would make these constants integrals of that profile. At the critical point, ask whether

$$\mathbb{E}\sqrt{D_n} - \frac{3}{2\sqrt{\pi}} \log n$$

converges. The  $O(1)$  result proved here does not imply convergence of the centered quantity. [Added 29 September 2026, batch 52: answered in Part IV for  $p > 1/2$ :  $n^{1/2-p}\mathbb{E}D_n^p \rightarrow m_p = \frac{3}{\sqrt{\pi}} \int_0^1 (z^2/(1+z^2))^{p-1} dz$ , so  $\mathbb{E}D_n/\sqrt{n} \rightarrow 3/\sqrt{\pi}$ , and the constants are indeed integrals of the bulk profile (theorem 38.3). The critical question stays open (section 47.6).]

### 32.3 The sharp total-variation constant and local corrections

The order  $n^{-1/2}$  is now proved sharp, but the interval in (134) leaves the leading constant open. Does  $\sqrt{n} d_{TV}(\mathcal{L}(D_n), \mathcal{L}(D))$  converge? Is the missing limiting mass beyond the finite support the whole leading discrepancy, or does an additional contribution arise inside the support? Separately, for each fixed  $k$ , determine the first correction to  $\mathbb{P}(D_n = k)$ . A pointwise expansion can be much sharper than the global total-variation bound and would give effective finite-size estimates for the first several adjacency thresholds.

### 32.4 Ascending and descending extreme jumps jointly

Define the largest ascent and largest descent by

$$M_n^+ = \max_i (\Pi_n(i+1) - \Pi_n(i))_+, \quad M_n^- = \max_i (\Pi_n(i) - \Pi_n(i+1))_+.$$

Determine their joint limiting behavior, allowing the possibility that one coordinate requires an extended state space or a different scale. The current clipping step mixes an old extreme jump with a newly created ascent, so its explicit dependence on the terminal endpoint should provide useful information. Independence must not be assumed.

### 32.5 Near-maximal jumps as a point process

Instead of retaining only the minimum deficit, retain the multiset

$$\{n - |\Pi_n(i+1) - \Pi_n(i)| : 1 \leq i < n\}$$

near the finite end of the positive integers. Determine its joint count law on every fixed finite set, including ties for the maximum jump. A decorated skeleton records where a finite number of new extreme candidates are created. It may therefore support a marked extension of the present generating-function calculation. A proof must also control unrecorded jumps inside the large hole.

### 32.6 Weighted Catalan models

Replace the uniform measure by weights tracking descents, leaves of the associated binary tree, or another statistic with an explicit bivariate Catalan generating function. Determine when an absorbing finite skeleton still exists, whether the moment threshold remains  $1/2$ , and which critical parameter values change the tail exponent. Both the operation probabilities and the endpoint clipping law would have to be recomputed; a formal substitution into (84) is not justified.

### 32.7 Other avoidance classes

Develop an analogous boundary analysis for other pattern classes, with particular attention to classes whose recursive specifications have several large-block types. Before treating a new pattern as a distinct case, account for the reversal and complement symmetries that preserve absolute adjacent differences. The difficult extension is not a mere renaming of a symmetry-equivalent class, but a new decomposition in which the competition between large adjacent jumps has a different structure.

### 32.8 An exact sampler with controlled work

The skeleton probability space is an exact description of the limit, but its cost has infinite expectation by (106). Sampling full finite decorations literally can therefore be expensive on rare runs. Design an exact sampler for  $D$  that manipulates only the endpoint information and clipping events it needs. Determine its expected bit complexity, and compare a state-based sampler with inversion of the rational coefficient sequence. Almost-sure termination alone is not a finite expected-cost theorem.

### 32.9 A formalization boundary

A feasible first formal target is the deterministic reconstruction lemma: define 132-avoidance, the maximum decomposition, decorated skeletons, and the endpoint state, then prove reconstruction and injectivity by induction. Next formalize the first-entry formula and the common-submeasure inequality. These steps isolate the combinatorial content from the later analytic transfer theorem. A complete formalization would still need countable probability, formal power series or analytic identities, and asymptotic coefficient transfer; the present package does not claim to supply them.

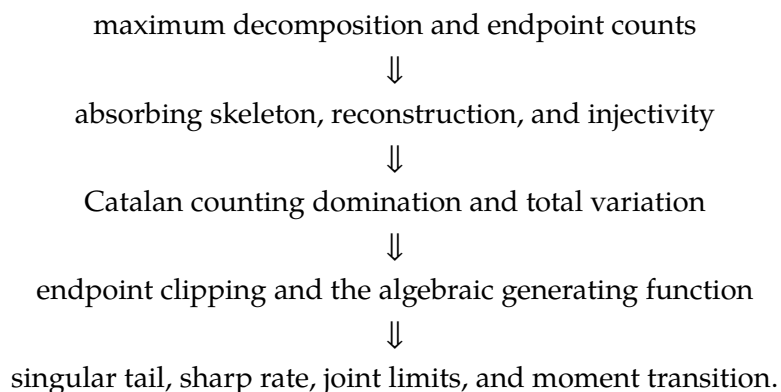
### 32.10 Reconnect to the fixed-bound spectral questions

The boundary law does not decide which of the two irreducible endpoint components controls the fixed- $m$  growth rate, nor the optimal defect between that rate and the Catalan scalar comparison rate discussed in Parts I and II (sections 9 and 19). Study whether a two-parameter endpoint system can interpolate between the fixed- $m$  spectral regime and the boundary regime here. Any proposed interpolation must specify its uniform error before taking a moving-bound limit; fixed-parameter rationality alone is insufficient.

## 33 Proof audit, scope, and conclusion

### 33.1 The dependency chain

The argument proceeds through explicit finite objects:



The only analytic coefficient-transfer input is standard singularity analysis, with the needed continuation and local expansion checked explicitly. No growth-constant estimate from the

Table 13: Claim-by-claim status of Part III (manuscript 02).

| Claim                                     | Status and limitation                                                                                                  |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Algebraic law for $D_n$                   | Written proof from finite counting and exact state equations; not formally verified.                                   |
| Sharp order $n^{-1/2}$ in total variation | Upper bound by common mass; lower bound by the limiting mass beyond finite support. Exact constant open.               |
| Uniform estimate for $m = n - d$          | Absolute estimate for $2 \leq d \leq n - 1$ ; relative equivalent only when $d \rightarrow \infty$ and $d = o(n)$ .    |
| Moment transition                         | Exact critical logarithmic coefficient and subcritical convergence; only two-sided orders above the critical exponent. |
| Finite verification                       | Exact rational and integer regression tests over specified finite ranges; not a substitute for the proofs.             |
| Novelty and priority                      | Targeted comparison with the cited papers and repository report; no exhaustive literature or priority certification.   |
| Bulk profile, exact mean constant         | Proposed research questions, not claimed results. [Answered in Part IV, added 29 September 2026.]                      |

existing ProveIt report (Parts I and II) is needed, and no general conditioned-tree convergence theorem is used as a black box.

### 33.2 What is resolved

The adjacency-bounded enumeration has a nontrivial joint-limit boundary at distance  $O(1)$  below  $n$ . Part III identifies the entire limiting profile there, gives a sharp-order convergence rate, and extends it uniformly to every growing sublinear deficit window. The algebraic law has a heavy enough tail to force a moment transition, so typical and average deficits live on different scales. These conclusions answer a specific missing regime in the repository’s program while leaving its fixed-bound spectral problems and the bulk finite-size profile separate.

The reusable mechanism is the common-mass comparison: a finite stopping skeleton of cost  $t$  has limiting weight  $4^{-t}$ , while the corresponding finite Catalan family has at least that weight. Once exceptional first entries are removed, the extreme statistic agrees exactly. This finite argument simultaneously supplies convergence, its quantitative rate, and a controlled bridge from limiting probabilities to moving-bound counts.

## 34 Exact coefficient recursions

This section and the next two were the manuscript’s Appendices A–C. They are numbered here as sections of Part III, so that Part I’s appendices keep their letters. The package computes all coefficients with rational arithmetic. To degree  $N$ , begin with

$$q(z) = \sum_{k=1}^N \frac{C_{k-1}}{4^k} z^k + O(z^{N+1}).$$

For a series  $a(z) = 1 + \sum_{k \geq 1} a_k z^k$ , the square root  $b(z) = 1 + \sum_{k \geq 1} b_k z^k$  is obtained from

$$b_k = \frac{1}{2} \left( a_k - \sum_{j=1}^{k-1} b_j b_{k-j} \right). \quad (144)$$

For a series  $a$  with nonzero constant term, its inverse is obtained from

$$b_0 = a_0^{-1}, \quad b_k = -a_0^{-1} \sum_{j=1}^k a_j b_{k-j}. \quad (145)$$

Use these recursions to compute  $r = \sqrt{1-q}$ ,  $H = q/(3-4q)$ ,  $\Upsilon = q(1-r)/(2(3-4q))$ , and  $J = (q + (1-z)\Upsilon)/(4(1-q)^2)$ . Their sum is  $G$ . Only the first  $N$  coefficients of each input are needed. Ordinary convolution gives  $O(N^2)$  rational arithmetic operations; this is not a bit-complexity bound, because numerators and denominators grow.

The independent clipping computation uses instead the endpoint counts (98) and their tails. Writing  $P(z) = P(1/4, z) = z/(1 + 2\sqrt{1-z})$ , it extracts coefficients of  $P$  without the formula for  $\Upsilon$ , subtracts finite first-entry counts, and sums over the possible finite  $R$ -increments. Agreement is therefore not merely two evaluations of the same line of code.

## 35 An algebraic certificate without branch ambiguity

One may describe  $G$  without nested radical notation by introducing  $q$  and  $r$  and imposing

$$4q^2 - 4q + z = 0, \quad q(0) = 0, \quad (146)$$

$$r^2 - (1-q) = 0, \quad r(0) = 1. \quad (147)$$

In addition,

$$8(3-4q)(1-q)^2 G = q(15-24q+8q^2-z-(1-z)r). \quad (148)$$

Equations (146)–(148), with the stated initial values, select the same analytic series as (84). Eliminating  $r$  first gives

$$\begin{aligned} & [q(15-24q+8q^2-z) - 8(3-4q)(1-q)^2 G]^2 \\ & - q^2(1-z)^2(1-q) = 0. \end{aligned} \quad (149)$$

The resultant of (149) and  $4q^2 - 4q + z$  eliminates  $q$ . Squaring can add branches, so the conditions at zero must be retained; an arbitrary root of the eliminated polynomial is not automatically the probability generating function. Part III claims degree at most four, not an independently proved irreducibility statement for a minimal polynomial.

## 36 Repository pin, literature scope and merge

### 36.1 The manuscript's pin and literature scope

The inspected repository head was

8d936ee2357f9decf78c2ecc7d9baf3100a6787d.

The relevant report is in the directory

SetTheory/Cardinals/docs/reports/  
enumerative-combinatorics/  
adjacency-bounded-132-avoiders/.

The inspected README blob was 7f56b878e420c9fb0a0df299f54f9c154c68780e; its LaTeX article blob was f24ea59acf0f9809462f7fb0f055e78d38d4740a. The README identifies two source parts, from September 19 and September 28, 2026, and explicitly says that no joint limit in  $n$  and  $m$  is claimed there. That exclusion motivates the present target; it is not presented as a named classical conjecture.

The literature comparison used the arXiv records and available HTML of Nadler and Mayama–Akita, the records of the two Janson papers, and targeted searches for extreme adjacent differences in 132-avoiders. The searches did not establish exhaustive coverage. In particular, absence of a matching result from the returned material is not evidence sufficient to certify priority. The full metadata ledger is retained in SOURCES.md (shipped as 03-largest-jump-SOURCES.md). No external PDF or source article is redistributed in this package.

Both blobs above are the ones this report still had when the manuscript was placed (section 21.1); in particular the README sentence quoted above is the one Part III re-scopes, and it now carries a dated pointer here, as does Part II’s research direction “Joint growth of length and adjacency bound” (section 19).

## 36.2 Where the merge had to choose

- *Placement.* Part III follows Part II and precedes Part I’s appendices, which contain no numbered equations, tables or figures; no number of Part I or Part II changed. The manuscript’s three appendices are printed as sections 34 to 36. Its title page and abstract are replaced by the opening note of Part III, the status box of section 21.1 (which prints its research-status box) and a paragraph of the report’s abstract; every claim of its abstract is stated in section 22.
- *Scope.* The manuscript describes its target as the joint limit the report’s README excluded. Here it is recorded as answering that question, and Part II’s corresponding research direction, only in the boundary regime  $m = n - d$  with  $d = o(n)$  (section 21.3); the README and Part II’s text keep their wording, with dated pointers.
- *Duplicates.* The maximum decomposition is Part I’s (section 2) and is credited there; the first-entry count is Part I’s (21), reindexed, and the manuscript’s different derivation is kept (remark 23.2). No other result of Part III is proved in Parts I or II; elementary Catalan identities such as  $\sum_{k \geq 1} C_{k-1} 4^{-k} = \frac{1}{2}$  are used in both. Remark 23.2’s comparison with Part II’s tail bound is the merge’s, as are section 21 and this list.
- *Notation.*  $\varepsilon_n \rightarrow \omega_n$ ,  $\mathcal{A} \rightarrow \Upsilon$ ,  $q_{1-\varepsilon} \rightarrow \kappa_{1-\varepsilon}$ ,  $C, C_0 \rightarrow C_{TV}$ ,  $c_p \rightarrow \underline{c}_p$  (table 11). The deficit  $D_n$ , its law  $D$ , the threshold  $d$  and the cutoff  $h$  keep the manuscript’s letters, because each collides only with objects of a different kind (polynomials, constants, two-argument counts) that are listed in the table. No normalization changed.
- *References to this report.* The manuscript cites Parts I and II as a repository report (its reference “Repo”); here those citations are internal cross-references. Its entries for Nadler and Mayama–Akita are Part I’s; its Janson and Flajolet–Odlyzko entries are added to the bibliography.
- *Reproduction.* The manuscript’s package listing and commands are replaced by the shipped names and the copy recipe (section 31.4); its article.tex, article.pdf, README.md and the PNG versions of its figures are not shipped.
- *Figures.* In the write phase the manuscript’s two figure PDFs were copied byte-identically into figures/, under their delivered names with the prefix 03-largest-jump- (figures 5 and 6).
- *Stale sentences elsewhere.* The abstract, Part I’s remark on the order of limits (section 1), Part II’s remark 11.2, its research direction on joint growth, its scope ledger (table 10) and Part I’s section B carry dated pointers to Part III; their original wording is kept.

## Part IV. The macroscopic jump law

Part IV was added on 29 September 2026 in batch 52 of ProveIt's incoming-report intake, merged from two manuscripts that were written independently of each other against the state of Parts I–III printed above. Part III determined the law of the largest-jump deficit  $D_n = n - M_n$  at bounded distance from the trivial bound  $m = n - 1$ , and asked in its Sections 32.1 and 32.2 for the macroscopic profile of  $D_n$  and the exact constants of its supercritical moments. Both manuscripts answer both questions, with different proofs. Part IV prints the shared results once, keeps both proofs, and adds what only one of them proved (section 37.1).

### 37 Provenance, conventions and the answer

#### 37.1 Two manuscripts, one addition

Part IV is built from two AI-assisted research manuscripts prepared for Vladimir Reshetnikov, both dated September 29, 2026:

- *Manuscript 10* of batch 52, *The Macroscopic Jump Law: Exact rare-event profiles and moment constants for random 132-avoiding permutations* (23 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Macroscopic_Jump.zip` (inner directory `ProveIt_Macroscopic_Jump/`, main file `article.tex`) in ProveIt commit 458ccfb80. It was written against ProveIt commit 1ee53d57d, whose root tree c772efcde0 it records. For the proofs and questions of Part III in contiguous form it also read the standalone source of Part III's manuscript, `article(20260929-161704).tex`, from Vladimir's file library; that file is not in the repository, and its own pin is Part III's 8d936ee23 (section 36).
- *Manuscript 11* of batch 52, *Macroscopic Deficits of the Largest Jump: A rare-event law and exact moment constants for 132-avoiding permutations* (24 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Macroscopic_Jump_Research.zip` (inner directory also `ProveIt_Macroscopic_Jump/`, main file `article.tex`) in ProveIt commit 11e1e9001. It was written against the same commit 1ee53d57d and quotes the blob aeadaf364b of this report's `article.tex`.

At 1ee53d57d this report's `article.tex` and `README.md` had the blobs aeadaf364b and 815242ed6, and both were unchanged at the placement commit a701d9098. Every statement either manuscript makes about this report therefore refers to Parts I–III as printed above. Neither manuscript saw the other. Their code, data and provenance notes are shipped beside the earlier parts' files with the prefixes `04-macroscopic-jump-` (manuscript 10) and `05-macroscopic-deficits-` (manuscript 11), continuing the report's local sequence after Part III's `03-largest-jump-`. Their manuscripts, READMEs and PDFs are not shipped and survive in the arrival commits; their checksum ledgers were verified at placement and retired. Their four figures were regenerated in the write phase (section 50.3).

Manuscript 10 is the base: it arrived first and covers more regimes. The two agree on every shared statement and neither contradicts Parts I–III. They contribute, in the order printed here:

- *both*: the macroscopic profile  $\Psi$  and the moving-bound equivalent for  $1/2 < m/n < 1$  (theorem 38.1), the deficit rare-event measure (theorem 38.2, last assertion), every supercritical moment constant with the mean and variance constants (theorem 38.3), a uniform finite-size tail envelope (proposition 40.3, one explicit and one non-explicit form), the first-entry and last-entry estimates of section 39, the matching with Part III's boundary tail and the linear vanishing at  $u = 1/2$  (section 43.1), the critical pole of the moment constants, and seven of the research questions of section 47;
- *manuscript 10 only*: the exact deficit recursion at the maximum (proposition 39.1), the stopping-word envelope, the marked rare-event measure recording the last entry and the renewal proof of the profile (section 41), the conditional three-channel law (corollary 43.1), the

power-biased and size-biased limits (theorem 38.4), polynomial lower bounds below the half-size cutoff (theorem 45.1), a uniform Catalan sampler with Monte Carlo illustrations (section 46.6), and its small exact examples (section 49);

- *manuscript 11 only*: the canonical threshold parser and its rare-scale cost tail (section 40.4), the second proof of the profile through fixed outer words (section 42), the universal outer-word and terminal-split law with its three endpoint modes (theorem 43.2, corollary 43.3, and proposition 43.4), the moment constants at  $p = 3/2$  and  $p = 4$  with a recurrence (section 44.2) and their large-order asymptotic (proposition 44.1), an exact shortcut for the endpoint recurrence, and exact counts up to  $n = 320$  (section 46.3).

Every result, proof, example, remark, limitation and question of both manuscripts is printed.

**Status (manuscripts 10 and 11).** The theorems of Part IV have written proofs, exact finite regression checks, and (manuscript 10) explicitly labelled Monte Carlo illustrations. They are not Lean- or Rocq-verified and have not been independently refereed. Publication priority has not been established: the advances are stated relative to the inspected repository report, whose explicit scope they extend. No asymptotic theorem from an unreferenced repository report is assumed: Part IV rederives the finite facts it uses and uses none of the asymptotic theorems of Parts I–III, and no fixed-bound spectral asymptotic is substituted into a varying-bound problem. Nothing in Part IV is formalized in any proof assistant.

### 37.2 Notation of Part IV

The symbols  $a_n^{(m)}$ ,  $C_n$ ,  $C$ ,  $Av_n(132)$ ,  $M(\pi)$ ,  $D(\pi)$ ,  $M_n$ ,  $D_n$ ,  $F(\pi)$ ,  $c_j = C_j/4^j$ , the operations  $R_\beta$ ,  $A$ ,  $T_r$ , the cost  $t$  and the cutoff  $h$  keep their meanings from Part III;  $D$  without index is Part III's limit law. Manuscript 10 also sets  $M = 0$  and  $D = 1$  for the one-element permutation, which makes its recursions exact. The two manuscripts name the same objects differently, and several of their letters are used in Parts I–III for other objects. Part IV fixes one name for each notion (table 14). In particular the threshold fraction is always  $u$  (manuscript 11 writes  $x$ ), the split fraction always  $y$  (manuscript 11 writes  $u$ ), and the profile always  $\Psi$ , the name Part III's question (143) gave it (manuscript 11 writes  $\Phi$ ).

No normalization changed: every renamed symbol denotes exactly its manuscript's quantity. Probability, expectation, total variation, law and cost are written  $\mathbb{P}$ ,  $\mathbb{E}$ ,  $d_{TV}$ ,  $\mathcal{L}$ , cost as in Part III, and variance  $\text{Var}$ . The letters  $a$ ,  $b$  (block sizes),  $N$  (a core size),  $\sigma$ ,  $\tau$ ,  $\beta$  (blocks),  $\eta$ ,  $\delta$ ,  $\epsilon$ ,  $\varepsilon$ , and  $r$  and  $q$  inside single proofs are local and declared where used. Equations are numbered consecutively with those of Parts I–III.

### 37.3 Which questions are answered, and how far

Part III's Section 32.1 asks whether  $\Psi(u) = \lim_n \sqrt{n} \mathbb{P}(D_n > \lfloor un \rfloor)$  exists for  $0 < u < 1$ , and for its value and endpoint behaviour. Its Section 32.2 asks whether  $n^{1/2-p} \mathbb{E} D_n^p$  converges for each  $p > 1/2$ , especially  $\mathbb{E} D_n / \sqrt{n}$ , and whether  $\mathbb{E} \sqrt{D_n} - \frac{3}{2\sqrt{\pi}} \log n$  converges. Part IV settles:

- the first question completely: the limit exists for every  $0 < u \leq 1$ , uniformly away from  $u = 0$ , and equals  $\frac{3}{\sqrt{\pi}} \frac{1-2u}{\sqrt{u(1-u)}}$  for  $u < 1/2$  and 0 for  $u \geq 1/2$ ; near  $u = 0$  it matches Part III's boundary tail, and it vanishes linearly at  $u = 1/2$  (theorem 38.1, section 43.1);
- the supercritical half of the second:  $n^{1/2-p} \mathbb{E} D_n^p \rightarrow m_p$ , an explicit integral, for every fixed  $p > 1/2$ ; in particular  $\mathbb{E} D_n \sim 3\sqrt{n/\pi}$  and  $\text{Var}(D_n) \sim \frac{3(4-\pi)}{4\sqrt{\pi}} n^{3/2}$  (theorem 38.3);
- Part III's remark 29.3 and Part II's research direction "Joint growth of length and adjacency bound" (section 19) for  $m = \lfloor \theta n \rfloor$  with  $1/2 < \theta < 1$ :  $a_n^{(\lfloor \theta n \rfloor)} / C_n \sim \frac{3(2\theta-1)}{\sqrt{\pi\theta(1-\theta)}} n^{-1/2}$ . The equivalent  $3/\sqrt{\pi(1-\theta)n}$ , which Part III declined to assert, is indeed false for every such  $\theta$ :

Table 14: Symbols of Part IV. “From” gives the manuscripts’ symbols (10, 11); the last column prints the tempting false reading.

| Symbol                                                | Meaning in Part IV                                                 | From                                | Not to be confused with                                                                        |
|-------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------|
| $L(\pi), L_n$                                         | last value $\pi_n$ ; that of $\Pi_n$                               | 10: $L$ ; 11: $L$                   | Part III’s state $L$ and its left block $L$ in $\pi = L n R$ (not used here), Part I’s $L_m$ . |
| $J(\pi), J_n$                                         | first deficiency $ \pi  - F(\pi)$ ; that of $\Pi_n$                | both                                | Part III’s series $J$ , Part II’s order $J$ .                                                  |
| $u; y; x$                                             | threshold fraction $d/n$ ; split fraction $a/n$ ; deficit variable | 10: $u, y, x$ ; 11: $x, u, x$       | Part II’s $u = 1/\Lambda$ .                                                                    |
| $\Psi(u)$                                             | the profile (150)                                                  | 10: $\Psi$ ; 11: $\Phi$             | Part I’s shift map $\Phi$ .                                                                    |
| $\mu; \mu_D$                                          | marked measure on $\mathcal{X}$ ; its deficit marginal             | 10: $\mu$ ; 11: $\mu$ ( $= \mu_D$ ) | the generic measures of (138).                                                                 |
| $\mathcal{X}, \Xi_n$                                  | $(0, 1] \times [0, 1]$ ; the pair $(D_n/n, L_n/n)$                 | 10: $E, Y_n$                        | Part I’s $E_m$ .                                                                               |
| $\rho, v, v(y)$                                       | deficit density; source functional; $\min(y, 1 - y)$               | 10                                  | —                                                                                              |
| $S, S^* f$                                            | endpoint map; $f \circ S$                                          | 10: $S, P f$                        | Part III’s $P(x, z); \mathbb{P}$ .                                                             |
| $r_a, \bar{r}_H$                                      | $C_a 4^{-a-1}$ ; $\sum_{a \leq H} r_a$                             | 10: $r_a, R_H$                      | Part II’s $r_m$ ; Part III’s $r$ ; the operations $R_\beta$ ; Part I’s $R_m$ .                 |
| $m_p$                                                 | moment constant, $p > 1/2$                                         | both: $\kappa_p$                    | Part III’s quantile $\kappa_{1-\epsilon}$ (index also in $(0, 1)$ ).                           |
| $\mathcal{I}(p); I(u)$                                | $\int_0^1 (z^2/(1+z^2))^{p-1} dz$ ; $\int_u^{1-u} \mathfrak{f}$    | 11: $I(p)$ ; 10: $I(u)$ , 11: $I_x$ | each other.                                                                                    |
| $\mathfrak{f}(y)$                                     | $[y(1-y)]^{-3/2}$                                                  | 11: $k(u)$                          | an index $k$ .                                                                                 |
| $H^\star, \mathbb{P}_n^\star$                         | size-biased distribution function and measure                      | 10: $H, \mathbb{P}_n^\star$         | Part III’s $H(z)$ .                                                                            |
| $\mathcal{W}, \mathcal{W}_R; w_t, w_t^R$              | outer-word series; their counts                                    | 11: $W, W_R; w_t, r_t$              | Part I’s matrix $W_m$ and $W_{U_m}$ , $W_{V_m}$ .                                              |
| $W_n, W$                                              | canonical outer word; its limit                                    | 11                                  | the series $\mathcal{W}$ .                                                                     |
| $T_{n,m}, T_W$                                        | canonical cost; $\text{cost}(W)$                                   | 11: $T_{n,m}, T$                    | Part III’s skeleton cost $T$ (which is also manuscript 10’s $T$ ).                             |
| $\text{len}(W)$                                       | number of operations of $W$                                        | 11: $K$                             | the constants $K$ .                                                                            |
| $\chi(w)$                                             | endpoint acceptance weight                                         | 11: $\alpha(w)$                     | Part I’s $\alpha_m$ .                                                                          |
| $\ell_n(w)$                                           | append threshold                                                   | 11: $r_n(w)$                        | $r_a$ .                                                                                        |
| $\mathcal{E}_{n,m}, \mathcal{E}_n(w)$                 | the events $M_n \leq m$ , and with canonical word $w$              | 11: $E_{n,m}, E_n(w)$               | Part I’s $E_m$ .                                                                               |
| $Q_{N,m}$                                             | raw terminal-pair count                                            | 11                                  | Part III’s $Q$ .                                                                               |
| $U_n, U^{(u)}$                                        | terminal split fraction; its limit                                 | 11: $U_n, U_x$ ; 10: $Y$            | —                                                                                              |
| $\xi, V, \zeta$                                       | channel, deficit, last-entry mark in corollary 43.1                | 10: $B, V, W$                       | $B(x); W_n$ .                                                                                  |
| $\mathcal{N}_s(u)$                                    | avoiders of size $s$ with first deficiency $\leq u$                | 11: $H_s(u)$                        | Part III’s $N_{N,f}$ ( $F = f$ exactly).                                                       |
| $\Omega_r(\theta)$                                    | conjectured constant of section 47.1                               | 10: $A_r(\theta)$                   | Part I’s $A_m(x)$ .                                                                            |
| $\underline{c}_{\theta,r}, \underline{c}'_{\theta,r}$ | constants of theorem 45.1                                          | 10: $c_{\theta,r}, c'_{\theta,r}$   | Part I’s $c_{k,p}$ ; Part III’s $\underline{c}_p$ .                                            |
| $K, K_f, K_\eta, \dots$                               | constants, absolute unless indexed                                 | both                                | Part I’s $K_{m,d}$ , Part II’s $K_m$ , Part III’s $\mathcal{K}$ .                              |

the true constant is smaller by the factor  $(2\theta - 1)/\sqrt{\theta} < 1$  (the merge's comparison).

It does not settle: the critical centered half-moment, which stays open (section 47.6); an equivalent for  $0 < \theta \leq 1/2$ , where only  $o(n^{-1/2})$  and polynomial lower bounds are proved (theorems 38.1 and 45.1); the window at  $m = n/2$ ; a local limit; the exact total-variation constant of Part III's corollary 28.2; and any interpolation with the fixed- $m$  regime of Parts I and II. Part IV refutes nothing in Parts I–III.

## 38 The questions and the main results

### 38.1 From a boundary distribution to macroscopic rare events

Recall from Part III that  $M(\pi) = \max_i |\pi_{i+1} - \pi_i|$  and  $D(\pi) = n - M(\pi)$  for  $\pi \in \text{Av}_n(132)$ ,  $n \geq 2$ , that  $\Pi_n$  is uniform on  $\text{Av}_n(132)$  with  $M_n = M(\Pi_n)$ ,  $D_n = D(\Pi_n)$ , and that (82) reads  $a_n^{(m)}/C_n = \mathbb{P}(M_n \leq m) = \mathbb{P}(D_n \geq n - m)$ . For a nonempty permutation write  $F(\pi) = \pi_1$ ,  $L(\pi) = \pi_{|\pi|}$  and  $J(\pi) = |\pi| - F(\pi)$ , and write  $F_n, L_n, J_n$  for their values at  $\Pi_n$ . An inequality  $D_n \geq un$  means  $D_n \geq \lceil un \rceil$ . All moment orders are fixed before  $n$  tends to infinity.

The finite-state, fixed- $m$  problem was developed by Nadler [1], who treated  $m = 2$ , and Mayama–Akita [2], who supplied endpoint states, rational generating functions for each fixed bound and growth-constant results; their maximum-position restriction is especially useful here, and their finite-state recursion is the basis of manuscript 11's exact numerical checks, not a new result of it. Part III proves a proper integer-valued boundary limit for  $D_n$ , a sharp-order total-variation bound, a limiting tail proportional to  $k^{-1/2}$ , a growing sublinear-deficit equivalent and a moment transition at  $p = 1/2$ . It explicitly does *not* infer a relative asymptotic at  $D_n$  of order  $n$  from its additive boundary estimate (remark 29.3), and its Sections 32.1 and 32.2 ask for the bulk profile and the exact supercritical moment constants. Those are the starting points of Part IV.

That distinction is essential. A boundary tail of the form  $c/\sqrt{k}$  does not by itself identify  $\mathbb{E}D_n/\sqrt{n}$ : events with  $D_n$  proportional to  $n$  contribute at precisely the same order. Part IV analyzes those events directly, rather than extrapolating the boundary approximation. On the ordinary probability scale  $D_n/n \rightarrow 0$ ; on the scale of probabilities multiplied by  $\sqrt{n}$ , macroscopic deficits remain visible, and it is this second scale that governs all moments of order greater than  $1/2$ . Part IV builds on the subject and elementary decompositions of Parts I–III but assumes none of their probabilistic or spectral theorems; this makes the dependency on the existing research particularly narrow and auditable.

### 38.2 The macroscopic profile and moving-bound enumeration

Define, for  $0 < u \leq 1$ ,

$$\Psi(u) = \begin{cases} \frac{3}{\sqrt{\pi}} \frac{1-2u}{\sqrt{u(1-u)}}, & 0 < u < \frac{1}{2}, \\ 0, & \frac{1}{2} \leq u \leq 1. \end{cases} \quad (150)$$

$\Psi$  is continuous on  $(0, 1]$ . The zero at  $1/2$  and beyond is part of the definition, not a claim that the finite probability vanishes there.

**Theorem 38.1** (Complete first-order macroscopic profile; both manuscripts). *For every  $0 < u \leq 1$ ,*

$$\sqrt{n} \mathbb{P}(D_n > un) \longrightarrow \Psi(u). \quad (151)$$

*More generally, for every integer sequence  $d_n$  with  $d_n/n \rightarrow u \in (0, 1)$ ,*

$$\sqrt{n} \mathbb{P}(D_n \geq d_n) \longrightarrow \Psi(u). \quad (152)$$

For each fixed  $\delta > 0$  the convergence in (151) is uniform in  $u \in [\delta, 1]$ , and so is that of  $u \mapsto \sqrt{n} \mathbb{P}(D_n \geq \lceil nu \rceil)$ . Consequently, for fixed  $1/2 < \theta < 1$ ,

$$\frac{a_n^{(\lfloor \theta n \rfloor)}}{C_n} \sim \frac{3(2\theta - 1)}{\sqrt{\pi\theta(1 - \theta)}} n^{-1/2}, \quad (153)$$

$$a_n^{(\lfloor \theta n \rfloor)} \sim \frac{3(2\theta - 1)}{\pi\sqrt{\theta(1 - \theta)}} \frac{4^n}{n^2}. \quad (154)$$

For fixed  $0 < \theta \leq 1/2$  the ratio is  $o(n^{-1/2})$  and the count is  $o(4^n/n^2)$ ; no positive asymptotic equivalent in that range is asserted.

Manuscript 10 proves (151) with uniformity on  $[\delta, 1]$  and both consequences; manuscript 11 proves the sequence form (152), compact uniformity and (153). The first proof (manuscript 10) solves a renewal equation for a marked measure (section 41); the second (manuscript 11) sums fixed outer words of a canonical threshold decomposition (section 42). The zero part of  $\Psi$  is a first-order *asymptotic* cutoff, not a finite combinatorial prohibition. The increasing permutation, for example, has  $M_n = 1$  and  $D_n = n - 1$ . Such permutations lie beyond the cutoff, but have negligible mass on the  $n^{-1/2}$  scale.

### 38.3 The marked measure behind the coefficient three

Put

$$\mathcal{X} = (0, 1] \times [0, 1], \quad \Xi_n = (D_n/n, L_n/n), \quad v(y) = \min(y, 1 - y).$$

The following locally finite measure is infinite near the missing boundary  $x = 0$  of the deficit coordinate; it is not a probability measure.

**Theorem 38.2** (Marked rare-event measure; manuscript 10, last assertion both). *For every continuous function  $f$  on  $[0, 1]^2$  that vanishes on some strip  $\{0 \leq x \leq \delta\}$  of the deficit coordinate,  $\delta > 0$ ,*

$$\sqrt{n} \mathbb{E} f(\Xi_n) \longrightarrow \mu(f), \quad (155)$$

$$\mu(f) = \frac{1}{4\sqrt{\pi}} \int_0^1 \frac{f(v(y), 0) + f(v(y), 1 - y) + f(v(y), 1)}{[y(1 - y)]^{3/2}} dy. \quad (156)$$

Equivalently,  $\sqrt{n} \mathcal{L}(\Xi_n)$  converges vaguely on  $\mathcal{X}$  to  $\mu$ . The deficit marginal of  $\mu$  has density

$$\rho(x) = \frac{3}{2\sqrt{\pi}} [x(1 - x)]^{-3/2} \mathbf{1}_{(0, 1/2)}(x). \quad (157)$$

That is, on the locally compact space  $(0, 1]$  the measures  $\mu_{D,n} = \sqrt{n} \mathcal{L}(D_n/n)$  converge vaguely to

$$\mu_D(dx) = \rho(x) dx = \frac{3}{2\sqrt{\pi}} \frac{\mathbf{1}_{(0, 1/2)}(x)}{[x(1 - x)]^{3/2}} dx, \quad (158)$$

and  $\mu_D([u, 1]) = \Psi(u)$  for  $u > 0$ . The measure  $\mu_D$  has infinite mass near zero and is not a probability measure.

The three summands describe a last entry close to the low end, a last entry in the interior, and a last entry close to the high end. Their equal weights are a theorem of the renewal calculation, not an independence assumption about endpoints. The last assertion is manuscript 11's Theorem 1.2, proved there from the profile (section 42.4); manuscript 10 obtains it as the deficit marginal (section 41.6).

### 38.4 Exact moment constants and the mean-carrying law

**Theorem 38.3** (Every supercritical moment; both manuscripts). *For every fixed real  $p > 1/2$ ,*

$$\mathbb{E}D_n^p \sim m_p n^{p-1/2}, \quad (159)$$

where

$$m_p = \frac{3}{2\sqrt{\pi}} \int_0^{1/2} x^{p-3/2} (1-x)^{-3/2} dx \quad (160)$$

$$= \frac{3}{\sqrt{\pi}} \int_0^1 \left( \frac{z^2}{1+z^2} \right)^{p-1} dz. \quad (161)$$

In particular,

$$\mathbb{E}M_n = n - \frac{3}{\sqrt{\pi}} \sqrt{n} + o(\sqrt{n}), \quad (162)$$

$$\text{Var}(D_n) = \text{Var}(M_n) \sim \frac{3(4-\pi)}{4\sqrt{\pi}} n^{3/2} = \frac{3}{\sqrt{\pi}} \left(1 - \frac{\pi}{4}\right) n^{3/2}. \quad (163)$$

The two printed forms of the variance constant are manuscript 10's and manuscript 11's; they are equal. Both manuscripts state both integral forms of  $m_p$ ; manuscript 10 adds the size-biased law below, which at  $p = 1$  carries the same information as the integral.

Define deficit-size bias by

$$\mathbb{P}_n^\star(B) = \frac{\mathbb{E}[D_n \mathbf{1}_B]}{\mathbb{E}D_n}. \quad (164)$$

This is a probability measure on the same finite permutation space.

**Theorem 38.4** (A uniform law under deficit-size bias; manuscript 10). *Under  $\mathbb{P}_n^\star$ ,*

$$\sqrt{\frac{D_n}{n - D_n}} \Rightarrow \text{Uniform}[0, 1]. \quad (165)$$

Equivalently, the limiting distribution function of  $D_n/n$  under this bias is

$$H^\star(x) = \begin{cases} 0, & x \leq 0, \\ \sqrt{x/(1-x)}, & 0 < x < 1/2, \\ 1, & x \geq 1/2. \end{cases} \quad (166)$$

Thus half the leading mean contribution comes from deficits above  $n/5$ :

$$\frac{\mathbb{E}[D_n; D_n > n/5]}{\mathbb{E}D_n} \longrightarrow \frac{1}{2}. \quad (167)$$

Because  $m_1 = 3/\sqrt{\pi}$ , manuscript 11's moment formula at  $p = 1$  and manuscript 10's size-biased law carry the same information: the size-biased density  $x\rho(x)/m_1$  is the derivative of  $H^\star$ , whose total mass one is the statement  $\mathcal{I}(1) = 1$  of section 44.2 (the merge's comparison).

### 38.5 What makes the constant three

Both manuscripts explain the coefficient 3 in  $\Psi$ , from different decompositions.

*Manuscript 11: outer words.* The combinatorial proof isolates one split into two macroscopic blocks, preceded by a finite outer word of two possible operations. One operation adds a finite block on the left of a maximum; the other appends a new maximum. The total critical weight of

all outer words is 4. However, words containing an append operation are accepted only when the terminal right block has a last entry near its high endpoint. That event has asymptotic probability  $1/2$ . Words without an append operation have total weight 2; words with one or more appends also have total weight 2. The accepted weight is therefore

$$2 + \frac{1}{2} \cdot 2 = 3. \quad (168)$$

Neither ignoring append operations nor treating every append as harmless gives the correct constant. The endpoint condition is not a secondary error term.

*Manuscript 10: a renewal with three endpoint channels.* Integrating the Catalan root-split density alone gives only the source term  $\nu$  of (202) (remark 41.2): small outer decorations recur, and appending a maximum can erase a rare deficit. The resulting renewal equation has the solution (156), whose three equally weighted summands are the three endpoint channels; section 49 separates the folding factor two from the three channels. (On tests of the deficit alone the source is a third of the answer; section 43.4 identifies the three channels with manuscript 11's three endpoint modes. Both observations are the merge's.)

### 38.6 Scope of the advance

The main profile and moment results answer the two explicit repository questions just identified. The marked endpoint limit, the size-biased uniform law, the lower bounds of section 45, and the conditional outer-word and split laws of section 43 are further results. All needed finite facts are rederived. In particular, no asymptotic claim from an unrefereed repository report is assumed in the proofs; the finite-state recurrence is used only for computations and is attributed to its prior source. Manuscript 10's proof is self-contained at the level of finite Catalan decompositions, elementary generating functions, Stirling's formula, and convergence of measures: an exact root identity eliminates the left block's internal jump statistic, and a uniform counting bound controls the otherwise dangerous mesoscopic remainder. Manuscript 11's proof is by exact Catalan counting, endpoint estimates, and a tail bound at the rare-event scale, a different asymptotic decomposition for a bound proportional to  $n$ ; in neither is a fixed-bound spectral asymptotic substituted into a varying-bound problem.

Neither manuscript claims the exact total-variation constant, a constant term in the critical half-moment, a local limit for individual macroscopic lattice masses, or the optimal scale below the half-size cutoff. The positive bulk equivalent concerns adjacency bounds  $m/n \in (1/2, 1)$ ; the smaller-bound bulk regime and the critical window at  $m = n/2$  remain open. Nor does this work resolve the fixed- $m$  component-ordering problem of Mayama–Akita.

## 39 Exact identities at the maximum and at the endpoints

### 39.1 The Catalan maximum decomposition

Both manuscripts start from the maximum decomposition of Part I (section 2), which Part III also rederives (section 23): every  $\pi \in \text{Av}_n(132)$  is uniquely

$$\pi = (\sigma + b) n \tau, \quad \sigma \in \text{Av}_a(132), \quad \tau \in \text{Av}_b(132), \quad a + b + 1 = n,$$

and every such pair produces an avoider. Hence, for the left and right block sizes of  $\Pi_n$  (manuscript 10),

$$\mathbb{P}(\text{left size } a, \text{ right size } b) = \frac{C_a C_b}{C_n}, \quad (169)$$

and, conditional on their sizes, the standardized blocks are independent and uniform. Summing gives  $C = 1 + xC^2$ . If  $\pi$  is  $m$ -bounded and  $b > 0$ , then  $a + 1 \leq m$  (manuscript 11's positional restriction), because the entry after  $n$  is at most  $b$ : this is Part I's (4). Both manuscripts rederive the decomposition and the restriction by the argument of Part I; it is not repeated.

### 39.2 The exact deficit recursion

**Proposition 39.1** (The exact deficit recursion; manuscript 10). *Let  $\pi = (\sigma + b)n\tau$  as above. If  $b \geq 1$ , then*

$$M(\pi) = \max\{n - F(\tau), M(\tau)\}, \quad (170)$$

$$D(\pi) = \min\{F(\tau), a + 1 + D(\tau)\}, \quad (171)$$

$$L(\pi) = L(\tau). \quad (172)$$

*If  $b = 0$  and  $n \geq 2$ , then*

$$D(\pi) = \min\{D(\sigma) + 1, L(\sigma)\}, \quad L(\pi) = n. \quad (173)$$

*For  $a > 0$ , the first-entry deficiency satisfies*

$$J(\pi) = n - F(\pi) = a + 1 - F(\sigma); \quad (174)$$

*for  $a = 0$ , it is zero.*

*Proof.* Suppose  $b \geq 1$ . The downward boundary jump from  $n$  to  $F(\tau)$  is  $n - F(\tau) \geq n - b = a + 1$ . Every jump inside the left block is at most  $a - 1$ , and the jump from the left block into  $n$ , when present, is at most  $a$ . Therefore none of these can be largest. Only the downward boundary and jumps internal to the right block remain, proving (170). Subtracting the maximum from  $n$  gives (171); the last value is unchanged. If  $b = 0$ , appending  $n$  introduces the single new jump  $n - L(\sigma)$ , and subtraction gives (173). Finally,  $F(\pi) = b + F(\sigma)$  when  $a > 0$ , proving (174).  $\square$

This is stronger than an asymptotic simplification. It explains why an arbitrary left decoration contributes only its Catalan count in the later analysis. The last endpoint cannot likewise be discarded: the minimum in (173) is exactly where a putative large deficit can be clipped away. (The recursion is the one-step form of the state transformations (108)–(109) of Part III, which hold there under a first-entry hypothesis on a large hole; (171) and (173) hold for every avoider.)

### 39.3 First entries: a uniformly exponential tail

The first-entry count

$$\#\{\pi \in \text{Av}_N(132) : F(\pi) = f\} = \frac{N - f + 1}{N} \binom{N + f - 2}{N - 1}, \quad 1 \leq f \leq N, \quad (175)$$

is Part III's (98), which is Part I's (21) reindexed (remark 23.2). Manuscripts 10 and 11 rederive it exactly as Part III does: the maximum decomposition gives the bivariate first-value series (95), the substitution  $(x, z) \mapsto (xu, u^{-1})$  gives the deficiency form (96), deficiency  $j$  has generating function  $(xC(x))^{j+1}$ , and Lagrange inversion applied to  $C = 1 + xC^2$  gives  $[x^k]C(x)^q = \frac{q}{2k+q} \binom{2k+q}{k}$ . The new content is a pair of bounds uniform in the size, which remain useful when the event being studied has probability only of order  $n^{-1/2}$ .

**Lemma 39.2** (A uniform first-deficiency bound; both manuscripts). *For  $n \geq 1$  and  $0 \leq j \leq n - 1$ ,*

$$\mathbb{P}(J_n = j) \leq (j + 1)2^{-j-1} \leq (j + 1)2^{-j}, \quad \mathbb{E}J_n \leq 8, \quad (176)$$

*and for every integer  $h \geq 0$ ,*

$$\mathbb{P}(J_n \geq h) \leq (h + 2)2^{-h}. \quad (177)$$

*In particular  $\mathbb{P}(J_n \geq h)$  is bounded by an absolute constant times  $(h + 1)2^{-h}$ , uniformly in  $n$ .*

Manuscript 10 states the weaker bound  $(j + 1)2^{-j}$ ,  $\mathbb{E}J_n \leq 8$  and the last sentence; manuscript 11 states  $(j + 1)2^{-j-1}$  and (177). The proofs are the same computation.

*Proof.* For  $n \geq 2$ , divide the count (175) with  $f = n - j$  by  $C_n$ :

$$\mathbb{P}(J_n = j) = (j+1) \frac{n+1}{2(2n-1)} \prod_{r=0}^{j-1} \frac{n-1-r}{2n-2-r}.$$

Indeed the ratio  $\binom{2n-2}{n-1} / \binom{2n}{n}$  is  $n/(2(2n-1))$ , and lowering the first upper index a further  $j$  times multiplies it by the product. The empty product is one, each product factor is at most  $1/2$ , and  $(n+1)/(2(2n-1)) \leq 1/2$ . Thus  $\mathbb{P}(J_n = j) \leq (j+1)2^{-j-1}$  (manuscript 10 bounds the prefactor  $(j+1)(n+1)/(2(2n-1))$  by  $j+1$  instead). Summing the infinite geometric-arithmetic tail gives  $\sum_{j \geq h} (j+1)2^{-j-1} = (h+2)2^{-h}$ . For  $n = 1$ , the deficiency is zero and every inequality holds directly. Finally  $\sum_{j \geq 0} j(j+1)2^{-j} = 8$ .  $\square$

### 39.4 Last entries: two endpoint modes

The last-entry count

$$\#\{\pi \in \text{Av}_N(132) : L(\pi) = \ell\} = C_{N-\ell} C_{\ell-1}, \quad 1 \leq \ell \leq N, \quad (178)$$

is Part III's (101), from the unique decomposition  $\pi = (\tau + \ell) \sigma \ell$  with  $|\tau| = N - \ell$ ,  $|\sigma| = \ell - 1$  ((100)); both manuscripts rederive it that way, noting that triples crossing the separated blocks cannot form 132.

**Lemma 39.3** (The last entry has two endpoint limits; both manuscripts). *As  $n \rightarrow \infty$ ,*

$$L_n/n \implies \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1. \quad (179)$$

*More precisely, for every fixed  $\eta \in (0, 1/2)$ , uniformly for integer  $r \in [\eta N, (1 - \eta)N]$ ,*

$$\mathbb{P}(L(\Pi_N) \geq r) = \frac{1}{2} + O_\eta(N^{-1/2}), \quad (180)$$

*and the probability that  $L(\Pi_N)/N$  belongs to  $[\eta, 1 - \eta]$  is  $O_\eta(N^{-1/2})$ .*

Manuscript 10 proves (179), manuscript 11 the quantitative form; the two proofs differ.

*Proof (manuscript 10).* For fixed  $j \geq 0$ , Stirling's formula and (178) give

$$\mathbb{P}(L_n = j+1) \longrightarrow C_j 4^{-j-1}, \quad \mathbb{P}(L_n = n-j) \longrightarrow C_j 4^{-j-1}.$$

Each family of limiting masses sums to  $\frac{1}{4}C(1/4) = 1/2$ . For a fixed endpoint cutoff the two families are disjoint for large  $n$ . Letting the cutoff tend to infinity leaves no asymptotic mass in the middle. This proves (179).  $\square$

*Second proof, with the uniform estimate (manuscript 11).* The distribution (178) is symmetric under  $\ell \mapsto N+1-\ell$ . For  $\eta N \leq \ell \leq (1-\eta)N$ , the Catalan bounds (181) below bound each probability by  $K_\eta N^{-3/2}$ . Summing at most  $N$  terms gives the middle-mass bound. Symmetry puts one half of the remaining mass at each end, proving (180) uniformly in the stated range.  $\square$

In particular, the first deficiency has an exponentially bounded tail, whereas the last entry converges after division by its size to an equal mixture of the endpoints zero and one. No independence between the two endpoints, or between  $D_n$  and  $L_n$ , has been asserted or will be needed. Later, the fact that  $D_n/n \rightarrow 0$  in probability will make its macroscopic value deterministic inside a large unconditioned block, which is a different argument.

### 39.5 Uniform Catalan estimates

Manuscript 11 records the following bounds. Stirling's formula and adjustment of finitely many small indices give absolute constants  $c_-, c_+ > 0$  such that

$$c_-(j+1)^{-3/2} \leq c_j \leq c_+(j+1)^{-3/2}, \quad c_j = \frac{1 + O(j^{-1})}{\sqrt{\pi} j^{3/2}} \quad (j \rightarrow \infty). \quad (181)$$

(Manuscript 10 proves  $c_h \leq (h+1)^{-3/2}$  directly, in the proof of lemma 40.2.) All constants denoted by  $K$  are absolute unless a subscript specifies otherwise; they can be enlarged from line to line. For  $N \geq n/2$ , (181) implies

$$\frac{C_N}{C_n} \leq K 4^{N-n}. \quad (182)$$

These inequalities are used as bounds, and the asymptotic formula only when all involved block sizes tend to infinity uniformly. The numbers  $c_j$  decrease strictly ((112)), so  $C_{n-t}/C_n \geq 4^{-t}$  for  $0 \leq t < n$ .

## 40 Tail envelopes at the rare-event scale

The proof of a rare-event limit needs an upper bound that survives multiplication by  $\sqrt{n}$ . Ordinary weak convergence of  $D_n/n$  is not enough. Both manuscripts obtain such a bound from finite counting, without using Part III's algebraic boundary law, by two different decompositions.

### 40.1 Stopping words and a deterministic deficit bound

Manuscript 10 uses Part III's three reconstruction operations  $R_\beta$ ,  $A$  and  $T_\tau$  (section 24), with costs  $|\beta| + 1$ , 1 and  $|\tau| + 1$ . It calls Part III's skeletons *stopping words*: a word in  $R_\beta, A$  followed by one  $T_\tau$ , applied from the terminal operation outward. It has total cost  $t$  and one unexpanded hole; filling that hole with a size- $N$  avoider produces size  $N + t$ .

**Lemma 40.1** (Disjoint families and a deterministic deficit bound). *Fix  $n$  and  $h < n/2$ . Reconstructions from different stopping words of cost  $t \leq h$  are disjoint, and the reconstruction map is injective in the size- $(n - t)$  hole. Every such reconstruction satisfies*

$$D(\pi) \leq t - 1. \quad (183)$$

*Proof.* Disjointness and injectivity are Part III's lemma 25.2; manuscript 10 rederives them by the same argument (the block containing the hole is larger than its finite sibling at every maximum decomposition, so the word and the hole are read off from the final permutation).

For (183): at the terminal, the jump from the maximum to the first value of  $\tau$  has size at least  $N + 1$ . Every outer operation preserves all existing internal differences, so the final maximum jump is still at least  $N + 1$ . Thus  $D \leq (N + t) - (N + 1) = t - 1$ . Notably, no first-entry condition on the hole is needed for this one-sided assertion, unlike the exact identity of Part III's lemma 25.1.  $\square$

### 40.2 An elementary bound for the exact cost tail

The stopping words of cost  $t$  are counted by Part III's  $b_t$ , with generating function  $B(x) = (C(x) - 1)^2$  ((103)): the numerator of  $x(C - 1)/(1 - xC - x)$  counts terminals and the denominator an arbitrary prefix. In particular  $B(1/4) = 1$ , so the weights  $b_t 4^{-t}$  form the probability law of Part III's cost  $T$ ; and  $b_t = C_{t+1} - 2C_t$  for  $t \geq 1$ ,  $b_0 = 0$  ((104)). Manuscript 10 rederives all of this.

**Lemma 40.2** (Exact tail and an elementary bound). *For every integer  $h \geq 0$ ,*

$$\mathbb{P}(T > h) = \frac{2(h+1)(2h+1)}{h+2} \frac{C_h}{4^h} \leq \frac{4}{\sqrt{h+1}}. \quad (184)$$

*Proof.* The equality is Part III's lemma 24.1, proved there, and in manuscript 10, from  $c_{j+1}/c_j = (2j+1)/(2j+4)$ ,  $\sum_{j=0}^h c_j = 2 - (2h+1)c_h$  and  $b_t 4^{-t} = 4c_{t+1} - 2c_t$  for  $t \geq 1$ . For the bound (manuscript 10),  $c_h \leq (h+1)^{-3/2}$  by induction, because

$$(2h+1)^2(h+2) \leq 4(h+1)^3.$$

The prefactor in (184) is at most  $4(h+1)$ , proving the bound.  $\square$

### 40.3 The uniform envelope

**Proposition 40.3** (Uniform finite-size deficit envelope; both manuscripts). *For  $1 \leq d \leq n$ , put  $h = \min\{d, \lfloor (n-1)/2 \rfloor\}$ . Then*

$$\mathbb{P}(D_n \geq d) \leq \mathbb{P}(T > h) \tag{185}$$

$$\leq \min \left\{ 1, \frac{4\sqrt{2}}{\sqrt{d}} \right\}. \tag{186}$$

Consequently  $D_n/n \rightarrow 0$  in probability and, for every  $0 < u \leq 1$ ,

$$\sqrt{n} \mathbb{P}(D_n > un) \leq 4\sqrt{2} u^{-1/2}. \tag{187}$$

Manuscript 10 proves the explicit form given here. Manuscript 11 proves  $\mathbb{P}(D_n \geq d) \leq K/\sqrt{d}$  for  $n \geq 2$ ,  $1 \leq d \leq n-1$ , with an unspecified absolute constant (corollary 40.8), by a different route.

*Proof.* By lemma 40.1, the disjoint short-word families have total probability  $\sum_{t \leq h} b_t C_{n-t}/C_n$  and all satisfy  $D_n < d$ . Since  $c_j$  decreases,  $C_{n-t}/C_n = 4^{-t} c_{n-t}/c_n \geq 4^{-t}$ . Their mass is therefore at least  $\mathbb{P}(T \leq h)$ , proving the first inequality. Because  $h+1 \geq d/2$ , lemma 40.2 proves the second. Use  $d = \lfloor un \rfloor + 1 > un$  to obtain the scaled bound, with the zero event handled separately when  $u = 1$ .  $\square$

The same stopping construction occurs in Part III's boundary-law proof. Its use here is deliberately weaker and cleaner: the preserved terminal jump supplies a tail bound directly, without reconstructing the exact limiting deficit or controlling an exceptional set of hole endpoints.

### 40.4 A canonical threshold parser

Manuscript 11 decomposes a bounded permutation differently. Its outer words use only the operations  $R_\beta$  (with  $\beta$  an arbitrary avoider, including  $R_\emptyset$  at cost one) and  $A$ , with Part III's costs; an *outer word* is a finite word in these decorated operations, including the empty word, written outermost operation first, so that reconstruction applies it from right to left. Its cost is the sum of operation costs, and every operation preserves avoidance.

Fix integers  $n, m$  with

$$n/2 \leq m \leq n-1, \quad d = n - m. \tag{188}$$

Let  $\mathcal{E}_{n,m}$  be the event  $M_n \leq m$ . For an  $m$ -bounded permutation, start at its maximum decomposition. As long as a child block has more than  $m$  entries, follow that block. A large right child records an  $R_\beta$  operation. A large left child must have an empty right sibling, by the positional restriction, and records an  $A$  operation. Stop at the first node whose two child blocks both have size at most  $m$ .

**Lemma 40.4** (Canonical parsing and reconstruction; manuscript 11). *Every permutation in  $\mathcal{E}_{n,m}$  has a unique representation*

$$w((\sigma + b) N \tau), \quad a = |\sigma|, \quad b = |\tau|, \tag{189}$$

where  $w$  is an outer word of cost  $t$ , and

$$0 \leq t \leq d-1, \quad N = n - t > m, \quad a + b = N - 1, \quad 0 \leq a, b \leq m. \quad (190)$$

The word, all its decorations, and the terminal pair are recoverable from the permutation. For any fixed word of cost  $t \leq d-1$ , reconstruction from terminal pairs satisfying (190) is injective. Whenever the reconstructed permutation is  $m$ -bounded, its canonical word is exactly the prescribed word.

*Proof.* At every stage the current block has more than  $m$  entries. Under (188), two children of more than  $m$  entries are impossible by their total size; for bounded permutations the positional restriction also rules out a nonempty right child next to a left child of more than  $m$  entries. The stated rule therefore gives a unique next operation whenever parsing continues. The followed size strictly decreases. At size  $m+1$  both children have at most  $m$  entries, so parsing has stopped by then. This proves  $N \geq m+1$  and  $t \leq d-1$ .

Each maximum decomposition recovers its standardized children, so it also recovers every decoration and the terminal pair. Conversely, the core in (189) and every intermediate core have more than  $m$  entries. An  $R$  decoration has size at most  $t-1 \leq d-2 < m$ , while an  $A$  operation has no right sibling. Hence the same parsing rule follows the reconstructed core at each operation and stops exactly at the specified terminal pair. This proves injectivity and the final assertion.  $\square$

*Remark 40.5* (The terminal pair is not a boundary-law skeleton). The absorbing skeleton of Part III (manuscript 10's stopping word) stops at a finite right block next to a giant left block. The threshold parser stops when *both* blocks lie below the adjacency threshold. Under a macroscopic deficit, these terminal blocks are themselves macroscopic. The two parsers serve different limits and must not be identified.

Let  $w_t$  count all individually decorated outer words of cost  $t$ , and let  $w_t^R$  count those containing no  $A$  operation. The generating function of one  $R$  operation is  $zC(z)$ , including  $R_\emptyset$  at cost one. Therefore

$$\mathcal{W}(z) := \sum_{t \geq 0} w_t z^t = \frac{1}{1 - zC(z) - z} = C(z)^2, \quad (191)$$

$$\mathcal{W}_R(z) := \sum_{t \geq 0} w_t^R z^t = \frac{1}{1 - zC(z)} = C(z). \quad (192)$$

Both identities follow from  $zC^2 = C - 1$ . In particular,

$$w_t = C_{t+1}, \quad w_t^R = C_t. \quad (193)$$

At  $z = 1/4$  these series converge, with  $\mathcal{W}(1/4) = 4$  and  $\mathcal{W}_R(1/4) = 2$ . This critical summability will allow the finite outer word to be summed without a spectral approximation. (Part III's cost series factors as  $B(z) = \mathcal{W}(z) \cdot z(C(z) - 1)$ : an outer word followed by a terminal  $T_\tau$ , so the two parsers share their outer operations. This remark is the merge's.)

#### 40.5 Tightness of the canonical cost

The main technical point is not merely that the outer word is usually short under some limiting construction. Its tail must be negligible *relative to a probability of order  $n^{-1/2}$* . The following exact majorant supplies that missing uniformity. Let

$$Q_{N,m} = \sum_{\substack{a+b=N-1 \\ 0 \leq a, b \leq m}} C_a C_b \quad (194)$$

count terminal pairs before checking the boundary jumps.

**Lemma 40.6** (Terminal split bound; manuscript 11). *For  $m < N \leq 2m + 1$ , with  $r = N - m \geq 1$ ,*

$$Q_{N,m} \leq \frac{KC_N}{\sqrt{r}}. \quad (195)$$

*Proof.* For  $r = 1$ , the full Catalan recurrence gives  $Q_{N,m} \leq C_N$ . For  $r \geq 2$ , every admissible pair has  $a, b \geq r - 1$ . By symmetry we may sum the half with  $a \leq b$  and multiply by two. On that half,  $b + 1 \geq N/2$ , and the Catalan bounds give

$$\frac{C_a C_b}{C_N} \leq \frac{K}{(a + 1)^{3/2}}.$$

The sum over  $a \geq r - 1$  is at most  $K/\sqrt{r}$ , by comparison with the integral of  $u^{-3/2}$ . This proves (195).  $\square$

On  $\mathcal{E}_{n,m}$ , write  $T_{n,m}$  for the cost of the canonical outer word. It need not be defined outside that event.

**Lemma 40.7** (Uniform tail of the canonical cost; manuscript 11). *Under (188), for every integer  $h \geq 0$ ,*

$$\mathbb{P}(\mathcal{E}_{n,m}, T_{n,m} > h) \leq K \left\{ \frac{1}{\sqrt{d}\sqrt{h+1}} + \frac{1}{d} \right\}. \quad (196)$$

Moreover,

$$\mathbb{P}(\mathcal{E}_{n,m}) \leq \frac{K}{\sqrt{d}}. \quad (197)$$

*Proof.* Discard every boundary constraint after applying the injective parser. For each cost  $t$ , there are  $w_t$  words and at most  $Q_{n-t,m}$  terminal pairs per word. Since  $n - t \geq m + 1 \geq n/2$ , lemma 40.6, (182) and (193) give

$$\begin{aligned} \mathbb{P}(\mathcal{E}_{n,m}, T_{n,m} > h) &\leq \frac{1}{C_n} \sum_{h < t \leq d-1} C_{t+1} Q_{n-t,m} \\ &\leq K \sum_{h < t \leq d-1} \frac{1}{(t+1)^{3/2} \sqrt{d-t}}. \end{aligned} \quad (198)$$

For  $t \leq d/2$ , the sum is at most  $Kd^{-1/2} \sum_{t > h} (t+1)^{-3/2}$ , hence at most  $K/\sqrt{d(h+1)}$ . For  $t > d/2$ , put  $r = d - t$ . This part is at most

$$Kd^{-3/2} \sum_{1 \leq r \leq d/2} r^{-1/2} \leq Kd^{-1}.$$

These bounds prove (196). Starting the same sum at  $t = 0$  proves (197), after enlarging  $K$  for  $d = 1$ .  $\square$

**Corollary 40.8** (An all-deficit envelope; manuscript 11). *For every  $n \geq 2$  and  $1 \leq d \leq n - 1$ ,*

$$\mathbb{P}(D_n \geq d) \leq \frac{K}{\sqrt{d}}. \quad (199)$$

*In particular,  $D_n/n \rightarrow 0$  in probability.*

*Proof, and second proof of proposition 40.3 up to the constant.* For  $d \leq n/2$ , apply (197) with  $m = n - d$ . For  $d > n/2$ , use monotonicity and the bound at  $\lfloor n/2 \rfloor$ . Since  $d \leq n$ , the resulting  $K/\sqrt{n}$  is at most a constant multiple of  $d^{-1/2}$ . Enlarge  $K$  to cover the small values of  $n$ .  $\square$

*Remark 40.9* (The order of truncations). If  $d_n/n \rightarrow u > 0$ , (196) yields

$$\limsup_{n \rightarrow \infty} \sqrt{n} \mathbb{P}(\mathcal{E}_{n,n-d_n}, T_{n,n-d_n} > h) \leq K_u(h+1)^{-1/2}.$$

Thus one first fixes a finite cost cutoff, takes  $n \rightarrow \infty$ , and only then lets the cutoff grow. Ordinary unscaled tightness would not justify this interchange. This estimate is the reason the finite-word calculation of section 42 determines the complete leading constant.

## 41 First proof of the profile: a renewal equation for the marked measure

This section and the next are manuscript 10's proof of theorem 38.2, and with it of theorem 38.1.

### 41.1 Test functions and the append projection

It is enough initially to consider a Lipschitz function  $f$  on  $[0, 1]^2$  that is zero whenever the deficit coordinate satisfies  $x \leq \delta$ , for some  $\delta > 0$ . Set

$$Z_n(f) = \sqrt{n} \mathbb{E} f(D_n/n, L_n/n). \quad (200)$$

By (187), this sequence is bounded, with a bound depending only on  $\delta$  and  $\|f\|_\infty$ . Define

$$S(x, \ell) = (\min(x, \ell), 1), \quad S^* f = f \circ S. \quad (201)$$

Then  $S^2 = S$ , hence  $(S^*)^2 = S^*$ . Also  $S^* f$  is Lipschitz and vanishes on the same low-deficit strip.

The macroscopic source functional is

$$v(f) = \frac{1}{8\sqrt{\pi}} \int_0^1 \frac{f(v(y), 0) + f(v(y), 1-y)}{[y(1-y)]^{3/2}} dy. \quad (202)$$

Its apparent endpoint singularities cause no problem: the numerator is zero when  $v(y) \leq \delta$ .

**Lemma 41.1** (Truncated renewal with a controlled remainder). *Put  $r_a = C_a 4^{-a-1}$ . For each admissible Lipschitz  $f$  there is a finite constant  $K_f$  such that, for every sufficiently large integer  $H$ ,*

$$\limsup_{n \rightarrow \infty} \left| Z_n(f) - \sum_{a=0}^H r_a Z_{n-a-1}(f) - \frac{1}{4} Z_{n-1}(S^* f) - v(f) \right| \leq K_f (H+1)^{-1/2}. \quad (203)$$

### 41.2 Proof of the truncated renewal

*Proof of lemma 41.1: small left blocks, the identity transition.* Fix  $H$  and use the exact root decomposition. If  $0 \leq a \leq H$  and  $b = n - a - 1$ , proposition 39.1 gives

$$D_n = \min\{b - J_b, a + 1 + D_b\}, \quad L_n = L_b$$

inside the conditional expectation. The distance between this normalized state and  $(D_b/b, L_b/b)$  is bounded by an absolute constant times  $(a + 1 + J_b)/n$ . By lemma 39.2, its expected value is  $O_H(n^{-1})$ . After multiplying by  $\sqrt{n}$ , the Lipschitz error tends to zero. Moreover

$$\frac{C_a C_b}{C_n} \sqrt{\frac{n}{b}} \longrightarrow C_a 4^{-a-1} = r_a.$$

Boundedness of  $Z_b(f)$  therefore gives the small-left contribution  $\sum_{a=0}^H r_a Z_{n-a-1}(f) + o(1)$ .

*An empty right block: the append transition.* For  $b = 0$ , the exact state differs from  $S(D_{n-1}/(n-1), L_{n-1}/(n-1))$  by  $O(n^{-1})$ . The coefficient  $(C_{n-1}/C_n)\sqrt{n/(n-1)}$  tends to  $1/4$ . Thus this contribution is  $\frac{1}{4}Z_{n-1}(S^*f) + o(1)$ . If  $1 \leq b \leq H$ , then  $D_n \leq F_b \leq H$  exactly, so its contribution is zero for all sufficiently large  $n$ .

*Two macroscopic blocks: the source term.* Choose a fixed  $0 < \eta < \delta/4$  and consider  $\eta n \leq a, b \leq (1 - \eta)n$ . Set  $y = a/n$ . The exact recursion and the endpoint lemmas give, uniformly in these sizes,

$$D_n/n - \min(a/n, b/n) \rightarrow 0 \quad \text{in probability,}$$

and the last coordinate is asymptotically 0 or  $b/n$ , each with probability  $1/2$ . Indeed  $J_b/b \rightarrow 0$  and  $D_b/b \rightarrow 0$  uniformly for  $b \geq \eta n$ , while  $L_b/b$  has the two-point limit of lemma 39.3. Because the other coordinates have deterministic limiting values, no endpoint independence is required. The uniformity follows directly from convergence of the corresponding scalar sequences as their integer index tends to infinity, and uniform continuity of  $f$  on a compact square.

Stirling's formula gives, uniformly on the same interval,

$$\frac{C_a C_b}{C_n} = \frac{1 + o(1)}{4\sqrt{\pi} n^{3/2}} [y(1-y)]^{-3/2}. \quad (204)$$

The replacement of  $b/n = 1 - y - 1/n$  by  $1 - y$  has vanishing effect. A Riemann sum therefore gives the source integral over  $[\eta, 1 - \eta]$ , which is exactly (202), since its integrand vanishes outside  $[\delta, 1 - \delta]$ .

*Mesoscopic left blocks: the indispensable error bound.* It remains to treat  $H < a < \eta n$ ; the case  $b < \eta n$  contributes zero because  $D_n \leq b < \delta n$ . On the support of  $f$ , the root identity forces

$$a + 1 + D_b > \delta n, \quad \text{hence} \quad D_b > \delta n/2$$

for large  $n$ . By proposition 40.3, its conditional probability is  $O_\delta(n^{-1/2})$ . In this range  $b$  is a fixed positive fraction of  $n$ , so the Catalan ratio also gives

$$\frac{C_a C_b}{C_n} \leq K r_a$$

for an absolute constant  $K$  depending at most on  $\eta$ . After multiplication by  $\sqrt{n}$ , the entire mesoscopic contribution is bounded by

$$K'_{f,\delta} \sum_{a>H} r_a = O_{f,\delta}((H+1)^{-1/2}).$$

Here the summable tail follows either from Stirling or from  $c_a \leq (a+1)^{-3/2}$  proved earlier. Combining all cases proves (203).  $\square$

*Remark 41.2* (Why a formal split heuristic is insufficient). Simply integrating the Catalan root-split density would give the source  $\nu$ , not the answer  $\mu$ . Small outer decorations recur, and appending a maximum can erase a rare deficit. Moreover, without the uniform tail envelope, the range  $H < a = o(n)$  cannot be discarded after multiplication by  $\sqrt{n}$ . The proof handles both issues explicitly.

### 41.3 Solving the renewal equation without assuming convergence

The Catalan identity gives

$$\sum_{a \geq 0} r_a = \frac{1}{4} C(1/4) = \frac{1}{2}. \quad (205)$$

Formally, a limit should satisfy  $\mu(f) = \frac{1}{2}\mu(f) + \frac{1}{4}\mu(S^*f) + \nu(f)$ . We now prove the limit exists rather than inserting this equation as an assumption.

*Proof of theorem 38.2 for Lipschitz tests.* First put  $g = S^*f$ , so that  $S^*g = g$ , and let  $\bar{Z}_g = 4\nu(g)$ . All  $Z_n(g)$  are bounded. Write  $\bar{r}_H = \sum_{a=0}^H r_a$  and  $e_g = \limsup_n |Z_n(g) - \bar{Z}_g|$ . Applying lemma 41.1 to  $g$  and subtracting  $\bar{Z}_g$  yields

$$e_g \leq (\bar{r}_H + \frac{1}{4})e_g + K_g(H+1)^{-1/2} + (\frac{1}{2} - \bar{r}_H)|\bar{Z}_g|.$$

Let  $H \rightarrow \infty$ . By (205),  $e_g \leq \frac{3}{4}e_g$ . Since  $e_g$  is finite, it is zero. Thus

$$Z_n(S^*f) \rightarrow 4\nu(S^*f). \quad (206)$$

Next set  $\bar{Z}_f = 2\nu(f) + 2\nu(S^*f)$  and  $e_f = \limsup_n |Z_n(f) - \bar{Z}_f|$ . The already established convergence of the projected sequence gives

$$e_f \leq \bar{r}_H e_f + K_f(H+1)^{-1/2} + (\frac{1}{2} - \bar{r}_H)|\bar{Z}_f|.$$

Letting  $H \rightarrow \infty$  now gives  $e_f \leq \frac{1}{2}e_f$ , hence  $e_f = 0$ . Therefore

$$\mu(f) = 2\nu(f) + 2\nu(S^*f). \quad (207)$$

Since  $f$  vanishes at deficit zero and  $v(y) \leq 1 - y$ ,

$$S^*f(v(y), 0) = f(0, 1) = 0, \quad S^*f(v(y), 1 - y) = f(v(y), 1).$$

Substituting into (202) proves precisely (156).  $\square$

#### 41.4 Extension to continuous tests

An admissible continuous  $f$  can be approximated uniformly by Lipschitz functions supported away from  $x = 0$ , using piecewise-linear approximation on a rectangular mesh and a Lipschitz cutoff in the first coordinate. The scaled finite measures have uniformly bounded mass on every strip  $x \geq \delta$  by (187), and the limiting integral has finite mass there. Consequently the approximation error in both integrals is bounded by a constant times the uniform approximation error. Letting that error tend to zero proves the full theorem.

#### 41.5 A reusable two-state resolvent

The calculation is an elementary renewal principle. If the total identity weight is  $w_{\text{id}}$ , the weight of an idempotent endpoint operator  $S^*$  is  $w_S$ , and  $w_{\text{id}} + w_S < 1$ , the analogous equation has the solution

$$\mu(f) = \frac{\nu(f)}{1 - w_{\text{id}}} + \frac{w_S}{(1 - w_{\text{id}})(1 - w_{\text{id}} - w_S)} \nu(S^*f). \quad (208)$$

The same bounded-sequence limsup proof establishes convergence whenever an analogue of lemma 41.1 is available. Here  $w_{\text{id}} = 1/2$  and  $w_S = 1/4$ , so both coefficients in (207) equal 2. This algebraic principle is not itself advertised as a new general theorem; the work is in proving the appropriate source and uniform remainder for this statistic.

#### 41.6 Evaluating the profile

For a test depending only on the deficit, all three summands in (156) agree. Folding  $y$  and  $1 - y$  onto  $(0, 1/2)$  gives the density (157). Thus

$$\mu\{x > u\} = \frac{3}{2\sqrt{\pi}} \int_u^{1/2} [x(1-x)]^{-3/2} dx \quad (0 < u < 1/2). \quad (209)$$

The derivative identity

$$\frac{d}{dx} \frac{1-2x}{\sqrt{x(1-x)}} = -\frac{1}{2[x(1-x)]^{3/2}} \quad (210)$$

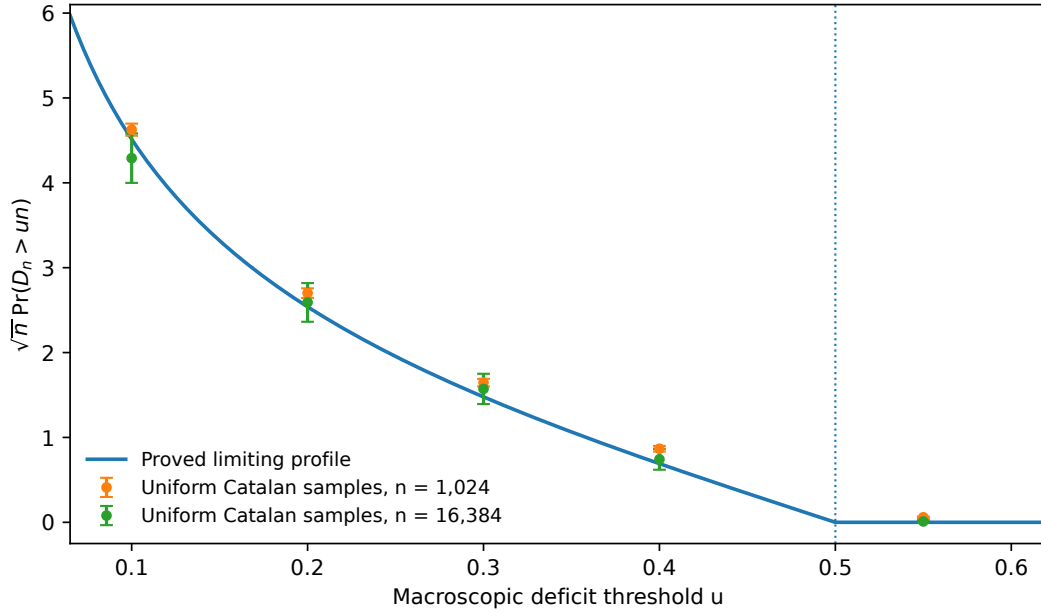


Figure 7: Manuscript 10: the proved macroscopic tail profile  $\Psi$  and uniform Catalan sampling illustrations at  $n = 1,024$  and  $16,384$ , from the shipped simulation data of table 16. Error bars show two estimated Monte Carlo standard errors; they do not bound finite-size bias and are not rigorous confidence certificates. In particular, positive finite-size tails beyond  $u = 1/2$  are compatible with the limiting zero there.

evaluates the integral as (150). The limiting measure has no mass at any fixed positive deficit, including  $1/2$ . Continuous upper and lower approximations to the tail indicator therefore prove (151).

For uniformity on  $[\delta, 1]$ , both the finite tails and  $\Psi$  are nonincreasing, and  $\Psi$  is continuous. Choose a finite mesh on which its oscillation is small, use pointwise convergence at the mesh points, and sandwich between adjacent points. This proves the uniform assertion in theorem 38.1. Finally (82), continuity of  $\Psi$ , and  $C_n \sim 4^n / (\sqrt{\pi} n^{3/2})$  yield the moving-bound formulas (153) and (154), including their floor conventions. (The sequence form (152) follows from (151) in the same way, by monotonicity and continuity of  $\Psi$ ; manuscript 11 proves it directly.)

## 42 Second proof of the profile: the canonical threshold decomposition

This section is manuscript 11's proof of theorem 38.1 and of the deficit part of theorem 38.2. It uses the parser of section 40.4 instead of the renewal equation.

### 42.1 The asymptotic contribution of one outer word

Fix a sequence  $d_n/n \rightarrow u \in (0, 1/2)$  and put  $m_n = n - d_n$ . In this subsection  $w$  is one fixed, fully decorated outer word, of fixed cost  $t$ . Define its endpoint acceptance weight by

$$\chi(w) = \begin{cases} 1, & w \text{ contains no } A, \\ 1/2, & w \text{ contains at least one } A. \end{cases} \quad (211)$$

Let  $\mathcal{E}_n(w)$  be the event that  $M_n \leq m_n$  and the canonical word is  $w$ . On this event, let  $a$  be the left size at the terminal split and define  $U_n = a/n$ . Write

$$\mathfrak{f}(y) = [y(1-y)]^{-3/2}, \quad I(u) = \int_u^{1-u} \mathfrak{f}(y) dy. \quad (212)$$

**Proposition 42.1** (Fixed-word contribution; manuscript 11). *For every continuous function  $g$  on  $[0, 1]$ ,*

$$\sqrt{n} \mathbb{E}[\mathbf{1}_{\mathcal{E}_n(w)} g(U_n)] \longrightarrow \frac{\chi(w)4^{-t}}{4\sqrt{\pi}} \int_u^{1-u} g(y) \mathfrak{k}(y) dy. \quad (213)$$

*The random variable inside the expectation is set to zero off  $\mathcal{E}_n(w)$ .*

*Proof of proposition 42.1: the endpoint step.* The terminal core has size  $N = n - t$  and is  $(\sigma + b)N\tau$ , with  $a + b = N - 1$  and  $a, b \leq m_n$ . Fix a small  $\eta > 0$  such that  $[u + \eta, 1 - u - \eta]$  is nonempty, and first consider splits with

$$u + \eta \leq a/n \leq 1 - u - \eta. \quad (214)$$

For large  $n$ , both  $a$  and  $b$  are bounded below by a positive multiple of  $n$ , and each lies below  $m_n$  by a positive multiple of  $n$ .

All internal jumps in the two terminal blocks are at most  $m_n$  simply because their sizes are at most  $m_n$ . The jump from the left block into  $N$  is at most  $a$ . The jump from  $N$  into the right block equals

$$N - F(\tau) = a + 1 + J(\tau). \quad (215)$$

By lemma 39.2, the fraction of terminal pairs for which  $J(\sigma)$  or  $J(\tau)$  exceeds a sufficiently small fixed multiple of  $n$  is exponentially small, uniformly over (214). On the complementary set, (215) is at most  $m_n$ .

The first deficiency of the terminal core is  $J(\sigma) + 1$ . An append operation increases a current first deficiency by one. An  $R_\beta$  operation either makes the first entry the new maximum, when  $\beta$  is empty, or makes its new first deficiency at most the operation cost, when  $\beta$  is nonempty. Therefore, throughout the fixed outer word, the current first deficiency is at most  $J(\sigma) + t + 1$ . Each new downward jump at an  $R$  operation is at most  $J(\sigma) + 2t + 1$ . Each finite decoration and its jump into the adjacent maximum have bounded size. These contributions are harmless on the same good set for large  $n$ .

If  $w$  contains no  $A$ , all constraints are now satisfied. Thus the accepted fraction of the  $C_a C_b$  terminal pairs is  $1 - o(1)$ , uniformly on (214).

Suppose instead that  $w$  contains an  $A$ . Apply the word from the inside out and let  $s_w$  be the total cost of the  $R$  operations encountered before the first such  $A$ . Those operations preserve the terminal last entry  $L(\tau)$ . The first append creates a jump bounded by  $m_n$  exactly when

$$L(\tau) \geq \ell_n(w) := N + s_w + 1 - m_n = d_n - t + s_w + 1. \quad (216)$$

After this append, the last entry is a maximum at least  $N$ . Subsequent  $R$  operations preserve that value, and every later append creates a jump of size at most  $t$ . Thus no further macroscopic last-entry condition is introduced.

In the interior range,  $\ell_n(w)/n \rightarrow u$ , while both  $\ell_n(w)$  and  $b - \ell_n(w)$  are bounded below by positive multiples of  $n$ . By lemma 39.3, the fraction satisfying (216) is  $1/2 + O_{u,\eta,w}(n^{-1/2})$ . Removing the exponentially small bad first-deficiency set changes this by  $o(1)$ . We have not assumed that the first and last entries are independent: deleting any event of probability  $o(1)$  changes the last-entry probability by at most  $o(1)$ .

We have proved that, uniformly over (214), the accepted fraction of terminal pairs is

$$\chi(w) + o(1). \quad (217)$$

By lemma 40.4 these are exactly the pairs counted by  $\mathcal{E}_n(w)$ , rather than merely a possibly overlapping construction.  $\square$

*Proof of proposition 42.1: counting and boundary strips.* For fixed  $t$  and  $a + b = n - t - 1$ , Stirling's formula gives, uniformly when  $a/n$  stays in a compact subset of  $(0, 1)$ ,

$$\frac{C_a C_b}{C_n} = \frac{4^{-(t+1)}}{\sqrt{\pi} n^{3/2}} \{f(a/n) + o(1)\}. \quad (218)$$

The exponent  $t + 1$  is important: one unit is spent at the terminal maximum in addition to the  $t$  units of outer cost. Combining (217) with (218), summing the interior splits, and using the Riemann-sum mesh  $1/n$ , gives the asserted limit with integral restricted to  $[u + \eta, 1 - u - \eta]$ .

All raw terminal splits have

$$d_n - t - 1 \leq a \leq m_n.$$

Thus their scaled range tends to  $[u, 1 - u]$ . In the two excluded strips, the number of terms is at most  $2\eta n + o(n)$ ; every normalized count is at most  $K_{u,w} n^{-3/2}$  because both block sizes remain macroscopic. The contribution of the strips after multiplication by  $\sqrt{n}$  is at most  $K_{u,w} \|g\|_\infty \eta + o(1)$ . Let  $\eta \downarrow 0$ . This proves (213), including the moving integer boundaries.  $\square$

*Remark 42.2* (Three distinct uniformities). The argument uses an exponentially uniform first-deficiency bound, a uniform half-mass estimate for the last entry away from both endpoints, and a Catalan asymptotic uniform on compact split intervals. The subsequent sum over words uses the separate rare-scale tail estimate of lemma 40.7. None of these four tasks can be replaced by a pointwise “typical block” statement.

## 42.2 Summing the endpoint-weighted words

*Second proof of theorem 38.1, for  $0 < u < 1/2$ .* Fix a cost cutoff  $h$ . There are finitely many decorated words of cost at most  $h$ . Apply proposition 42.1 with  $g = 1$  and sum over them:

$$\sqrt{n} \mathbb{P}(\mathcal{E}_{n,m_n}, T_{n,m_n} \leq h) \longrightarrow \frac{I(u)}{4\sqrt{\pi}} \sum_{\text{cost}(w) \leq h} \chi(w) 4^{-\text{cost}(w)}.$$

By lemma 40.7, the limsup of the omitted scaled probability is at most  $K_u(h + 1)^{-1/2}$ . It tends to zero as  $h \rightarrow \infty$ . The total weighted mass is, by (191)–(192),

$$\begin{aligned} \sum_w \chi(w) 4^{-\text{cost}(w)} &= \mathcal{W}_R(1/4) + \frac{1}{2} \{\mathcal{W}(1/4) - \mathcal{W}_R(1/4)\} \\ &= 2 + \frac{1}{2}(4 - 2) = 3. \end{aligned} \quad (219)$$

Thus the limit is  $3I(u)/(4\sqrt{\pi})$ . To evaluate  $I(u)$ , use (210): the expression being differentiated changes sign under  $y \mapsto 1 - y$ . Consequently,

$$I(u) = \frac{4(1 - 2u)}{\sqrt{u(1 - u)}}, \quad (220)$$

and the resulting limit is exactly (150).  $\square$

## 42.3 The edge at one half and compact uniformity

*Completion of the second proof of theorem 38.1.* Let  $d_n/n \rightarrow u \geq 1/2$ . For any fixed  $y < 1/2$ , eventually  $d_n \geq \lceil yn \rceil$ . Therefore

$$\limsup_{n \rightarrow \infty} \sqrt{n} \mathbb{P}(D_n \geq d_n) \leq \Psi(y).$$

Letting  $y \uparrow 1/2$  gives zero because  $\Psi(y) \rightarrow 0$ . This proves the pointwise assertion at and above one half.

For compact uniformity, fix  $\delta > 0$  and consider the nonincreasing functions  $f_n(u) = \sqrt{n} \mathbb{P}(D_n \geq \lfloor nu \rfloor)$  on  $[\delta, 1]$ . They converge pointwise to the continuous nonincreasing function  $\Psi$ . Given  $\varepsilon > 0$ , choose a finite partition of  $[\delta, 1]$  on which the oscillation of  $\Psi$  in each interval is at most  $\varepsilon$ . Pointwise convergence holds at every partition point, simultaneously for sufficiently large  $n$ . Monotonicity sandwiches  $f_n$  between its endpoint values on each interval, so the uniform error is at most  $3\varepsilon$ . This proves the claim. Substitution  $d_n = n - \lfloor \theta n \rfloor$  in (82) gives (153) and the stated smaller-bound estimate.  $\square$

#### 42.4 The deficit measure

*Second proof of the last assertion of theorem 38.2.* For  $0 < x_1 < x_2 \leq 1$ , the mass of  $[x_1, x_2]$  is the difference of the two tail probabilities, with the corresponding integer rounding. Hence theorem 38.1 gives convergence of these interval masses to  $\Psi(x_1) - \Psi(x_2)$ . On every  $[x_1, 1]$ , the total masses are uniformly bounded by corollary 40.8. Approximating continuous functions there by interval step functions proves weak convergence on compact subintervals whose endpoints are not atoms of the limit. This is vague convergence on  $(0, 1]$ .

Differentiation, by (210), gives

$$-\Psi'(x) = \frac{3}{2\sqrt{\pi}} [x(1-x)]^{-3/2} \quad (0 < x < 1/2).$$

There is no atom at  $1/2$ , because  $\Psi$  is continuous there and is zero to its right. These facts identify (158). Its density is asymptotic to  $3x^{-3/2}/(2\sqrt{\pi})$  at zero, so its total mass is infinite, while its restriction to  $[x_1, 1]$  is finite for every  $x_1 > 0$ .  $\square$

### 43 The ends of the profile and the conditional geometry

#### 43.1 Matching the boundary, and the edge at one half

As  $u \downarrow 0$ ,

$$\Psi(u) = \frac{3}{\sqrt{\pi}} u^{-1/2} \left( 1 - \frac{3}{2}u - \frac{5}{8}u^2 + O(u^3) \right). \quad (221)$$

(Manuscript 10 prints the expansion to  $O(u^2)$ , manuscript 11 to  $O(u^3)$ .) The leading term matches Part III's boundary equivalent  $\mathbb{P}(D_n \geq d) \sim 3/\sqrt{\pi d}$  when  $d \rightarrow \infty$  and  $d = o(n)$  (theorem 22.2), and its limiting tail constant  $3/\sqrt{\pi}$  ((88)): with  $u = d/n$ ,  $\Psi(u)/\sqrt{n} \approx 3/\sqrt{\pi d}$ . This is a consistency check between two proven regimes, not a proof of a uniform two-parameter expansion down to arbitrary  $u = u_n \rightarrow 0$ , and not an illicit substitution of a moving  $u = d/n$  into a fixed- $u$  theorem. Theorem 38.1 gives compact *absolute* uniformity away from zero; it does not by itself give relative error estimates when  $u \rightarrow 0$ .

At the other endpoint, if  $u = 1/2 - \varepsilon$  with  $\varepsilon \downarrow 0$ , then

$$\Psi(u) = \frac{12}{\sqrt{\pi}} \varepsilon + O(\varepsilon^3). \quad (222)$$

The profile vanishes linearly at one half. This signals the need for a smaller probability scale or a separate transition window, not an absence of bounded permutations on the other side: a shrinking neighbourhood of  $1/2$  may interact with a smaller finite-size scale that the first-order theorem does not determine. (Both manuscripts; the two readings are theirs.)

#### 43.2 Three equally weighted endpoint channels

Fix  $0 < u < 1/2$  and condition on  $D_n > un$ . The split integral  $I(u)$  of (220) satisfies

$$I(u) = \int_u^{1-u} [y(1-y)]^{-3/2} dy = \frac{4(1-2u)}{\sqrt{u(1-u)}}, \quad (223)$$

and  $\Psi(u) = 3I(u)/(4\sqrt{\pi})$ . The following is a sampling description of the limiting conditional law, not a claim that a unique finite tree split has already been identified (manuscript 10; but see theorem 43.2).

**Corollary 43.1** (Three equally weighted endpoint channels; manuscript 10). *Let  $U^{(u)}$  have density  $I(u)^{-1}[y(1-y)]^{-3/2}$  on  $(u, 1-u)$ , and independently let  $\xi$  be uniform on {low, interior, high}. Define*

$$V = \min(U^{(u)}, 1 - U^{(u)}), \quad \zeta = \begin{cases} 0, & \xi = \text{low}, \\ 1 - U^{(u)}, & \xi = \text{interior}, \\ 1, & \xi = \text{high}. \end{cases}$$

Then

$$\mathcal{L}((D_n/n, L_n/n) \mid D_n > un) \implies \mathcal{L}(V, \zeta). \quad (224)$$

The limiting last-entry law has atoms of mass  $1/3$  at each of  $0$  and  $1$ ; its remaining mass  $1/3$  has the displayed interior density. Also  $F_n/n \rightarrow 1$  in probability under the same conditioning.

*Proof.* Restrict theorem 38.2 to  $x > u$  and divide by the positive limiting mass  $\Psi(u) = 3I(u)/(4\sqrt{\pi})$ . The three equal summands yield the sampling description. For the first value, lemma 39.2 makes  $\mathbb{P}(J_n > \varepsilon n)$  exponentially small, whereas the conditioning event has probability asymptotic to  $\Psi(u)n^{-1/2}$ . Their ratio tends to zero.  $\square$

### 43.3 Universal outer word and macroscopic split

The canonical parser of section 40.4 yields more than the scalar tail. Fix  $u \in (0, 1/2)$ ,  $d_n/n \rightarrow u$ , and  $m_n = n - d_n$ . Condition on  $\mathcal{E}_{n,m_n}$ . Let  $W_n$  be the canonical outer word and  $U_n$  the terminal left size divided by  $n$ .

**Theorem 43.2** (Universal outer word and macroscopic split; manuscript 11). *Conditionally on  $\mathcal{E}_{n,m_n}$ , the pair  $(W_n, U_n)$  converges in law to an independent pair  $(W, U^{(u)})$ , where*

$$\mathbb{P}(W = w) = \frac{\chi(w)4^{-\text{cost}(w)}}{3} \quad (225)$$

and  $U^{(u)}$  has density

$$f_u(y) = \frac{\mathfrak{f}(y)}{I(u)} \mathbf{1}_{[u, 1-u]}(y). \quad (226)$$

The word space is the countable discrete space of individually decorated finite words. Its limiting law is independent of  $u$ .

*Proof.* For each fixed word, divide (213) by  $\mathbb{P}(\mathcal{E}_{n,m_n}) \sim 3I(u)/(4\sqrt{\pi n})$ . This gives the product law on that word and every continuous test function of  $U_n$ . The normalized version of (196), together with the positive denominator constant, gives tightness of the word costs. There are finitely many words at each bounded cost, so the word variables are tight on the discrete space. The split variable lies in a compact interval. Finite-word convergence plus this tightness proves joint convergence. Equation (219) verifies that (225) is normalized.  $\square$

The split law  $U^{(u)}$  is the same in corollary 43.1 and theorem 43.2: manuscript 10's split variable (its  $Y$ ) and manuscript 11's (its  $U_x$ ) have the same density.

**Corollary 43.3** (A heavy-tailed cost but a short word; manuscript 11). *Let  $T_W = \text{cost}(W)$ . Then*

$$\mathbb{P}(T_W = t) = \frac{C_t + C_{t+1}}{6 \cdot 4^t} \quad (t \geq 0), \quad (227)$$

$$\mathbb{E} z^{T_W} = \frac{C(z/4) + C(z/4)^2}{6}, \quad (228)$$

$$\mathbb{P}(\text{len}(W) = k) = \frac{(1/2)^k + (3/4)^k}{6} \quad (k \geq 0), \quad (229)$$

$$\mathbb{E} \text{len}(W) = \frac{7}{3}, \quad \mathbb{P}(W \text{ has no } A) = \frac{2}{3}. \quad (230)$$

Furthermore,

$$\mathbb{P}(T_W > h) \sim \frac{5}{3\sqrt{\pi h}}, \quad \mathbb{E} T_W = \infty. \quad (231)$$

These expectation statements concern the limiting word law; convergence of finite- $n$  word expectations is not asserted.

*Proof.* At cost  $t$ , there are  $C_t$  words without  $A$  and  $C_{t+1} - C_t$  with at least one  $A$ . Sum (225) to obtain (227), and then use (191)–(192) for its probability generating function.

The critical total weight of one  $R$  operation is  $1/2$ , and that of one  $A$  is  $1/4$ . At length  $k$ , all words have weight  $(3/4)^k$ , whereas those without  $A$  have weight  $(1/2)^k$ . The accepted weight is half the sum of these two quantities. Division by three gives (229). The geometric-series identities give

$$\mathbb{E} \text{len}(W) = \frac{1}{6} \left\{ \frac{1/2}{(1 - 1/2)^2} + \frac{3/4}{(1 - 3/4)^2} \right\} = \frac{7}{3}.$$

The no- $A$  probability is  $\mathcal{W}_R(1/4)/3 = 2/3$ . Finally, (181) and  $C_{t+1}/C_t \rightarrow 4$  imply  $\mathbb{P}(T_W = t) \sim 5/(6\sqrt{\pi} t^{3/2})$ . Summing this regularly varying power tail gives (231) and infinite mean cost.  $\square$

The word typically has only a few operations, but one decoration can be large. This explains why summability alone was insufficient for the rare-scale interchange: the limiting cost has no finite first moment. (The limiting word cost  $T_W$  is not Part III's skeleton cost  $T$ , whose tail is  $\sim 4/\sqrt{\pi h}$  by lemma 24.1.)

*Three equally weighted endpoint modes (manuscript 11).* Call the terminal right block's last-entry mode *high* when  $L(\tau) > b/2$  and *low* otherwise. In the interior of the split range, the low and high modes each have unconditioned probability tending to one half. The proof of proposition 42.1 therefore also yields the following table under  $\mathcal{E}_{n,m_n}$ :

| Outer word          | Terminal last-entry mode | Limiting probability |
|---------------------|--------------------------|----------------------|
| No append           | Low                      | 1/3                  |
| No append           | High                     | 1/3                  |
| At least one append | High                     | 1/3                  |
| At least one append | Low                      | 0                    |

For a fixed no-append word, each mode has mass  $4^{-t}/6$  in the conditional law. For a fixed word containing an append, only the high mode survives, also with mass  $4^{-t}/6$ . Summing over their respective critical word weights, both equal to two, proves the table. Boundary strips are removed exactly as in proposition 42.1. This is another interpretation of the factor three.

**Proposition 43.4** (Geometric meaning of the rare-event measure; manuscript 11). *Conditionally on  $\mathcal{E}_{n,m_n}$ ,*

$$\frac{D_n}{n} - \min(U_n, 1 - U_n) \longrightarrow 0 \quad \text{in probability.} \quad (232)$$

Consequently, for  $y \geq u$ ,

$$\mathbb{P}(D_n/n \geq y \mid \mathcal{E}_{n,m_n}) \longrightarrow \frac{\Psi(y)}{\Psi(u)}, \quad (233)$$

with the right side zero for  $y \geq 1/2$ .

*Proof.* First restrict to a bounded outer cost and an interior split interval. By corollary 40.8, a uniform block of size  $a$  has maximum jump  $a - o(n)$  in probability, uniformly when  $a$  is proportional to  $n$ ; the same is true for the right block of size  $b$ . Deleting a set of such terminal pairs of relative probability  $o(1)$  remains harmless after imposing the endpoint acceptance event, whose limiting fraction is either one or one half.

The terminal downward jump is  $a + 1 + J(\tau) = a + o(n)$  by lemma 39.2, and the upward jump into the terminal maximum is at most  $a$ . Every outer  $R$  jump is  $o(n)$  on the same good endpoint set. If an append occurs, acceptance selects the high last-entry mode, and lemma 39.3 gives  $L(\tau) = b - o(n)$  in conditional probability. Its first append jump is therefore  $a + o(n)$ ; later append jumps are bounded by the fixed outer cost. Hence the global maximum jump is  $\max(a, b) + o(n)$ , and the deficit is  $\min(a, b) + o(n)$ . Since  $(a + b)/n \rightarrow 1$ , this proves (232) under the two restrictions. Remove the boundary strips and then the cost cutoff by theorem 43.2 and lemma 40.7.

The map  $y \mapsto \min(y, 1 - y)$  sends (226) to density  $2\mathfrak{f}(y)/I(u)$  on  $[u, 1/2]$ . Its upper tail is  $\Psi(y)/\Psi(u)$  by (220), proving (233). The same tail statement also follows directly by taking the ratio in theorem 38.1.  $\square$

#### 43.4 How the two descriptions of the factor three match

The two manuscripts explain the coefficient 3 by three equally weighted alternatives, and the alternatives correspond. On the terminal split, with  $y = a/n$  the left fraction, manuscript 11's modes and manuscript 10's channels match as follows. Without an append, the whole permutation ends with the terminal right block's last entry, since  $R$  operations preserve the last entry; the low mode puts  $L_n/n$  near 0 (manuscript 10's *low* channel), and the high mode puts  $L_n/n$  near  $b/n \approx 1 - y$  (its *interior* channel, the mark  $1 - U^{(u)}$ ). With an append, the last entry is a maximum at least  $N = n - t$ , so  $L_n/n \rightarrow 1$  (its *high* channel). The deficit is  $\min(y, 1 - y)$  in all three cases (proposition 43.4 and the proof of lemma 41.1). Manuscript 10's corollary does not identify a canonical split; manuscript 11's parser supplies one, and corollary 43.1's interior mark is then its high-mode, no-append terminal block. This correspondence is the merge's reading of the two proofs; neither manuscript states it, and no joint limit theorem for  $(W_n, U_n, L_n/n)$  is claimed.

### 44 Moment constants and size-biased limits

#### 44.1 Uniform integrability at exactly the required threshold

*Proof of theorem 38.3 (both manuscripts).* For  $p > 0$ , the tail formula for a nonnegative bounded random variable gives the layer-cake identity

$$n^{1/2-p} \mathbb{E} D_n^p = p \int_0^1 x^{p-1} \sqrt{n} \mathbb{P}(D_n > nx) dx. \quad (234)$$

Using a weak rather than a strict inequality changes only the integer threshold by at most one and has no effect on the pointwise limits. The bound (187) dominates the integrand by  $4\sqrt{2} p x^{p-3/2}$  (manuscript 10). Equivalently (manuscript 11), by corollary 40.8 the scaled tail is at most  $Kx^{-1/2}$  for  $nx \geq 1$ , and for  $nx < 1$  the trivial bound  $\sqrt{n} \leq x^{-1/2}$  gives the same envelope, so the integrand

is at most  $Kpx^{p-3/2}$ . This is integrable at zero exactly when  $p > 1/2$ . Dominated convergence and theorem 38.1 therefore prove

$$n^{1/2-p} \mathbb{E} D_n^p \longrightarrow p \int_0^{1/2} x^{p-1} \Psi(x) dx. \quad (235)$$

Integration by parts is legitimate, because the boundary term at zero vanishes,  $x^p \Psi(x) = O(x^{p-1/2}) \rightarrow 0$ , and  $\Psi(1/2) = 0$ ; with (210) it gives (160). The constant is finite and positive. For the alternative formula, substitute

$$x = \frac{z^2}{1+z^2}, \quad 0 < z < 1, \quad dx = \frac{2z}{(1+z^2)^2} dz.$$

The integrand  $x^{p-3/2}(1-x)^{-3/2} dx$  becomes  $2(z^2/(1+z^2))^{p-1} dz$ , proving (161). At  $p = 1$  the integral is one, so  $m_1 = 3/\sqrt{\pi}$ . At  $p = 2$ ,

$$\int_0^1 \frac{z^2}{1+z^2} dz = [z - \arctan z]_0^1 = 1 - \frac{\pi}{4}.$$

Since  $(\mathbb{E} D_n)^2 \sim m_1^2 n = o(n^{3/2})$ , subtraction from the second moment gives  $\text{Var}(D_n) \sim \mathbb{E} D_n^2 \sim m_2 n^{3/2}$ , which is (163). The identity  $M_n = n - D_n$  gives the mean formula (162) and equality of the variances.  $\square$

The convergence proof is not a formal interchange of limits. The finite counting envelope supplies exactly the near-zero domination needed for the moment integral; vague convergence alone would not do so.

## 44.2 Closed forms

It is convenient to set

$$I(p) = \int_0^1 \left( \frac{z^2}{1+z^2} \right)^{p-1} dz, \quad m_p = \frac{3}{\sqrt{\pi}} I(p). \quad (236)$$

Direct integration gives (manuscript 11; manuscript 10 prints the orders 1, 2, 3):

| $p$ | $I(p)$           | $m_p$                                                  |
|-----|------------------|--------------------------------------------------------|
| 1   | 1                | $3/\sqrt{\pi}$                                         |
| 3/2 | $\sqrt{2} - 1$   | $\frac{3}{\sqrt{\pi}}(\sqrt{2} - 1)$                   |
| 2   | $1 - \pi/4$      | $\frac{3}{\sqrt{\pi}}(1 - \frac{\pi}{4})$              |
| 3   | $5/4 - 3\pi/8$   | $\frac{3}{\sqrt{\pi}}(\frac{5}{4} - \frac{3\pi}{8})$   |
| 4   | $3/2 - 15\pi/32$ | $\frac{3}{\sqrt{\pi}}(\frac{3}{2} - \frac{15\pi}{32})$ |

For example,

$$m_1 = 1.6925687506\dots, \quad m_2 = 0.3632283625\dots$$

For  $p = 2$  the integral is  $\int_0^1 t^2/(1+t^2) dt = 1 - \pi/4$ ; for  $p = 3$ , expand  $t^4/(1+t^2)^2 = 1 - 2/(1+t^2) + 1/(1+t^2)^2$ . For any positive integer  $p$ , partial fractions in (161) reduce the constant to rational numbers and  $\pi$ , followed by the common factor  $3/\sqrt{\pi}$  (manuscript 10). For every  $p > 1/2$  with  $p \neq 1$ , integration of

$$\frac{d}{dz} \frac{z^{2p-1}}{(1+z^2)^{p-1}} = (2p-1) \left( \frac{z^2}{1+z^2} \right)^{p-1} - 2(p-1) \left( \frac{z^2}{1+z^2} \right)^p$$

over  $[0, 1]$  yields the recurrence (manuscript 11)

$$\mathcal{I}(p+1) = \frac{(2p-1)\mathcal{I}(p) - 2^{1-p}}{2(p-1)}. \quad (237)$$

At  $p = 1$  this identity degenerates; one must not divide by zero or use it to obtain  $\mathcal{I}(2)$  without an additional calculation.

The integral in (160) can also be written  $\frac{3}{2\sqrt{\pi}}B_{1/2}(p-1/2, -1/2)$ , an incomplete beta function. The second parameter is negative, but the defining integral stops at  $1/2$ , not at its singular endpoint  $1$ , so it is convergent for every  $p > 1/2$  (manuscript 10).

### 44.3 The two limits of the constant function

**Proposition 44.1** (Two limits of the constant function; manuscript 11, the first also manuscript 10). *As  $p \downarrow 1/2$  and as  $p \rightarrow \infty$ , respectively,*

$$m_p \sim \frac{3}{\sqrt{\pi}(2p-1)} = \frac{3}{2\sqrt{\pi}} \frac{1}{p-1/2}, \quad (238)$$

$$m_p \sim \frac{6}{\sqrt{\pi} p} 2^{-p}. \quad (239)$$

*These are statements about the constants after taking the fixed- $p$  limit in  $n$ , not joint limits with  $p = p(n)$ .*

*Proof.* Near zero the integrand of  $\mathcal{I}(p)$  is  $z^{2p-2}(1+z^2)^{1-p}$ . For  $p$  in a fixed right neighbourhood of  $1/2$ , subtracting  $z^{2p-2}$  leaves an integrable function with uniformly bounded integral: the factor difference is  $O(z^2)$ . Hence  $\mathcal{I}(p) = (2p-1)^{-1} + O(1)$ , proving (238). (Manuscript 10's argument is the same: the only divergent contribution is the integral of  $t^{2p-2}$  near zero, and the remaining factor tends regularly to one there.)

For large  $p$ , write  $\varphi(z) = \log(z^2/(1+z^2))$ . It is strictly increasing on  $(0, 1]$ , with  $\varphi(1) = -\log 2$  and  $\varphi'(1) = 1$ . For fixed  $\eta > 0$ , the integral over  $[0, 1-\eta]$  is exponentially smaller than  $2^{-(p-1)}$ . On  $[1-\eta, 1]$ , substitute  $v = (p-1)(1-z)$  and use  $\varphi(1-v/(p-1)) = \varphi(1) - v/(p-1) + O(v^2/(p-1)^2)$ . A one-sided derivative bound near one supplies an integrable exponential majorant. Dominated convergence yields  $\mathcal{I}(p) \sim 2^{-(p-1)}/(p-1)$ , which is (239).  $\square$

The pole in (238) is consistent with the logarithmic critical moment (92) of Part III. It does not prove convergence of its centered  $O(1)$  remainder. The proof of theorem 38.3 deliberately stops at  $p > 1/2$ : its integrable envelope fails at the critical order, dominated convergence fails at  $p = 1/2$ , and vague convergence alone cannot repair that failure. That question remains separate (section 47.6).

### 44.4 General power bias and the uniform transformation

For  $p > 1/2$ , define (manuscript 10)

$$\mathbb{P}_{n,p}^*(B) = \frac{\mathbb{E}[D_n^p \mathbf{1}_B]}{\mathbb{E}D_n^p}.$$

The same domination argument, first away from zero and then with the cutoff removed, proves

$$\mathcal{L}_{\mathbb{P}_{n,p}^*}(D_n/n) \implies \frac{x^p \rho(x)}{m_p} dx, \quad 0 < x < 1/2. \quad (240)$$

To be explicit about the cutoff removal, (187) and layer cake bound the contribution from  $D_n/n \leq \varepsilon$  by a constant times  $\varepsilon^{p-1/2}$ , uniformly in  $n$ .

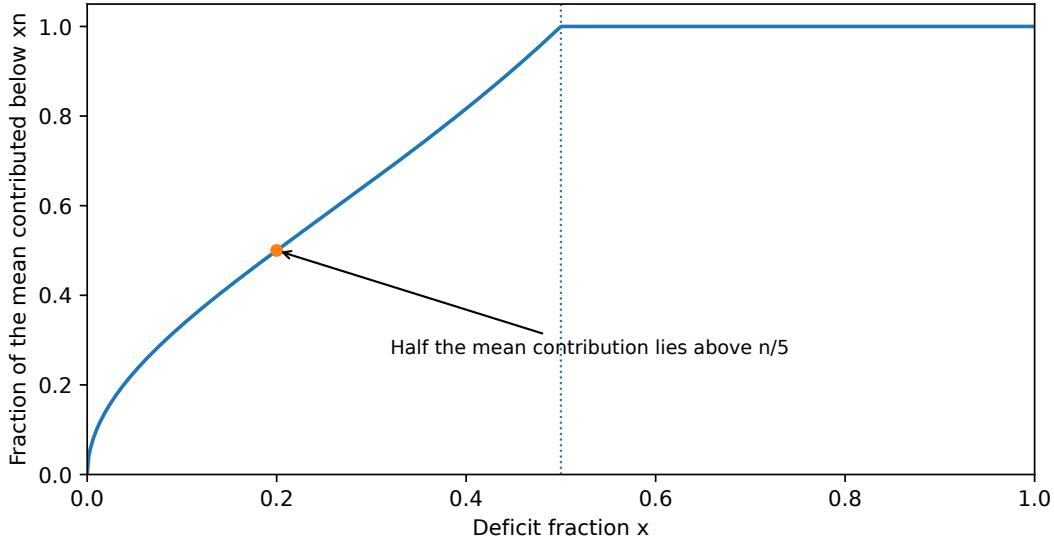


Figure 8: Manuscript 10: the limiting fraction  $H^*(x)$  of the mean deficit contributed by permutations with  $D_n \leq xn$ . This is a size-biased distribution, not the ordinary distribution of  $D_n/n$ , which converges to a point mass at zero.

*Proof of theorem 38.4.* For  $p = 1$ , the limiting density simplifies to

$$\frac{1}{2}x^{-1/2}(1-x)^{-3/2} = \frac{d}{dx} \sqrt{\frac{x}{1-x}}.$$

Its integral over  $(0, 1/2)$  is one. Thus its distribution function is (166); applying the increasing change of variable  $t = \sqrt{x/(1-x)}$  proves (165). At  $x = 1/5$ , the value is  $1/2$ , giving (167).  $\square$

## 45 Constructive lower bounds beyond the half-size cutoff

The vanishing first-order profile below  $M_n/n = 1/2$  should not be mistaken for exponential suppression. A simple explicit family gives polynomial lower bounds in every fixed proportional-bound regime (manuscript 10).

**Theorem 45.1** (A many-block lower bound; manuscript 10). *Fix  $0 < \theta < 1$  and an integer  $r \geq 1$  such that  $(r+1)\theta > 1$ . There is a constant  $\underline{c}_{\theta,r} > 0$  such that, for all sufficiently large  $n$ ,*

$$\mathbb{P}(M_n \leq \lfloor \theta n \rfloor) \geq \underline{c}_{\theta,r} n^{-r/2}. \quad (241)$$

*Equivalently,  $a_n^{(\lfloor \theta n \rfloor)} \geq \underline{c}'_{\theta,r} 4^n n^{-(r+3)/2}$  for a positive constant  $\underline{c}'_{\theta,r}$ .*

*Proof.* Put  $\theta_r = 1/(r+1) < \theta$ . Choose  $\eta > 0$  sufficiently small that  $\theta_r - r\eta > 0$  and  $\theta_r + r\eta < \theta$ . Choose  $r$  integers  $a_1, \dots, a_r$  in intervals  $|a_i/n - \theta_r| < \eta$ , and put

$$b = n - r - \sum_{i=1}^r a_i.$$

After slightly shrinking the intervals if necessary, there are at least  $cn^r$  choices, every component size is bounded between two positive multiples of  $n$ , and all  $a_i, b$  are at most  $(\theta - \varepsilon)n$  for a fixed  $\varepsilon > 0$ .

Take arbitrary left decorations  $\beta_i \in \text{Av}_{a_i}(132)$  and a terminal core  $\sigma \in \text{Av}_b(132)$ . Form

$$\pi = R_{\beta_1} R_{\beta_2} \cdots R_{\beta_r}(\sigma).$$

The maximum decomposition along the right child recovers this fixed-length chain and every decoration uniquely. Restrict each component to have first deficiency at most  $\varepsilon n/3$ . By lemma 39.2, the excluded proportion in each component is exponentially small in  $n$ .

For an inner right subtree, (174) says that its first deficiency is either the deficiency of its left decoration plus one, or, at the final step, the deficiency of the core. The downward jump introduced by  $R_{\beta_i}$  is therefore at most  $a_i + 1 + \varepsilon n/3 + 1 < \theta n - 1$  for large  $n$ . By (170), no internal jump in a left decoration can exceed that boundary jump. The core itself has maximum jump at most  $b - 1 < \theta n$ . Thus all these good reconstructions obey the required integer bound.

For each length vector, their number is  $(1 - o(1))C_{a_1} \cdots C_{a_r} C_b$ , uniformly on the chosen compact size intervals. Stirling's formula gives

$$\frac{C_{a_1} \cdots C_{a_r} C_b}{C_n} \geq c'' n^{-3r/2},$$

the fixed exponential factor is  $4^{-r}$  because the component sizes sum to  $n - r$ . Summing over at least  $cn^r$  disjoint length vectors proves (241).  $\square$

For  $1/3 < \theta \leq 1/2$ , this gives a lower bound of order  $n^{-1}$ , while theorem 38.1 gives  $o(n^{-1/2})$ . The exact order and profile between those bounds are not proved here. For nonreciprocal  $\theta$ , the smallest admissible  $r$  is  $\lceil 1/\theta \rceil - 1$ ; at a reciprocal endpoint the strict condition requires a larger  $r$ . (For  $\theta > 1/2$ ,  $r = 1$  gives the order  $n^{-1/2}$  of (153); the merge's remark.)

## 46 Exact checks, exact data and sampling

### 46.1 Exact finite checks of manuscript 10

The standard-library program `code/04-macroscopic-jump-verify.py` exhaustively checks all 82,499 avoiding permutations of sizes 1 through 11 generated by the Catalan decomposition. For each it compares the direct adjacent maximum with the exact root recurrence of proposition 39.1. It checks both endpoint counting formulas and the finite tail envelope at every threshold in those sizes. For sizes 1 through 8, a separate three-index test applied to all ordinary permutations independently verifies the generated avoidance class. It additionally checks the exact stopping-cost tail (184) for  $h = 0, \dots, 100$ , and verifies the sampler's rotation multiplicities and full statistic histograms on all 2,352 degree words of sizes 1 through 6. The outcome is recorded in `data/04-macroscopic-jump-verification.json`; the exact finite deficit histograms for  $n = 1, \dots, 11$  are retained as integer counts in `data/04-macroscopic-jump-exact_small_distributions.csv`. These tests are finite regression evidence, not a substitute for any asymptotic proof.

### 46.2 Exact finite and symbolic checks of manuscript 11

Manuscript 11's package contains a standard-library integer implementation of the endpoint recurrence, a separate exhaustive generator, a canonical parser and reconstructor, and exact-count data. The verification (`code/05-macroscopic-deficits-verify.py`) was executed with the following scope:

| Check                                                            | Executed cases |
|------------------------------------------------------------------|----------------|
| Avoiders examined, sizes 2 through 9                             | 6,916          |
| Total counts versus exhaustive enumeration, every $1 \leq m < n$ | 36             |
| Cumulative endpoint states versus enumeration, sizes through 7   | 644            |
| Canonical parsing and exact reconstruction checks                | 18,738         |
| Positive-cost coefficients of both word identities               | 60             |

The first- and last-entry formulas were also checked against the same exhaustive classes. The report is retained in `data/05-macroscopic-deficits-verification.json`. These are finite checks of the combinatorial objects and implementation. They do not establish the universal analytic limits; the written proofs do that work.

A separate SymPy script (`code/05-macroscopic-deficits-verify_symbolic.py`) checked eleven groups of exact symbolic identities, including the derivative of  $\Psi$ , the split-integral primitive, the moment substitution and recurrence, the displayed moment integrals, and the word generating functions. Its executed report is retained in `data/05-macroscopic-deficits-symbolic_verification.json`; these checks do not verify limit interchanges or analytic hypotheses.

### 46.3 The exact recurrence used for manuscript 11's data

Manuscript 11 computes  $a_n^{(m)}$  exactly with Part I's endpoint counts  $T_{u,v}^{(m)}(n)$  (proposition 2.1): the number of  $m$ -bounded avoiders of length  $n$  with  $n - \pi_1 \leq u$  and  $n - \pi_n \leq v$ . A negative threshold gives zero,  $T_{u,v}^{(m)}(1) = 1$  for nonnegative thresholds, and thresholds above  $n - 1$  can be capped at  $n - 1$ . Let  $\mathcal{N}_s(u) = \#\{\pi \in \text{Av}_s(132) : s - \pi_1 \leq u\}$ ; its values follow by summing (175), and  $\mathcal{N}_0(u) = 1$  for  $u \geq 0$  and zero for  $u < 0$ . The known endpoint recurrence of Mayama–Akita is

$$T_{u,v}^{(m)}(n) = \sum_{k=1}^{\min(m,n-1)} c_{k,u} T_{m-k,v-k}^{(m)}(n-k) + T_{u-1,m-1}^{(m)}(n-1), \quad (242)$$

where  $c_{1,u} = 1$  and  $c_{k,u} = \mathcal{N}_{k-1}(u-1)$  for  $k \geq 2$ : this is the coefficient form of Part I's (8), and  $c_{k,u}$  is Part I's (5). Manuscript 11 rederives it from the maximum bijection, as Part I does, to make its computational certificates self-contained; the derivation is not repeated. When  $n \leq m + 1$ , the adjacency constraint is vacuous, and the last-entry formula gives the useful exact shortcut (manuscript 11)

$$T_{u,v}^{(m)}(n) = \mathcal{N}_{n-1}(u-1) + \sum_{q=1}^{\min(v,n-1)} C_{n-q-1} \mathcal{N}_q(u) \quad (n \geq 2, n \leq m+1). \quad (243)$$

For (243), consider a final value  $\ell = n - q$ . When  $q \geq 1$ , the last-entry decomposition has a high initial block of size  $q$ , which contains the first entry, and a lower block of size  $n - q - 1$ . The whole first deficiency is exactly the first deficiency in the high block. The count is therefore  $C_{n-q-1} \mathcal{N}_q(u)$ , while the last threshold is equivalent to  $q \leq v$ . The separate case  $q = 0$  appends the maximum and contributes  $\mathcal{N}_{n-1}(u-1)$ .

Memoization of (242), using (243) whenever possible, produces the retained counts. The implementation caps vacuous thresholds and memoizes smaller-size states. It uses Python arbitrary-precision integers throughout. It is intended as a transparent verifier and moderate-size data generator, not as a claim of a new optimal counting algorithm. Every count and every Catalan number in the CSV files is an exact integer. Floating point is used only for the displayed ratios and figures.

### 46.4 The bulk profile is approached slowly

The data in table 15 do not show rapid convergence, and the first column is not even monotone at these sizes. No monotonicity or finite-size error rate is claimed. This makes it especially important that the factor three was obtained from the endpoint calculation, not fitted to numerical values. The full integer certificates for all ten cases are in `data/05-macroscopic-deficits-bulk_counts.csv`.

Table 15: Manuscript 11: exact normalized counts  $\sqrt{n} a_n^{(n-d)}/C_n$  and their proved limits  $\Psi(d/n)$ . Here  $d/n$  is exactly the displayed fraction. A dash means that case was not computed.

| $n$   | $d/n = 1/5$ | $d/n = 3/10$ | $d/n = 2/5$ |
|-------|-------------|--------------|-------------|
| 40    | 2.794323    | 1.960484     | 1.291077    |
| 80    | 2.823905    | 1.897805     | 1.178743    |
| 160   | 2.806836    | 1.821163     | 1.071493    |
| 320   | —           | 1.746102     | —           |
| Limit | 2.538853    | 1.477395     | 0.690988    |

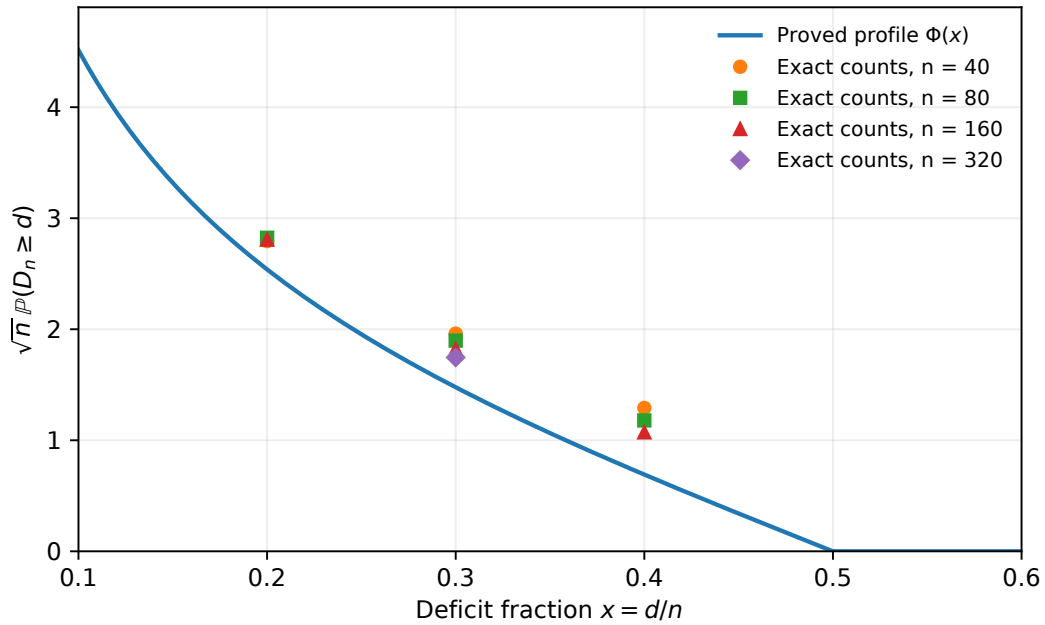


Figure 9: Manuscript 11: the proved limiting profile (labelled  $\Phi(x)$  in the figure,  $\Psi(u)$  here) and exact finite-size values  $\sqrt{n} \mathbb{P}(D_n \geq d)$  against  $x = d/n$ . The visible separation is real finite-size behaviour, not a confidence interval or an error estimate for the theorem. Values at or beyond  $1/2$  are zero only in the limiting profile.

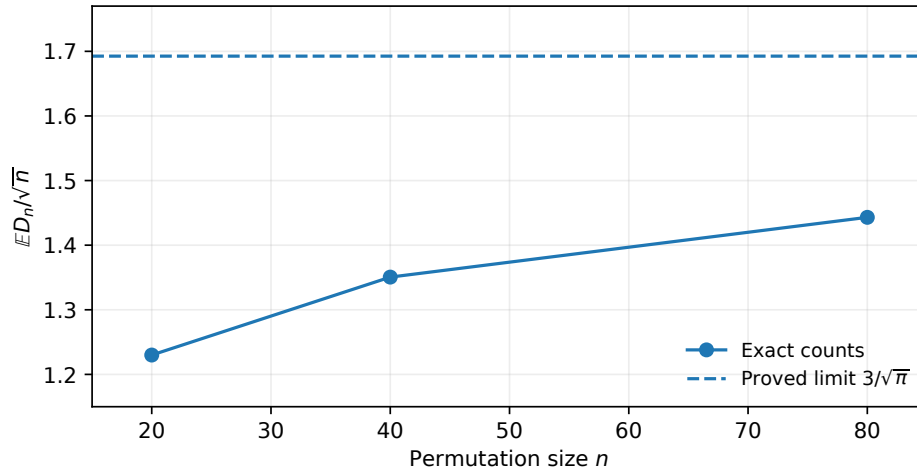


Figure 10: Manuscript 11: exact normalized mean deficits at three finite sizes. The dashed line is the analytically proved constant  $3/\sqrt{\pi}$ ; it was not estimated from these three data points.

#### 46.5 Moment data without a fitted constant

For integer-valued positive  $D_n$ ,

$$\mathbb{E}D_n = \sum_{d=1}^{n-1} \mathbb{P}(D_n \geq d),$$

$$\mathbb{E}D_n^2 = \sum_{d=1}^{n-1} (2d - 1) \mathbb{P}(D_n \geq d).$$

These identities allow the moments to be computed from exact bounded counts at every  $m = 1, \dots, n - 1$ . The retained results (manuscript 11) are:

| $n$   | $\mathbb{E}D_n/\sqrt{n}$ | $\mathbb{E}D_n^2/n^{3/2}$ |
|-------|--------------------------|---------------------------|
| 20    | 1.229950                 | 0.498514                  |
| 40    | 1.350574                 | 0.495311                  |
| 80    | 1.443085                 | 0.479054                  |
| Limit | 1.692569                 | 0.363228                  |

The second column of moments is a *second moment*, not a variance. Its leading constant equals that of the variance only asymptotically, because the squared mean is of smaller order. The complete rational values are retained as numerator and denominator integers in `data/05-macroscopic-deficits-moment_counts.csv`.

#### 46.6 Uniform Catalan sampling without constructing permutations

Manuscript 10's C++ sampler (code/04-macroscopic-jump-simulate.cpp) uses a binary-tree preorder word with  $n$  symbols +1 (internal nodes) and  $n + 1$  symbols −1 (leaves). A uniform shuffle of these symbols is rotated immediately after the first occurrence of its global minimum partial sum. The resulting word has nonnegative partial sums before its final −1 and encodes a full ordered binary tree.

The elementary cycle lemma says each tree word has exactly  $2n + 1$  preimages under this rotation map. One can see uniqueness of the valid rotation from the integer partial sums and

Table 16: Manuscript 10: numerical illustrations of the moment limits, from its shipped simulation data (section 46.8). Parenthesized values are one estimated Monte Carlo standard error for the mean column, not a bound on finite-size bias. The second-moment column is a sample estimate and is not presented as a certified interval.

| $n$          | Samples | $\widehat{\mathbb{E}D_n}/\sqrt{n}$ | $\widehat{\mathbb{E}D_n^2}/n^{3/2}$        |
|--------------|---------|------------------------------------|--------------------------------------------|
| 64           | 300,000 | 1.41333 (0.00250)                  | 0.48329                                    |
| 256          | 200,000 | 1.55368 (0.00486)                  | 0.44632                                    |
| 1,024        | 100,000 | 1.62981 (0.01027)                  | 0.41261                                    |
| 4,096        | 50,000  | 1.69621 (0.02139)                  | 0.40232                                    |
| 16,384       | 25,000  | 1.66394 (0.04260)                  | 0.37599                                    |
| Proved limit |         | $3/\sqrt{\pi} \approx 1.69257$     | $3(4 - \pi)/(4\sqrt{\pi}) \approx 0.36323$ |

total sum  $-1$ . A nontrivial period is impossible: it would make  $-1$  an integer multiple of the sum of one shorter period. Thus all  $2n + 1$  rotations are distinct. Uniformity of the shuffle, followed by the constant preimage count, proves uniformity of the Catalan tree. The maximum decomposition is a bijection between these trees and 132-avoiders.

The tree is then evaluated bottom-up using proposition 39.1; no length- $n$  permutation array is required. Each subtree stores only its size, first entry, last entry, and largest jump. One sample uses  $O(n)$  time and  $O(n)$  working memory. The implementation uses a fixed pseudorandom seed, so its retained output is reproducible with the stated implementation. Different standard-library implementations of `std::shuffle` may produce different sample streams even with the same engine seed.

#### 46.7 What the numerics do and do not say

The retained simulation has 675,000 pseudorandom samples across five sizes. It illustrates approach to the profile and moment constants, but finite-size bias is visible, especially in the second moment and near the cutoff. The theorems are established by the proofs, not by a fit to table 16 or table 15. The data are neither outward-rounded interval certificates nor a claimed convergence-rate estimate.

#### 46.8 Rebuilding and rerunning

Part IV's files are

|                                                 |                                                                                         |
|-------------------------------------------------|-----------------------------------------------------------------------------------------|
| code/04-macroscopic-jump-verify.py              | Standard-library exact finite checks                                                    |
| code/04-macroscopic-jump-simulate.cpp           | C++17 uniform Catalan-tree sampler                                                      |
| code/04-macroscopic-jump-plot_results.py        | Its two figures (NumPy, Matplotlib)                                                     |
| code/04-macroscopic-jump-build.sh               | Manuscript 10's three-pass build                                                        |
| data/04-macroscopic-jump-*                      | Recorded check, exact histograms<br>n = 1..11, 675,000 samples,<br>delivery build audit |
| 04-macroscopic-jump-SOURCES.md                  | Manuscript 10's provenance ledger                                                       |
| code/05-macroscopic-deficits-model.py           | Exact endpoint counts, generator                                                        |
| code/05-macroscopic-deficits-verify.py          | Exhaustive combinatorial checks                                                         |
| code/05-macroscopic-deficits-verify_symbolic.py | SymPy identity checks                                                                   |
| code/05-macroscopic-deficits-numerics.py        | Exact bulk and moment counts                                                            |
| code/05-macroscopic-deficits-plot_results.py    | Its two figures                                                                         |
| code/05-macroscopic-deficits-build.sh           | Manuscript 11's latexmk build                                                           |
| data/05-macroscopic-deficits-*                  | Recorded checks, exact counts,                                                          |



## 47.2 Reciprocal thresholds and the critical window at $m = n/2$

*Manuscript 10.* Determine the order of  $\mathbb{P}(M_n \leq n/2)$  and the transition when  $M_n \leq n/2 + s_n$ .

*Manuscript 11.* Find the probability scale and the transition function when  $m = n/2 + s_n$  and  $s_n = o(n)$ . The linear vanishing in (222) is a constraint on any crossover formula, but does not determine the size of  $s_n$ , the width of the window, or the probability exactly at its centre. More generally, do the  $\theta = 1/r$  mark distinct polynomial or logarithmic transitions? The discarded terminal boundary strips, negligible at a fixed bulk fraction, may become part of the main term here. Rounding effects, parity and the terminal maximum's unit cost must be retained at a lattice scale rather than replaced by a continuum approximation too early.

## 47.3 A uniform relative boundary-to-bulk approximation

*Manuscript 11.* Find a formula with relative error tending to zero in a range simultaneously covering  $d \rightarrow \infty$ ,  $d = o(n)$ , and  $d/n$  bounded inside  $(0, 1/2)$ . A useful goal is an explicit error bound that exposes dependence on both  $d$  and  $n - 2d$ . The proof here truncates the word cost before taking the limit. A quantitative version would keep the slowly decaying word tail, the endpoint acceptance correction, and the two split-boundary strips in one estimate. Compact absolute uniformity is not enough when the main profile itself tends to zero or infinity.

## 47.4 A local macroscopic limit

*Both manuscripts.* For  $d_n/n \rightarrow x \in (0, 1/2)$  (manuscript 11), or uniformly on compact subsets of  $(0, 1/2)$  with  $d_n = \lfloor xn \rfloor$  (manuscript 10), is it true that

$$n^{3/2} \mathbb{P}(D_n = d_n) \longrightarrow \frac{3}{2\sqrt{\pi}} [x(1-x)]^{-3/2}?$$

This is consistent with theorem 38.2, but it is not a consequence of it: the vague limit controls intervals, not individual lattice masses, and differencing two tail estimates whose errors are merely  $o(n^{-1/2})$  does not control a probability of order  $n^{-3/2}$ . A proof requires a local count or substantially sharper error control. A local theorem would substantially strengthen coefficient asymptotics and numerical certification.

## 47.5 The exact total-variation constant

*Both manuscripts.* Part III proves a sharp order but leaves the constant in  $d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D))$  open (its Section 32.3). Determine whether  $\sqrt{n} d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D))$  converges, and compute the limit if it exists. The limiting boundary mass beyond  $n/2$  suggests a possible contribution  $3\sqrt{2}/\sqrt{\pi}$ , but neither manuscript asserts that this is the full constant (manuscript 10). The bulk measure identifies one source of mass at large deficits. Total variation also depends on signed finite-size corrections at small and intermediate deficits, so the answer cannot be obtained just by integrating the unsigned bulk density; the distinction between a weak rare-event measure and a signed correction measure is essential (manuscript 11). Establishing it requires control of the signs and aggregate size of discrepancies inside the finite range, which the present weak rare-event limit does not provide.

## 47.6 Centered critical and subcritical corrections

*Manuscript 10.* Does

$$\mathbb{E}\sqrt{D_n} - \frac{3}{2\sqrt{\pi}} \log n$$

converge, and what is its limit? Can one identify the leading correction to  $\mathbb{E}D_n^p$  for  $0 < p < 1/2$ ? These questions require a quantitative matching of the fixed-deficit law and the macroscopic measure near the singular boundary  $x = 0$ . (The first is the critical part of Part III's Section 32.2, which Part IV leaves open.)

## 47.7 Second-order moment asymptotics

*Both manuscripts.* Determine the next term after  $3\sqrt{n/\pi}$  in  $\mathbb{E}D_n$ , the next term in the second moment, and the next terms in  $\mathbb{E}D_n^p$  for fixed  $p > 1/2$ . The numerical tables show significant finite-size corrections, and the finite data do not establish whether the next mean term is constant, logarithmic, or of another order. The next contribution can mix a boundary correction with a bulk correction. An expansion for a fixed positive deficit fraction alone need not be integrable down to zero. Matched subtraction of the small- $u$  singular terms of (221) is a possible route, but must come with a uniform remainder. A correct expansion must separate finite outer decorations, mesoscopic tails, and repeated macroscopic split events, rather than extrapolating a few simulated sizes.

## 47.8 A geometric conditional limit for the whole permutation

*Manuscript 10.* Corollary 43.1 describes two marked statistics. Can one construct a conditioned tree or permutation limit under  $D_n > un$ , with an actual canonical macroscopic split and a finite outer decoration process? Such a result could determine the position of the largest jump and the joint sizes of the largest ascent and descent. The sampling representation of the limiting pair is not, by itself, such a full geometric theorem. [Merge note: manuscript 11's parser supplies the canonical split and its theorem 43.2 the limiting law of the finite outer word jointly with the split; the question is re-scoped to the full conditioned permutation, the position of the largest jump and the joint largest ascent and descent, which remain open. Compare Part III's Section 32.4.]

## 47.9 Conditional cost moments and mesoscopic decorations

*Manuscript 11.* Under  $D_n \geq d_n$  with  $d_n/n \rightarrow u \in (0, 1/2)$ , determine the growth of  $\mathbb{E} \text{cost}(W_n)$  and the distribution of the largest outer decoration. The limiting word has finite mean length  $7/3$  but infinite mean cost. Hence convergence in law from theorem 43.2 does not determine these expectations. A joint theory of the word cost and a second large split may describe how the infinite limiting expectation is truncated at finite  $n$ .

## 47.10 A uniform bridge to shrinking adjacency fractions

*Manuscript 10.* The current result fixes  $\theta > 1/2$ , while the fixed- $m$  spectral program fixes  $m$  before letting  $n$  grow. Determine regimes with  $m \rightarrow \infty$ ,  $m/n \rightarrow 0$ . The lower-bound hierarchy suggests that the required number of large components itself grows, so a fixed-depth renewal expansion may cease to be uniform. (Compare Part III's Section 32.10.)

## 47.11 Weighted refinements

*Manuscript 11.* Extend the rare-event profile to avoiders weighted by descents, or record additional statistics jointly with the terminal split. The constant three arose from three equally weighted endpoint modes. Under a nonuniform weight those modes may no longer be equal, and the critical operation masses will change. A correct extension should derive those weights from the refined generating functions instead of replacing Catalan numbers by a new sequence without rechecking the endpoint law. (Compare Part III's Section 32.6.)

## 47.12 Other classes, statistics and endpoint operators

*Manuscript 10.* Identify statistics on other pattern-avoiding classes whose root recursion has an endpoint projection like  $S$ . If the source is a Catalan split measure but the finite endpoint transition has a different matrix, which rare-event exponents and coefficients persist? The useful target is a proved uniform-remainder theorem, not merely a formal matrix inverse. *Manuscript 11.* Identify other recursively decomposable classes in which an extreme adjacency statistic has a tight ordinary limit but a nontrivial  $n^{1/2}$ -weighted macroscopic measure. The proof suggests a

mechanism: a summable critical outer-word series, a square-root Catalan split law, and a finite endpoint acceptance matrix. These are hypotheses to verify in each class, not a universal theorem already proved. Symmetries preserving absolute adjacent differences give immediate equivalent formulations, but a genuinely new class needs a separate decomposition.

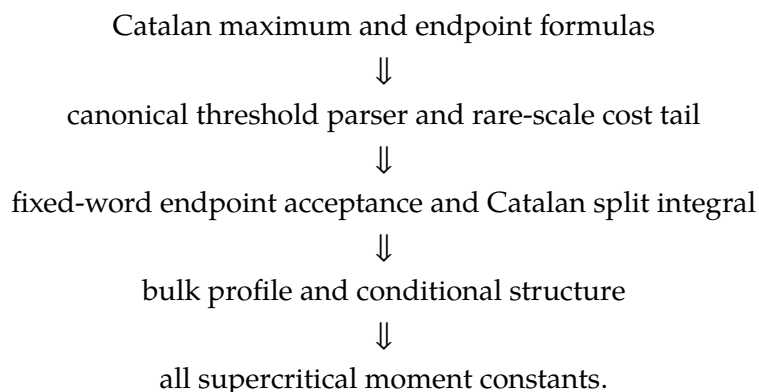
### 47.13 Formalization, certified and efficient computation

*Manuscript 10.* A Lean development could first formalize the finite decomposition, proposition 39.1, the endpoint counts, and the stopping-word injection. The analytic layer would then add the uniform tail envelope, the Riemann-sum source theorem, and the bounded-sequence contraction argument. The last two steps deserve separate reusable interfaces. No claim is made that these declarations already exist in the repository or that the current Python checks constitute a formal proof. *Manuscript 11.* Formalize the threshold parser and the rare-scale tail argument, and improve the exact endpoint implementation for large linear bounds. Natural proof-assistant milestones are the reconstruction bijection, the word generating-function identities, the convolution tail estimate, and the fixed-word endpoint acceptance lemma. The dominated-convergence step should then expose its precise  $p > 1/2$  hypothesis. For computation, a batched treatment of all bounds could avoid rebuilding overlapping endpoint states. Neither a new formal library nor an optimal complexity claim is supplied.

## 48 Dependency audit, scope and conclusion

### 48.1 The two dependency chains

Manuscript 11's proof dependency is



Manuscript 10's replaces the second and third lines by the exact deficit recursion with the stopping-word envelope, and the truncated renewal equation with its contraction proof. Neither depends on a fixed- $m$  exponential growth approximation, an unproved component ordering, a presumed uniform spectral gap, or the asymptotic theorems of Parts I–III. The finite-state recurrence is used for computations and is explicitly attributed to its prior source.

### 48.2 What is resolved

The advances claimed are deliberately specific. The positive linear-bound range  $m/n \in (1/2, 1)$  now has an explicit relative equivalent. Every fixed moment order above the threshold  $1/2$  now has a proved leading constant, including the previously undetermined mean and variance constants. The conditional rare event has an explicit two-block geometry and a universal outer-word law. The entire linear-bound problem is not claimed solved, because smaller bounds are only shown to have lower-order probability, with polynomial lower bounds (manuscript 11's summary, extended by manuscript 10's lower bounds).

Table 17: Claim-by-claim status of Part IV (manuscripts 10 and 11).

| Claim                                                                                     | Basis and limitation                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exact root identity (10)                                                                  | Direct finite comparison of all possible adjacent jumps; independently checked on all avoiders through size 11.                                                                                 |
| Uniform tail envelope (both)                                                              | An injective stopping-word reconstruction and exact Catalan coefficient identities (10); the canonical parser and a convolution bound (11).                                                     |
| Marked rare-event measure (10)                                                            | Endpoint limits, an explicit mesoscopic remainder bound, and a contraction proof of convergence.                                                                                                |
| Bulk profile for $0 < d/n < 1/2$ (both)                                                   | Two written proofs, each including rare-scale tail control.                                                                                                                                     |
| Scaled zero limit at and above $d/n = 1/2$ (both)                                         | Written monotonicity proof; no finer equivalent.                                                                                                                                                |
| All fixed- $p > 1/2$ moment constants (both)                                              | Evaluation of that measure and a proved uniform integrable envelope for moment integrals.                                                                                                       |
| Conditional word and split laws (11); three channels (10)                                 | Written joint convergence proof (11); a sampling description of the limiting marked pair (10).                                                                                                  |
| Deficit-size-biased uniform law (10)                                                      | A change of variable in the limiting weighted density; not the unweighted permutation law.                                                                                                      |
| Sub-half-size regimes (10, 11)                                                            | Constructive polynomial lower bounds only (10), $o(n^{-1/2})$ (both); matching upper bounds and equivalents remain open.                                                                        |
| Computations (both)                                                                       | Exact integer finite tests and exact symbolic checks (11); explicitly labelled Monte Carlo illustrations (10); rounded displays of exact counts (11). No asymptotic claim rests on computation. |
| New Lean or Rocq declarations                                                             | None supplied or claimed.                                                                                                                                                                       |
| Novelty, review and formal status                                                         | New relative to the inspected report's stated scope; not an exhaustive priority assessment, external review or a proof-assistant verification.                                                  |
| Local limit, critical window, smaller-bound bulk, centered half-moment, exact TV constant | Open.                                                                                                                                                                                           |

### 48.3 Conclusion

*Manuscript 10.* The main obstruction in passing from a boundary law to exact moments is not integration but missing probability mass at macroscopic distances. The present work identifies that mass. A two-block Catalan source, modified by a finite endpoint renewal, produces the explicit measure (156); its deficit marginal gives the complete first-order profile and all supercritical constants. The proof also isolates a useful methodological lesson. A finite exact identity can remove an apparently necessary subtree statistic. Once this identity is combined with a uniform counting envelope, the mesoscopic remainder becomes summable and a heuristic renewal equation becomes a convergent one. The equal three-channel endpoint mixture and the size-biased uniform law then emerge from an elementary explicit calculation.

*Manuscript 11.* The fixed-deficit law and the macroscopic deficit measure describe different parts of the same random statistic. The former governs ordinary probability and low moments; the latter accounts for rare configurations large enough to determine high moments. A canonical threshold parser makes that second regime accessible without importing a nonuniform spectral estimate. Its summable but heavy-tailed outer structure contributes an endpoint-weighted mass of three, and the remaining Catalan split integral can be evaluated exactly. This turns an order-of-magnitude moment statement into a complete family of explicit leading constants while leaving clear, nontrivial questions at the critical window and beyond it.

## 49 Small exact examples and normalization checks

This section was manuscript 10's Appendix B. For  $n = 3$ , the five avoiders are 123, 213, 231, 312, 321. Their deficits are respectively 2, 1, 1, 1, 2. For example,  $\pi = 213$  is an append operation on 21:

$$D(213) = \min\{D(21) + 1, L(21)\} = \min\{2, 1\} = 1.$$

For  $\pi = 231$ , the nonempty right block is 1 and the left size is one:

$$D(231) = \min\{F(1), 2 + D(1)\} = 1.$$

These examples illustrate why the append state needs the last value while a nonempty right block removes the left block's internal maximum.

For the marked source, the total tail integral is

$$\int_u^{1-u} [y(1-y)]^{-3/2} dy = 4 \frac{1-2u}{\sqrt{u(1-u)}}.$$

Multiplying by the source coefficient  $1/(4\sqrt{\pi})$  gives the unmarked two-block source. The full renewal contributes a factor three, so its tail is exactly (150). The folding factor two and the three endpoint channels are distinct factors; conflating them would produce an incorrect constant.

Finally, the size-biased density integrates to one because

$$\frac{1}{2} \int_0^{1/2} x^{-1/2} (1-x)^{-3/2} dx = \left[ \sqrt{\frac{x}{1-x}} \right]_0^{1/2} = 1.$$

This independently checks the normalization linking the marked source, the mean constant, and the uniform-law transformation.

## 50 Repository pins, literature scope and merge

### 50.1 Manuscript 10's pin and literature scope

This subsection was manuscript 10's Appendix A. The inspected repository was pinned at commit

1ee53d57de253d16cdaad79d1f54bbd95d76d682

(root tree c772efcde0cc2db7902005598190606da6cb338c). The relevant directory is

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/adjacency-bounded-132-avoiders/.
```

Its README at that pin describes the merger of three reports. The third, *The Largest Jump Is Almost Maximal*, was placed with the 03-largest-jump- prefix and appears as Part III of the combined article. The README explicitly excludes a bulk joint limit and exact leading supercritical moment constants. The original standalone source, `article(20260929-161704).tex`, was also consulted through Vladimir's file library to inspect the questions and proofs in contiguous form; its own reference pin is 8d936ee23, not manuscript 10's.

Manuscript 10 uses the same notation for the largest-jump deficit, credits the stopping-word construction, and answers the designated bulk and moment questions. Its proof does not require the earlier algebraic probability generating function, a total-variation theorem, or any fixed-bound spectral asymptotic. Those statements are contextual results, not unverified premises imported into the present argument.

A targeted literature check used the arXiv records of Nadler and Mayama–Akita, the latter's available HTML, and searches for largest adjacent differences in 132-avoiding permutations. No matching macroscopic formula was established from the retrieved material. This is not an exhaustive search or evidence sufficient to certify priority. External papers and earlier repository manuscripts are not redistributed. The full ledger is 04-macroscopic-jump-SOURCES.md.

## 50.2 Manuscript 11's pin and literature scope

Manuscript 11 inspected the repository snapshot

1ee53d57de253d16cdaad79d1f54bbd95d76d682,

the report at the repository-relative path

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/
adjacency-bounded-132-avoiders/article.tex,
```

with Git blob aeadaf364b083f7e25a475bd42b777cf926769b9, including the report's statements about its unresolved bulk and moment questions. It builds on the report's subject and elementary decompositions, but does not assume its probabilistic or spectral theorems. The primary external records inspected were the arXiv version-one records and HTML texts of [1, 2], accessed September 29, 2026. Targeted searches for maximum adjacent differences and largest jumps in 132-avoiders were also made. These checks locate the contribution relative to the inspected materials; they do not certify first-in-literature priority or exclude independent unpublished work. The full ledger is 05-macroscopic-deficits-SOURCES.md.

## 50.3 Where the merge had to choose

- *Placement.* Part IV follows Part III and precedes Part I's appendices, which contain no numbered equations, tables or figures; no number of Parts I–III changed. Manuscript 10's two appendices are printed as section 49 and the first subsection of section 50; manuscript 11's Appendix A (the recurrence) is section 46.3 and its Appendix B (reproduction) is replaced by section 46.8. Both title pages and abstracts are replaced by the opening note of Part IV, the status box of section 37.1 (which prints both research-status boxes) and a paragraph of the report's abstract; every claim of both abstracts is stated in section 38.

- *Base and order.* Manuscript 10 is the base. Its sections are printed in its order, with manuscript 11's parser and tightness sections beside its envelope (section 40), manuscript 11's proof of the profile after manuscript 10's (section 42), and manuscript 11's conditional structure, closed forms and exact data beside the corresponding material of manuscript 10.
- *Duplicates between the manuscripts.* Printed once, credited to both: the profile theorem (manuscript 10's Theorem 1.1 with manuscript 11's sequence form, Theorem 1.1 there), the deficit measure (manuscript 11's Theorem 1.2 as the last assertion of theorem 38.2), the moment theorem (both Theorems 1.3; one proof, with both domination arguments), the first-entry bound (lemma 39.2, both constants), the small- $u$  and half-edge expansions, the critical pole (proposition 44.1), and seven research questions. Kept as second routes: manuscript 11's uniform last-entry estimate (second proof of lemma 39.3), its all-deficit envelope (corollary 40.8, a second proof of proposition 40.3 up to the constant), its proof of the profile (section 42) and of the deficit measure (section 42.4). The constants checked equal: the mean constant  $3/\sqrt{\pi}$ ; the variance constants  $3(4 - \pi)/(4\sqrt{\pi})$  and  $(3/\sqrt{\pi})(1 - \pi/4)$ ; manuscript 10's size-biased form and manuscript 11's moment formula (the size-biased law has total mass  $\mathcal{I}(1) = 1$ , and manuscript 10's integral forms equal manuscript 11's  $m_p$  at  $p = 3/2$  and  $p = 3$  numerically and in general by the substitution of theorem 38.3); and the small- $u$  tail, which is Part III's  $3/\sqrt{\pi d}$ .
- *Duplicates with Parts I–III.* The maximum decomposition and positional restriction (Part I, section 2), the first- and last-entry counts (Part III, (98) and (101)), Part III's stopping words with their cost series, exact cost tail and disjointness (section 24, lemmas 24.1 and 25.2), and the endpoint recurrence (Part I, proposition 2.1) are printed once, in the earlier parts, and credited; both manuscripts rederive them by the same arguments, which are therefore not reprinted. What the manuscripts add to them (the uniform bounds of lemmas 39.2 and 39.3, the deficit bound (183), the bound in (184), the shortcut (243)) is printed with its proof.
- *Notation.* Every rename is listed in table 14: manuscript 11's  $\Phi \rightarrow \Psi$ ,  $x \rightarrow u$  (threshold),  $u \rightarrow y$  (split),  $\kappa_p \rightarrow m_p$  (both),  $I(p) \rightarrow \mathcal{I}(p)$ ,  $I_x \rightarrow I(u)$ ,  $k(u) \rightarrow \mathfrak{k}(y)$ ,  $W, W_R \rightarrow \mathcal{W}, \mathcal{W}_R$ ,  $r_t \rightarrow w_t^R$ ,  $T \rightarrow T_W$ ,  $K \rightarrow \text{len}(W)$ ,  $\alpha(w) \rightarrow \chi(w)$ ,  $r_n(w) \rightarrow \ell_n(w)$ ,  $E_{n,m}, E_n(w) \rightarrow \mathcal{E}_{n,m}, \mathcal{E}_n(w)$ ,  $U_x \rightarrow U^{(u)}$ ,  $H_s(u) \rightarrow \mathcal{N}_s(u)$ ,  $\mu \rightarrow \mu_D$ , its core  $\rho$  in the operation definitions (not printed), its mode indicator  $H_n$  (written out) and its  $T_n^{(m)}(u, v) \rightarrow T_{u,v}^{(m)}(n)$  (Part I's form); manuscript 10's  $E \rightarrow \mathcal{X}$ ,  $Y_n \rightarrow \Xi_n$ ,  $P \rightarrow S^*$ ,  $R_H \rightarrow \bar{r}_H$ ,  $h (= Pf) \rightarrow g$ ,  $c_h, c_f \rightarrow \bar{Z}_g, \bar{Z}_f$ ,  $E_h, E_f \rightarrow e_g, e_f$ , the resolvent weights  $r, a \rightarrow w_{\text{id}}, w_S$ ,  $H \rightarrow H^*$ ,  $\mathbb{P}^* \rightarrow \mathbb{P}^*$ , its corollary's  $Y, B, W \rightarrow U^{(u)}, \xi, \zeta$ ,  $q \rightarrow \theta_r$  in theorem 45.1,  $c_{\theta,r} \rightarrow \underline{c}_{\theta,r}$ ,  $A_r(\theta) \rightarrow \Omega_r(\theta)$ , and the left and right blocks  $L, R$  (written out). No normalization changed. Manuscript 10's  $\mathbb{P}(D_n > un)$  and manuscript 11's  $\mathbb{P}(D_n \geq \lceil xn \rceil)$  are kept as each stated them.
- *References to this report.* Both manuscripts cite Parts I–III as a repository report (their references “Repo” and “ProveIt”); here those citations are internal cross-references, including Part III's Sections 32.1 and 32.2, whose headings received labels in the merge. Their entries for Nadler and Mayama–Akita are Part I's.
- *Research questions.* Manuscript 10's ten directions and manuscript 11's ten questions are printed as thirteen subsections of section 47; seven pairs asked the same question and are printed once with both texts. Manuscript 10's question on a geometric conditional limit is re-scoped by a merge note, because manuscript 11's theorem 43.2 supplies part of it.
- *The merge's own text.* Section 37, section 43.4, the comparison with  $3/\sqrt{\pi(1 - \theta)n}$  in section 37.3, the  $m_p$  column of the table of closed forms, the sentences and parentheses explicitly marked as the merge's, the attributions “(manuscript 10)”, “(manuscript 11)”, “(both)”, and this list are the merge's.
- *Reproduction.* The manuscripts' package listings and commands are replaced by the shipped names and the copy recipes (section 46.8 and the README); their `article.tex`, `article.pdf`, `README.md` and `SHA256SUMS` are not shipped.
- *Figures.* The four figure PDFs were regenerated in the write phase by the delivered plot scripts,

run unchanged on copies of the delivered layouts with Matplotlib 3.10.8 and pdf . font type 42, so that they embed TrueType rather than Type 3 fonts; the delivered PDFs had Type 3 DejaVu and STIX fonts. The plotted data are the shipped data. The PNG previews in figures/ are the delivered files.

- *Stale sentences elsewhere.* The abstract, Part II's research direction on joint growth and its scope ledger (table 10), Part III's scope paragraph (section 21.3), its remark 29.3, the remark after corollary 30.1, its audit table (table 13), its research questions on the bulk profile and the moment constants, and Part I's section B carry dated pointers to Part IV; their original wording is kept.

## A Exact small-bound algebra

For  $m = 3$ , the certified generating function is  $A_3(x) = P_3(x)/Q_3(x)$ , where

$$\begin{aligned} P_3(x) &= 1 - x - x^2 + x^3 + 4x^4 - 4x^6 - 3x^7 - 3x^8 \\ &\quad - 5x^9 - 3x^{10} + 2x^{11} + 2x^{12}, \\ Q_3(x) &= (x - 1)(x + 1)(x^5 - x^4 - x^3 + x^2 - 2x + 1) \\ &\quad \cdot (x^6 + x^5 + x^4 + 2x^3 + x^2 - 1). \end{aligned}$$

For  $m = 4$ , the denominator factors as

$$\begin{aligned} Q_4(x) &= (x - 1) D_U(x) D_V(x), \\ D_U(x) &= x^{10} + x^8 + x^7 - 6x^5 - x^4 - x^3 - x^2 - x + 1, \\ D_V(x) &= x^{12} - x^{11} - x^{10} - x^9 + 5x^8 + 4x^7 + 4x^6 \\ &\quad + 4x^5 + 5x^4 + 3x^3 + x^2 - 1. \end{aligned}$$

Both displayed full denominators have constant term one. The numerator of  $A_4$ , listed from constant term upwards, has coefficient vector

$$(1, -1, -1, -1, 2, 9, 5, -3, -14, -10, -13, -23, -31, -11, 21, 32, 21, 1, -3, -5, -3, -3).$$

It has degree 21 and  $Q_4$  has degree 23. Thus the usual coefficient recurrence from  $Q_4 A_4 = P_4$  is valid for all  $n \geq 23$  when written using 23 preceding terms. The complete ascending coefficient arrays, full state certificates, and exact checks are in the JSON files.

## B Reproduction and certificate formats

The archive is arranged as follows:

|                                   |                                               |
|-----------------------------------|-----------------------------------------------|
| article.tex, article.pdf          | Main editable source and report               |
| README.md, requirements.txt       | Reproduction instructions                     |
| code/model.py                     | Exact endpoint model and combinatorics        |
| code/compute.py                   | Numerical proposals and exact symbolic solver |
| code/verify.py                    | Standard-library independent exact verifier   |
| code/plot_results.py              | Recreates the two separate figures            |
| data/sequences.csv                | $n = 0..200$ , $m = 1..12$                    |
| data/growth_constants.csv         | Components for $m = 2..20$                    |
| data/perron_certificates.json     | Rational bounds and positive integer vectors  |
| data/symbolic_certificate_m*.json | Polynomial state certificates, $m = 1..4$     |
| data/generating_functions.json    | Reduced rational functions, $m = 1..4$        |
| data/large_m_scalar_bounds.csv    | Scalar evaluations through $m = 1000000$      |

[Added 28 September 2026, batch 38: Part II's code, data and figures carry the prefix 02-centered-remainder- and are listed in section 20.2. Added 29 September 2026, batch 43: Part III's carry the prefix 03-largest-jump- and are listed in section 31.4. Added 29 September 2026, batch 52: Part IV's carry the prefixes 04-macroscopic-jump- and 05-macroscopic-deficits- and are listed in section 46.8.]

The primary reproducibility command needs no third-party Python packages:

```
python code/verify.py
```

To regenerate numerical and symbolic proposals, install the packages in requirements.txt and run

```
python code/compute.py --max-m 20 --symbolic-max 4 --large-scalars
python code/verify.py
python code/plot_results.py
```

The generator deliberately raises an exception if it cannot convert a numerical proposal to a valid exact certificate. A floating-point result alone is never accepted as a proof of a spectral inequality.

To build the report from the supplied figures:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The third pass settles page references and the table of contents if needed. The TeX source uses standard packages from a current full TeX Live installation; no font files are distributed.

The exact recurrence uses  $(m + 1)^2$  endpoint states and at most  $m + 1$  terms per state. Computing all coefficients through length  $N$  therefore uses  $O(Nm^3)$  integer multiply-add operations, after coefficient precomputation, and  $O(Nm^2)$  stored integers in the supplied straightforward implementation. These are arithmetic-operation counts, not bit-complexity bounds. Exact values have growing bit length. A rolling buffer can reduce storage when the complete coefficient table is not needed.

Perron certificates store integers  $p, D$  for each radius endpoint  $p/D$ , and a positive integer vector in the state order generated by the verifier. Symbolic certificates store all polynomial coefficients in ascending degree. The constant term of each common denominator is checked to be one. The certificate identities prove infinitely many coefficients because the formal system has a unique solution; the finite coefficient checks are additional consistency tests.

## C Source status and proof audit

The literature check was performed on September 19, 2026. The inspected primary items are the arXiv v1 versions listed below, including Nadler’s Conjecture 2 and Mayama–Akita’s Theorem 4.14 and Problem P3. The latter paper’s displayed numerical table and statement that its component comparison was unproved were also inspected in the PDF. Targeted searches used the paper titles and identifiers together with strict monotonicity, growth constants, and asymptotic refinements. No later solution was located in that search. This establishes the specific source-based status used here; it is not an assertion that every possible publication has been searched.

The logical dependencies of the new claims are particularly short. The strict-growth theorem uses the exact endpoint recurrence, the two irreducible components, their root characterization, the proper-principal Perron lemma, and the explicit shift of thresholds. This report includes proofs of those prerequisites. The asymptotic theorem instead uses the maximum-decomposition majorant, the injective skew-block construction, the first-entry Catalan formula, and the truncated-tail lemma. In particular, no numerical fit, unproved component-ordering pattern, or formal passage from finite checks to all  $m$  is used.

The work has not been formalized in a proof assistant. The integer and polynomial certificates are executable verification of the stated finite algebraic claims; they should not be described as a formal verification of the analytic limiting argument or of the all- $m$  theorem.

## References

- [1] N. Nadler, *On 132-Avoiding Permutations with an Adjacency Constraint*, arXiv:2604.22135v1, submitted April 24, 2026. <https://arxiv.org/abs/2604.22135>. Conjecture 2 is in Section 5.2.

- [2] T. Mayama and D. Akita, *Finite-state enumeration of adjacency-constrained 132-avoiding permutations*, arXiv:2605.23519v1, submitted May 22, 2026. <https://arxiv.org/abs/2605.23519>. Relevant locations: Theorem 2.4, Proposition 2.7, Theorems 4.12 and 4.14, and Problem P3 in Section 5.
- [3] S. Janson, *Simply generated trees, conditioned Galton–Watson trees, random allocations and condensation*, arXiv:1112.0510, 2011. <https://arxiv.org/abs/1112.0510>. Cited in Part III as background only.
- [4] S. Janson, *Patterns in random permutations avoiding the pattern 132*, arXiv:1401.5679, 2014. <https://arxiv.org/abs/1401.5679>. Cited in Part III as background only.
- [5] P. Flajolet and A. Odlyzko, *Singularity analysis of generating functions*, SIAM Journal on Discrete Mathematics **3**(2) (1990), 216–240. DOI: 10.1137/0403019. Manuscript 02 records that it did not inspect the PDF of this paper (03-largest-jump-SOURCES.md).